1.1.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 5, use the following scenario to identify the expression, equation, or inequality described.

> To support a local senior citizens center, a student club sent a flier home to the n students in the school. The flier said, "Please bring in money to support the senior citizens center. Paper money and coins accepted!" Their goal is to raise T dollars.

Choose the expression, equation, or inequality, from the list, that describes the given quantity.

1. The dollar amount the club would have if they reached half of their goal.

a. $T - 0.5n$
b. $0.5n$
c. $0.25n$
d. $0.5T$
e. $T + 50$

2. The dollar amount the club would have if every student at the school donated 50 cents to the cause.

a. $T - 0.5n$
b. $0.5n$
c. $0.25n$
d. $0.5T$
e. $T + 50$

3. The dollar amount the club could donate if they made $50 more than their goal.

a. $T - 0.5n$
b. $0.5n$
c. $0.25n$
d. $0.5T$
e. $T + 50$

4. The dollar amount the club would still need to raise to reach its goal after every student at the school donated 50 cents.

a. $T - 0.5n$
b. $0.5n$
c. $0.25n$
d. $0.5T$
e. $T + 50$

5. The dollar amount the club would have if half of the students at the school each gave 50 cents.

a. $T - 0.5n$
b. $0.5n$
c. $0.25n$
d. $0.5T$
e. $T + 50$

6. Each of the 10 students in the baking club made 2 chocolate cakes for a fundraiser. They all used the same recipe, using c cups of flour in total.

Write an expression that represents the amount of flour required for one cake.

a. $\frac{2c}{10}$
b. $c - 20$
c. $\frac{c}{20}$
d. $20c$

For questions 7 - 8, determine if each is an expression or an equation that represents the English phrase

1.2.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Translate to an equation: Which equation means nine more than x is equal to 52?

a. $\frac{52}{x} = 9$
b. $9 - x = 52$
c. $9x = 52$
d. $x + 9 = 52$

2. Translate to an equation: Which equation means ten less than m is −14?

a. $-14 - m = 10$
b. $m - 10 = -14$
c. $m - 10 = 14$
d. $10 - m = -14$

3. Translate to an equation: Which equation means the sum of y and −30 is 40?

a. $y + (-30) = 40$
b. $y + 40 = -30$
c. $y = 40 + (-30)$
d. $-30y = 40$

4. Translate to an equation: For a family birthday dinner, Celeste bought a turkey that weighed 5 pounds less than the one she bought for Thanksgiving. The birthday turkey weighed 16 pounds. Which equation represents how much the Thanksgiving turkey weighed?

a. $-x + 5 = 16$
b. $x + 5 = 16$
c. $x - 5 = 16$
d. $5 - x = 16$

5. Translate to an equation: Which equation represents 45% of 120?

a. $x = 45 + 120$
b. $x = 0.45(120)$
c. $x = 4.5(120)$
d. $x = 45(120)$

6. 250% of 65 can also be written as:

a. $x = 250(65)$
b. $x = 25(65)$
c. $x = 0.25(65)$
d. $x = 2.5(65)$

7. Translate to an equation: Geneva treated her parents to dinner at their favorite restaurant. The bill was $74.25. Geneva wants to leave 16% of the total bill as a tip. How much should the tip be?

a. $t = 16 + (74.25)$
b. $t = 1.6(74.25)$
c. $t = 0.16(74.25)$
d. $t = 16(74.25)$

8. Large cheese pizzas cost $5 each, and large one-topping pizzas cost $6 each.
Choose the equation that represents the total cost, T, of c large cheese pizzas and d large one-topping pizzas.

a. $T = 6c - 5d$
b. $T = 5c - 6d$

1.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 2, use the following scenario to identify the equation described.

Two different telephone carriers offer the following plans that a person is considering. Company A has a monthly fee of $20 and charges $0.05/minute for calls. Company B has a monthly fee of $5 and charges $0.10/minute for calls.

1. Find the model for the entire cost, c, when Company A's plan is used for m minutes.

a. $c = 5 + 0.05m$
b. $c = 20 + 0.10m$
c. $c = 5 + 0.10m$
d. $c = 20 + 0.05m$

2. Find the model for the entire cost, c, when Company B's plan is used for m minutes.

a. $c = 5 + 0.05m$
b. $c = 20 + 0.10m$
c. $c = 5 + 0.10m$
d. $c = 20 + 0.05m$

3. If a carpenter sawed a 10-ft board into two sections and one section was n ft long, how long would the other section, S, be in terms of n?

a. $S = \frac{10}{2}n$
b. $S = 10 - 2n$
c. $S = 10 - n$
d. $S = \frac{10}{n}$

Use the table below for problems 4 and 5:

Tyler needs to complete this table for his consumer science class. He knows that 1 tablespoon contains 3 teaspoons and that 1 cup contains 16 tablespoons.

Number of Teaspoons	Number of Tablespoons	Number of Cups
a. ____	b. ____	2
36	12	c. ____
d. ____	48	3

4. For each of the missing spaces in the table above, a-d, select the missing value that helps complete the table.

a. _____

a. 144
b. 96
c. 64
d. 32
e. $\frac{3}{4}$

b. _____

a. 144
b. 96

1.4.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Is $y = \frac{4}{3}$ a solution of $9y + 2 = 6y + 3$?

a. Yes
b. No

2. Is $y = \frac{7}{5}$ a solution of $5y + 3 = 10y - 4$?

a. Yes
b. No

3. Is $y = \frac{5}{3}$ a solution of $6y + 10 = 12y$?

a. Yes
b. No

4. Is $u = -\frac{1}{2}$ a solution of $8u - 1 = 6u$?

a. Yes
b. No

5. Tickets to the museum cost \$4.50 for children and \$9.00 for adults. The equation $63 = 4.5c + 9a$ represents the total cost for c children's tickets and a adult tickets. How many children's tickets were sold if 4 adult tickets were sold?

For questions 6 and 7, use the following scenario: T-shirts are on sale for \$12.50. Isla has a coupon for \$10.00 off.

6. Choose the equation that represents the total cost, c, if she buys n T-shirts.

a. $c = 12.5(10n)$
b. $c = 12.5(n - 10)$
c. $c = 10n - 12.5$
d. $c = 12.5n - 10$

7. How many T-shirts can she buy if she spends all of her \$40?

8. An artist is selling children's crafts. Necklaces cost \$2.25 each, and bracelets cost \$1.50 per each. Select **three** combinations of necklaces and bracelets that the artist could sell for exactly \$12.00

a. 5 necklaces and 1 bracelet
b. 2 necklaces and 5 bracelets
c. 3 necklaces and 3 bracelet
d. 4 necklaces and 2 bracelets
e. 3 necklaces and 5 bracelets
f. 6 necklaces and no bracelets
g. No necklaces and 8 bracelets

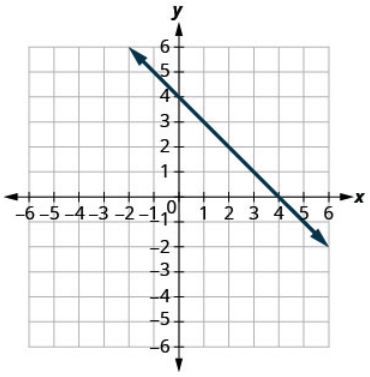
9. Volunteer drivers are needed to bring 80 students to the championship baseball game. Drivers either have cars, which can seat 4 students, or vans, which can seat 6 students. The equation $4c + 6v = 80$ describes the relationship between the number of cars, c, and number of vans, v, that can transport exactly 80 students. Select **four** statements that are true about the situation.

a. If 12 cars go, then 2 vans are needed.
b. $c = 14$ and $v = 4$ and are a pair of solutions to the equation.
c. If 6 cars go and 11 vans go, there will be extra space.
d. 10 cars and 8 vans isn't enough to transport all the students.
e. If 20 cars go, no vans are needed.
f. 8 vans and 8 cars are numbers that meet the constraints in this situation.

1.5.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the graph for questions 1 - 2.



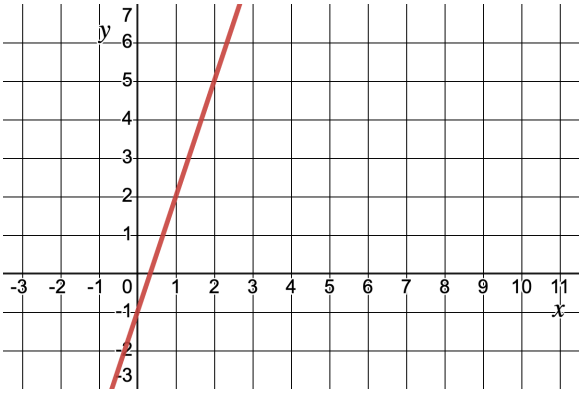
1. Which of the following ordered pairs are on the line? Select **three** answers.

a. $(0, 4)$
b. $(2, 2)$
c. $(-2, 6)$
d. $(-1, 3)$
e. $(-4, 0)$

2. Which of the following ordered pairs are solutions to the equation? Select **three** solutions.

a. $(0, 4)$
b. $(2, 2)$
c. $(-2, 6)$
d. $(-1, 3)$
e. $(-4, 0)$

For questions 3 and 4, use the graph of $y = 3x - 1$ as shown.



3. Which of the following ordered pairs are on the line? Select **three** solutions.

a. $(0, -1)$
b. $(1, 2)$
c. $(2, 5)$
d. $(1, 1)$
e. $(0, 1)$

4. Which of the following ordered pairs are solutions to the equation? Select **three** solutions.

a. $(0, -1)$
b. $(1, 2)$

1.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. **Select three** equations that have the same solution as the equation $3x - 12 = 24$.

   a. $15x - 60 = 120$
   b. $3x = 12$
   c. $3x = 36$
   d. $x - 4 = 8$
   e. $12x - 12 = 24$

2. **Select three** equations that have the same solution as $2x - 5 = 15$.

   a. $2x = 10$
   b. $2x = 20$
   c. $2(x - 5) = 15$
   d. $2x - 20 = 0$
   e. $4x - 10 = 30$
   f. $15 = 5 - 2x$

3. **Select four** of the equations that are equivalent to: $x + 0.4x - 4.2 = 12.8$.

   a. $1.4x - 4.2 = 12.8$
   b. $1.4x = 8.6$
   c. $1.4x = 17$
   d. $14x - 42 = 128$
   e. $10x + 4x - 42 = 12.8$
   f. $14x = 170$

4. **Select two** of the following that are equivalent to $3x - 6 = 12$.

   a. $3x = 6$
   b. $3(x - 2) = 12$
   c. $x - 2 = 4$
   d. $x - 2 = 12$
   e. $x - 6 = 4$

5. Identify which step the mistake is made in finding an equivalent equation to $2(x - 4) = 16$.

   Step 1: $2x - 4 = 16$
   Step 2: $2x = 20$
   Step 3: $x = 10$

   a. Step 1
   b. Step 2
   c. Step 3

6. Identify which step the mistake is made in finding an equivalent equation to $2x + 3x - 4 = 20$.

   Step 1: $5x - 4 = 20$
   Step 2: $5x = 16$
   Step 3: $x = 16/5$

   a. Step 1
   b. Step 2
   c. Step 3

7. Identify which step the mistake is made in finding an equivalent equation to $7x - 4x - 5 = 16$.

   Step 1: $3x - 5 = 16$
   Step 2: $3x = 21$

1.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 3, choose an equivalent equation.

1. Which equation is equivalent to $3x + 6 = 4x + 7$?

a. $3x = 4x + 7$
b. $3x - 1 = 4x$
c. $9x = 4x + 7$
d. $3x + 18 = 4x + 7$
e. $7x = 1$

2. Which equation is equivalent to $3(x + 6) = 4x + 7$?

a. $3x = 4x + 7$
b. $3x - 1 = 4x$
c. $9x = 4x + 7$
d. $3x + 18 = 4x + 7$
e. $7x = 1$

3. Which equation is equivalent to $4x + 3x = 7 - 6$?

a. $3x = 4x + 7$
b. $3x - 1 = 4x$
c. $9x = 4x + 7$
d. $3x + 18 = 4x + 7$
e. $7x = 1$

4. **Select three** equations that have 0 as a solution.

a. $2x + 10 = 4x + 10$
b. $4x = 9x$
c. $2x - 3 + 4x = 4 - 7$
d. $5x + 7 = 0$
e. $3(x - 2) = 9$

5. Which equation is equivalent to $6(x - 3) = 24$?

a. $6x = 27$
b. $6x = 21$
c. $x - 3 = 4$
d. $6x - 3 = 24$

6. Which of the following sets of equations are equivalent to each other?

a. $6x - 4 = 11$ and $6x = 7$
b. $5x - 2 = 7x + 4$ and $12x - 2 = 4$
c. $7(x + 2) = 21$ and $7x + 2 = 21$
d. $10(x + 9) = 50$ and $2(x + 9) = 10$

7. Using the distributive property, choose which equation would be an equivalent equation to $3(x - 2) = 7$.

a. $3x = 9$
b. $x - 2 = 4$
c. $3x - 6 = 7$
d. $3x - 2 = 7$

1.8.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4 use the problem below:

> Priya is buying raisins and almonds to make trail mix. Almonds cost $5.20 per pound and raisins cost $2.75 per pound. Priya spent $11.70 buying almonds and raisins. The relationship between pounds of almonds, pounds of raisins, and the total cost is represented by the equation $5.2a + 2.75r = 11.7$.

How many pounds of raisins did Priya buy if she bought the following amount of almonds?

1. 2 pounds of almonds

a. 0.473
b. 0.946
c. 2.75
d. 1.192

2. 1.06 pounds of almonds

a. 2.811
b. 5.5
c. 1.689
d. 2.25

3. 0.64 pounds of almonds

a. 8.371
b. 3.044
c. 1.911
d. 9.94

4. a pounds of almonds

a. $r = \frac{11.7-5.2a}{2.75}$
b. $r = \frac{2.75}{11.7-5.2a}$
c. $r = \frac{11.7-2.75a}{5.2}$
d. $r = \frac{11.7+5.2a}{275}$

For questions 5 – 6 use the scenario below:

> A car traveled 180 miles at a constant rate.

5. For each of the missing spaces in the table, a-d, select the missing value that completes the table to show the rate at which the car was traveling if it completed the same distance in each number of hours.

Travel Time (Hours)	Rate of Travel (Miles Per Hour)
5	a. ____
4.5	b. ____
3	c. ____
2.25	d. ____

a._____

1.9.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For numbers 1 – 9, use the following scenario:
A car has a 16-gallon fuel tank. When driven on a highway, it has a gas mileage of 30 miles per gallon. The gas mileage (also called "fuel efficiency") tells us the number of miles the car can travel for a particular amount of fuel (one gallon of gasoline, in this case). After filling the gas tank, the driver got on a highway and drove for a while.

1. How many miles has the car traveled if it has 15 gallons left in the tank?
2. How many miles has the car traveled if it has 10 gallons left in the tank?
3. How many miles has the car traveled if it has 2.5 gallons left in the tank?
4. Choose the equation that represents the relationship between the distance the car has traveled in miles, d, and the amount of gas left in the tank in gallons, x.

   a. $x = \frac{30}{d-480}$
   b. $d = 30(16 - x)$
   c. $x = \frac{d-480}{-30}$
   d. $d = 16(30 - x)$

5. How many gallons are left in the tank when the car has traveled 90 miles?
6. How many gallons are left in the tank when the car has traveled 246 miles?
7. Choose an equation that makes it easier to find the amount of gas left in the tank, x, if we know the car has traveled d miles.

   a. $x = \frac{30}{d-480}$
   b. $d = 30(16 - x)$
   c. $x = \frac{d-480}{-30}$
   d. $d = 16(30 - x)$

8. Noah was helping to collect the entry fees at his school's football game. Student entry costs $2.75 each and adult entry costs $5.25 each. By the end of the game, Diego had collected $281.25.

Select two equations that could represent the relationship between the number of students, s, the number of adults, a, and the dollar amount collected at the game.

   a. $281.25 - 5.25a = 2.75s$
   b. $a = 53.57 - \frac{2.75}{5.25}s$
   c. $x - 2 = 4$
   d. $x - 2 = 12$
   e. $x - 6 = 4$

1.10.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 4, use the problem and graph below.

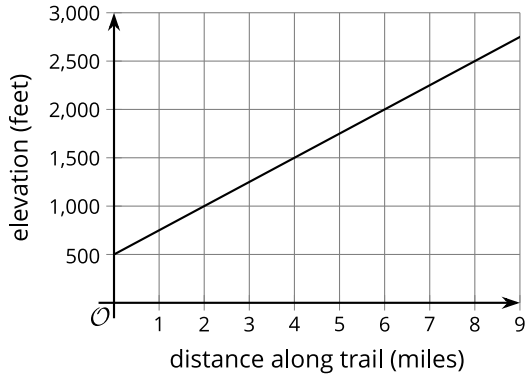
Andre bought a new bag of cat food. The next day, he opened it to feed his cat. The graph shows how many ounces were left in the bag on the days after it was bought.



1. How many ounces of food were in the bag 12 days after Andre bought it?
2. How many days did it take for the bag to contain 16 ounces of food?
3. How much did the bag weigh before it was opened?
4. Approximately many days did it take for the bag to be empty?

For 5 - 7, use the problem and graph below.

A group of hikers is progressing steadily along an uphill trail. The graph shows their elevation (or height above sea level), in feet, at each distance from the start of the trail, in miles.

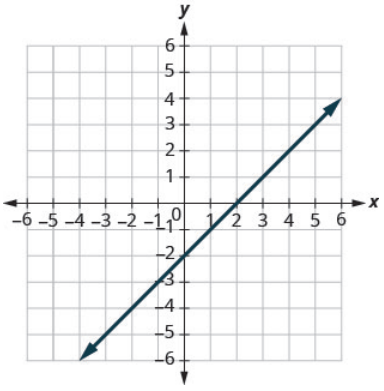


5. What is the slope of the graph?
6. What does the slope tell us about this situation?
   a. Only the elevation (feet).
   b. Only the distance along the trail (miles).
   c. The change in miles as the elevation increases.
   d. The change in elevation as the hikers hike more miles.
7. What is the equation that represents the relationship between the hikers' distance from the start of the trail, x, and their elevation, y?
8. A kindergarten teacher bought \$21 worth of stickers and cardstock for his class. The stickers cost \$1.50 a sheet and the cardstock cost \$3.50 per pack. The equation $1.5s + 3.5c = 21$ represents the relationship between sheets

1.11.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For numbers 1 – 2, use the graph below:



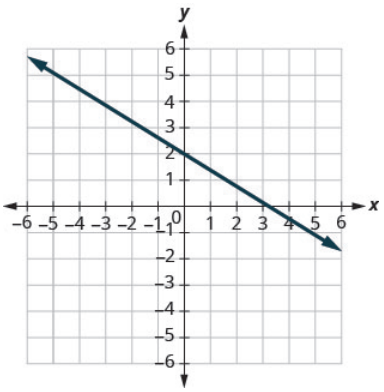
1. What are the coordinates of the x-intercept of the graph?

a. $(-2, 0)$
b. $(0, -2)$
c. $(2, 0)$
d. $(0, 2)$

2. What are the coordinates of the y-intercept of the graph?

a. $(-2, 0)$
b. $(0, -2)$
c. $(2, 0)$
d. $(0, 2)$

For numbers 3 – 4, use the graph below:



3. What are the coordinates of the x-intercept of the graph?

a. $(3, 0)$
b. $(0, 3)$
c. $(2, 0)$
d. $(0, 2)$

4. What are the coordinates of the y-intercept of the graph?

a. $(-2, 0)$
b. $(0, -2)$
c. $(2, 0)$
d. $(0, 2)$

1.12.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

In 1 – 2, find the equation of a line with given slope and y-intercept. Write the equation in slope-intercept form.

1. slope 3 and y-intercept $(0, 5)$

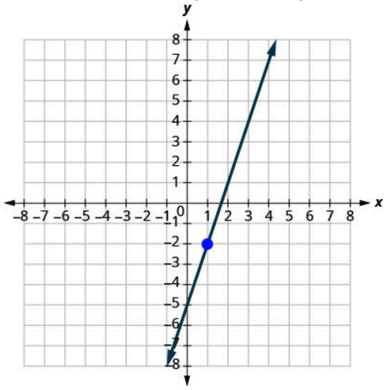
a. $y = 0x + 3$
b. $y = 3x + 5$
c. $y = 3x + 0$
d. $y = 5x + 3$

2. slope -3 and y-intercept $(0, -1)$

a. $y = 0x - 3$
b. $y = -x + 0$
c. $y = -3x - 1$
d. $y = -x - 3$

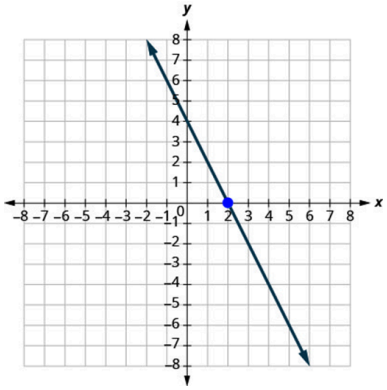
In 3 – 4, Choose the slope-intercept form of the equation of the line, shown in each graph.

3.



a. $y = 3x - 5$
b. $y = -5x + 3$
c. $y = 5x - 3$
d. $y = -3x + 5$

4.



a. $y = \frac{1}{2}x + 2$
b. $y = \frac{1}{2}x - 4$
c. $y = 2x + 4$
d. $y = -2x + 4$

5. Choose the equation of a line with the given slope and containing the given point that best represents the line in slope-intercept form.

1.13.5 Practice

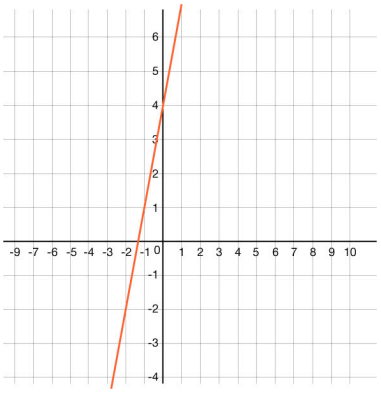
Complete the following questions to practice the skills you have learned in this lesson.

1. What is the slope of the line that goes through the points in the given table of values?

x	1	2	3	4
y	6	11	16	21

2. In the equation, $y = 4x + 2$, what is the value of the y-intercept?

3. Identify the slope in the given graph.



4. Which of the following is an equation for the table of values?

x	0	1	2	3
y	2	7	12	17

a. $y = 5x - 2$
b. $y = 7x - 2$
c. $y = 5x + 2$
d. $y = 2x + 5$

5. For the equation $y = -2x + 3$, what is the value when $x = 2$?

6. Jonah started the month with $40 and earned $15 each week walking dogs for neighbors. How much does he have after w weeks?

a. $y = 40(15w)$
b. $y = 40w + 15$
c. $y = 40w$
d. $y = 15w + 40$

7. Write the slope of the given table.

x	0	2	4
y	5	11	17

1.14.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

In questions 1 – 5, choose an equation of a line parallel to the given line and contains the given point. The choices are in slope-intercept form.

1. Which line is parallel to $y = 4x + 2$ and contains the point $(1, 2)$?

a. $y = -\frac{1}{4}x + 2$
b. $y = -\frac{1}{4}x - 2$
c. $y = 4x - 2$
d. $y = 4x + 2$

2. Which line is parallel to $y = -3x - 1$ and contains the point $(2, -3)$?

a. $y = \frac{1}{3}x - 3$
b. $y = \frac{1}{3}x + 3$
c. $y = -3x - 3$
d. $y = -3x + 3$

3. Which line is parallel to $2x - y = 6$ and contains the point $(3, 0)$?

a. $y = 2x - 6$
b. $y = -2x$
c. $y = -2x + 6$
d. $y = 2x$

4. Which line is parallel to $x = -4$ and contains the point $(-3, -5)$?

a. $y = -4$
b. $x = -5$
c. $y = -5$
d. $x = -3$

5. Which line is parallel to $y = 5$ and contains the point $(2, -2)$?

a. $y = 2$
b. $y = -2$
c. $x = 5$
d. $x = 2$

In questions 6 – 10, find an equation of a line that is perpendicular to the given line and contains the given point. The choices are in slope-intercept form.

6. Which line is perpendicular to $y = -2x + 3$ and contains the point $(2, 2)$?

a. $y = \frac{1}{2}x + 1$
b. $y = \frac{1}{2}x + 3$
c. $y = -\frac{1}{2}x + 3$
d. $y = -2x + 6$

7. Which line is perpendicular to $y = -x + 5$ and contains the point$(3, 3)$?

a. $y = -x + 3$
b. $y = x + 5$
c. $y = x$
d. $y = -x + 6$

8. Which line is perpendicular to $y = \frac{3}{4}x - 2$ and contains the point $(-3, 4)$?

a. $y = \frac{3}{4}x + \frac{17}{4}$
b. $y = \frac{3}{4}x + 4$
c. $y = -\frac{4}{3}x + 4$

1.15.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. If y varies directly as x and $y = 14$, when $x = 3$, find the equation that relates x and y.

2. The price, P, that Eric pays for gas varies directly with the number of gallons, g, he buys. It costs him $50 to buy 20 gallons of gas. Choose the equation that relates P and g.

   a. $P = \frac{g}{2.5}$
   b. $G = \frac{2.5}{p}$
   c. $P = \frac{2.5}{g}$
   d. $P = 2.5g$

3. Joseph is traveling on a road trip. The distance, d, he travels before stopping for lunch varies directly with the speed, v, he travels. He can travel 120 miles at a speed of 60 mph. How many miles would he travel before stopping for lunch at a rate of 65 mph?

4. The mass of a liquid, M, varies directly with its volume, V. A liquid with mass 16 kilograms has a volume of 2 liters. What is the volume, in liters, of this liquid if its mass is 128 kilograms?

5. The length, L, that a spring stretches varies directly with a weight, W, placed at the end of the spring. When Sarah placed a 10-pound watermelon on a hanging scale, the spring stretched 5 inches. What weight of watermelon, in pounds, would stretch the spring 6 inches?

6. The maximum load a beam, M, will support varies directly with the square of the diagonal of the beam's cross-section, d^2. A beam with diagonal 6 inch will support a maximum load of 108 pounds. What load, in pounds, will a beam with a 10-inch diagonal support?

7. The number of calories, c, burned varies directly with the amount of time, t, spent exercising. Arnold burned 312 calories in 65 minutes exercising. How many calories would he burn if he exercises for 90 minutes?

2.1.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Access the Desmos guide PDF (https://openstax.org/r/desmos-guide) for tips on solving problems with the Desmos graphing calculator.

In questions 1 – 4, determine if the following points are solutions to the given system of equations.

1. Determine if the point is a solution to the system of equations: $\begin{cases} 2x - 6y = 0 \\ 3x - 4y = 5 \end{cases}$

a. $(1, \frac{3}{4})$
b. $(\frac{8}{7}, \frac{6}{7})$
c. $(-3, 4)$
d. $(3, 1)$

2. Determine if the point is a solution to the system of equations: $\begin{cases} x - 6y = 0 \\ 3x - 4y = 5 \end{cases}$

a. $(1, \frac{3}{4})$
b. $(\frac{8}{7}, \frac{6}{7})$
c. $(-3, 4)$
d. $(3, 1)$

3. Determine if the point is a solution to the system of equations: $\begin{cases} x + y = 2 \\ y = \frac{3}{4}x \end{cases}$

a. $(1, \frac{3}{4})$
b. $(\frac{8}{7}, \frac{6}{7})$
c. $(-3, 4)$
d. $(3, 1)$

4. Determine if the point is a solution to the system of equations: $\begin{cases} 4x + 4y = 7 \\ y = \frac{3}{4}x \end{cases}$

a. $(1, \frac{3}{4})$
b. $(\frac{8}{7}, \frac{6}{7})$
c. $(-3, 4)$
d. $(3, 1)$

5. Solve the system of equations by graphing: $\begin{cases} 3x + y = -3 \\ 2x + 3y = 5 \end{cases}$

a. $(4, -1)$
b. $(-3, 6)$
c. $(0, -3)$
d. $(-2, 3)$

Access multimedia content (http://openstax.org/books/algebra-1/pages/2-1-5-practice)

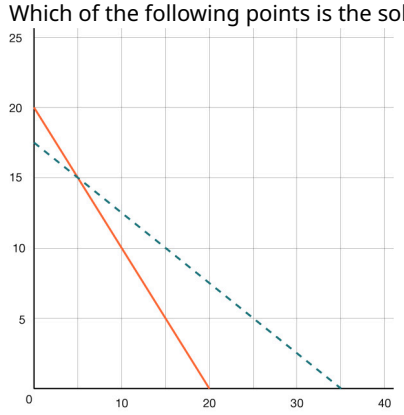
6. Solve the system of equations by graphing: $\begin{cases} y = x + 2 \\ y = -2x + 2 \end{cases}$

a. $(-2, 0)$
b. $(0, 2)$

2.2.5 Practice

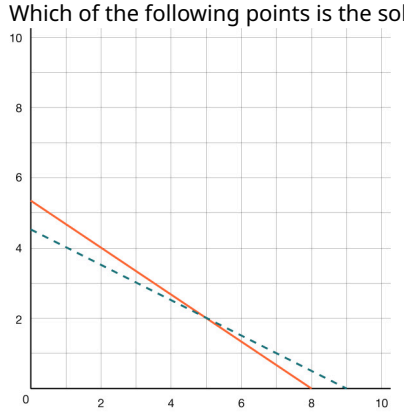
Complete the following questions to practice the skills you have learned in this lesson.

1. Which of the following points is the solution to the system of equations that is graphed?



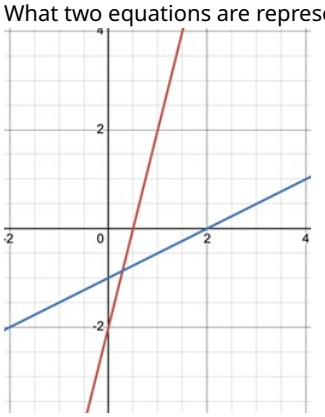
a. $(5, 15)$
b. $45, 0)$
c. $(0, 20)$
d. $(20, 0)$

2. Which of the following points is the solution to the system of equations that is graphed?



a. $(5.2, 0)$
b. $(8, 0)$
c. $(2, 5)$
d. $(5, 2)$

3. What two equations are represented by the graph? Select **two**.



2.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Lin is solving this system of equations:

$$\begin{cases} 6x - 5y = 34 \\ 3x + 2y = 8 \end{cases}$$

She starts by rearranging the second equation to isolate the y variable: $y = 4 - 1.5x$. She then substitutes the expression $4 - 1.5x$ for y in the first equation, as shown:

$6x - 5(4 - 1.5x) = 34$
$6x - 20 - 7.5x = 34$

$-1.5x = 54$
$x = -36$

$y = 4 - 1.5x$

$y = 4 - 1.5 \cdot (-36)$

$y = 58$

Does Lin's solution of $(-36, 58)$ makes both equations in the system true?

a. No; Her solution does not make both equations true.
b. yes; Her solution makes both equations true.

2. Which of the following is true of Lin's solution?

a. Lin's solution of $(-36, 58)$ is correct.
b. Lin made an addition error. The correct answer is $x = 0$ and $y = 4$, or $(0, 4)$.
c. Lin made a multiplication error. The correct answer is $x = 9$ and $y = 4$, or $(9, 4)$.
d. Lin made a multiplication error. The correct answer is $x = 4$ and $y = -2$, or $(4, -2)$.

In questions 3 –9, solve the systems of equations by substitution.

3. $\begin{cases} 2x + y = -2 \\ 3x - y = 7 \end{cases}$

a. $(2, 6)$
b. $(1, -4)$
c. $(0, -2)$
d. $(-1, 0)$

4. $\begin{cases} x - 3y = -9 \\ 2x + 5y = 4 \end{cases}$

a. $(-3, 2)$
b. $(0, 3)$
c. $(2, 0)$
d. $(-6, 1)$

5. $\begin{cases} -2x + 2y = 6 \\ 2y = -6x + 14 \end{cases}$

a. $(-3, 0)$
b. $(2, 1)$
c. $(0, 7)$
d. $(1, 4)$

2.4.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 8, solve the systems of equations using the elimination method.

1. $\begin{cases} 5x + 2y = 2 \\ -6x - 2y = 0 \end{cases}$

a. $(-2, 6)$
b. $(2, -4)$
c. $(0, 1)$
d. $(-3, 9)$

2. $\begin{cases} 2x - 5y = 7 \\ -15x + 5y = -85 \end{cases}$

a. $(-4, -3)$
b. $(6, 1)$
c. $(5, -2)$
d. $(7, 4)$

3. $\begin{cases} 6x - 10y = -18 \\ 25x + 10y = 80 \end{cases}$

a. $(4, -2)$
b. $(0, 8)$
c. $(7, 6)$
d. $(2, 3)$

4. $\begin{cases} -6x - 16y = 6 \\ 6x + 15y = -9 \end{cases}$

a. $(7, -2)$
b. $(1, -1)$
c. $(-9, 3)$
d. $(-14, 5)$

5. $\begin{cases} -15x - 40y = -335 \\ 15x + 9y = 180 \end{cases}$

a. $(6, 10)$
b. $(17, 2)$
c. $(9, 5)$
d. $(12, 0)$

6. $\begin{cases} x - 3y = -9 \\ x + \frac{5}{2}y = 2 \end{cases}$

a. $(-9, 0)$
b. $(2, 0)$
c. $(0, 3)$
d. $(-3, 2)$

2.5.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, add the equations in the system to find a new equation.

1. $\begin{cases} 6x - 5y = -1 \\ -6x - 3y = -39 \end{cases}$

a. $-8x = -40$
b. $-8y = -40$
c. $-2y = -40$
d. $2y = -38$

2. $\begin{cases} 5x - 3y = -1 \\ -6x + 3y = -6 \end{cases}$

a. $-11x = -7$ or $11x = 7$
b. $-6x = -7$ or $6x = 7$
c. $-x = -7$ or $x = 7$
d. $x = -7$ or $-x = 7$

3. $\begin{cases} 4x - 3y = 3 \\ -4x - 10y = 62 \end{cases}$

a. $-13x = 65$
b. $-8y = 65$
c. $-8x = 65$
d. $-13y = 65$

4. $\begin{cases} 6x + 16y = -6 \\ -6x - 10y = 6 \end{cases}$

a. $6y = 12$ or $y = 6$
b. $12x = 0$ or $x = 0$
c. $6x = 0$ or $x = 0$
d. $6y = 0$ or $y = 0$

For questions 5 – 10, solve each system of equations.

5. $\begin{cases} 6x + 27y = -12 \\ -6x - 26y = 14 \end{cases}$

a. $(-20, 4)$
b. $(7, -1)$
c. $(-11, 2)$
d. $(2.5, -1)$

6. $\begin{cases} 3x + 4y = 8 \\ -3x + 6y = 42 \end{cases}$

a. $(0, 7)$
b. $(-4, 5)$
c. $(-14, 0)$
d. $(-10, 2)$

7. $\begin{cases} 5x - 9y = 68 \\ -5x + 6y = -47 \end{cases}$

a. $(1, -7)$

2.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Solve each system of equations to practice the skills you have learned in this lesson.

1. $\begin{cases} -2x + 5y = -42 \\ 7x + 2y = 30 \end{cases}$

a. $(6, -6)$
b. $(1, -8)$
c. $(11, -4)$
d. $(0, 15)$

2. $\begin{cases} 6x - 5y = -34 \\ 2x + 6y = 4 \end{cases}$

a. $(-9, -4)$
b. $(-4, 2)$
c. $(1, 8)$
d. $(2, 0)$

3. $\begin{cases} 5x - y = -2.6 \\ -4x - 6y = 1.4 \end{cases}$

a. $(-0.5, 0.1)$
b. $(-3, -12)$
c. $(0.5, 0.1)$
d. $(7, -4.9)$

4. $\begin{cases} 7x - 2y = 3 \\ 4x + 5y = 3.25 \end{cases}$

a. $(3, 9)$
b. $(-1, -5)$
c. $(0.5, 0.25)$
d. $(1, 2)$

5. $\begin{cases} 5x + 9y = 16 \\ x + 2y = 4 \end{cases}$

a. $(8, -2)$
b. $(-8, 6)$
c. $(3.2, 0)$
d. $(-4, 4)$

6. $\begin{cases} \frac{3}{2}x - \frac{1}{4}y = 5 \\ \frac{1}{3}x + \frac{1}{6}y = 2 \end{cases}$

a. $(-4, -5)$
b. $(-\frac{1}{5}, \frac{2}{3})$
c. $(1, -5)$
d. $(4, 4)$

7. $\begin{cases} \frac{1}{3}x + \frac{1}{9}y = 2 \\ -\frac{1}{2}x + \frac{4}{5}y = -3 \end{cases}$

a. $(-2, -5)$
b. $(8, -5)$
c. $(7, -3)$
d. $(6, 0)$

2.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 8, solve each system of equations to determine if there is one solution, no solution, or infinitely many solutions.

1. $\begin{cases} 5x - 3y = 7 \\ x - 0.6y = 1.4 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

2. $\begin{cases} -x + 2y = -1 \\ 5x - 10y = 6 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

3. $\begin{cases} 7x + 6y = 2 \\ -28x - 24y = -8 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

4. $\begin{cases} -0.2x + 0.4y = 0.6 \\ x - 2y = -3 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

5. $\begin{cases} -0.1x + 0.2y = 0.6 \\ 5x - 10y = 1 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

6. $\begin{cases} 3x + 6y = 11 \\ 2x + 4y = 9 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

7. $\begin{cases} -x + 2y = 4 \\ 2x - 4y = 1 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

8. $\begin{cases} 3x - 2y = 5 \\ -9x + 6y = -15 \end{cases}$

a. infinitely many solutions
b. no solution
c. one solution

9. Which system of equations has infinitely many solutions?

2.8.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 2, use the following scenario:

> A runner is training for a marathon. She runs more than 4 days each week. Each day, she runs between 12 and 15 miles. The marathon is 3 months away at the most.

1. What does the inequality $d > 4$ represent?

a. the number of days the runner runs each week
b. the number of months away the marathon is
c. the number of miles the runner runs each day
d. the number of miles in a marathon

2. What does the inequality $12 \leq m \leq 15$ represent?

a. the number of days the runner runs each week
b. the number of months away the marathon is
c. the number of miles the runner runs each day
d. the number of miles in a marathon

For questions 3 – 8, write an inequality to represent the constraint given.

3. There are less than 4 patients, p, in the waiting room.

a. $p \geq 4$
b. $p \leq 4$
c. $p > 4$
d. $p < 4$

4. There are no more than 6 cats, c, in the yard.

a. $c \geq 6$
b. $c < 6$
c. $c \leq 6$
d. $c > 6$

5. There are no less than 7 markers, m, in the drawer.

a. $m \geq 7$
b. $m < 7$
c. $m \leq 7$
d. $m > 7$

6. The basket has between 12 and 19 eggs, e.

a. $12 \leq e \leq 19$
b. $e \leq 19$
c. $12 \geq e$ and $e \geq 19$
d. $12 > e$ and $e > 19$

7. The total sales of tickets to a show is more than \$68. An adult ticket, a, costs \$8 and a children's ticket, c, is \$3.

a. $8a + 3c > 68$
b. $8a + 3c < 68$
c. $8a + 3c \geq 68$
d. $8a + 3c \leq 68$

8. There needs to be at least 1 pound of clay, c, for every 4 students, s.

a. $c \leq \frac{s}{4}$

2.9.8 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here is an inequality: $\frac{7x+6}{2} \leq 3x + 2$

Select **two** of the values that are a solution to the inequality.

a. $x = -3$
b. $x = -2$
c. $x = -1$
d. $x = 0$
e. $x = 1$
f. $x = 2$
g. $x = 3$

2. Which number line is represented by the inequality $x \leq 6$?

a.



b.



c.



d.



For questions 3 – 4, use the following information:

Scooter rental company A charges $9 per hour plus a $11 initiation fee. Scooter rental company B charges $15 per hour.

3. A customer writes the following inequality to represent the situation: $9h + 11 > 15h$.
Which of the following is true about the given inequality?

a. h represents the number of scooters rented.
b. Company A costs the same as Company B.
c. Company A costs less than Company B.
d. Company A costs more than Company B.

4. For which number of hours does Company B cost less?

a. 4 hours
b. 3 hours
c. 2 hours
d. 1 hour

For questions 5 – 7, use the following information:

Cycle Rentals offers 20 membership points for each day that a customer rents a bike. Bikes Unlimited offers 5 membership points for each day that a customer rents a bike plus 60 points for joining the membership club.

5. The equation $20x = 5x + 60$ represents a possible constraint about the situation. Solve for x. x = ___.

6. Which shop offers more membership points if you rent a bike for longer than the constraint you found in the previous question?

a. Cycle Rentals
b. Bikes Unlimited

2.10.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Xander is loading a pallet with boxes that each weighs 45 pounds. The pallet can safely support no more than 900 pounds. How many boxes can he safely load onto the pallet?
   ___ boxes or less

2. The elevator in Yehire's apartment building has a sign that says the maximum weight is 2100 pounds. If the average weight of one person is 150 pounds, how many people can safely ride the elevator?
   a maximum of ___ people

3. Jaden is looking at apartments with three of his friends. They want the monthly rent to be no more than $2360. If the roommates split the rent evenly among the four of them, what is the maximum rent each will pay?
   $___

4. Melody got a $20 gift card for the coffee shop. Her favorite iced drink costs $3.79. What is the maximum number of drinks she can buy with the gift card?
   ___ drinks

5. Teegan likes to play golf. He has budgeted $60 next month for the driving range. It costs him $10.55 for a bucket of balls each time he goes. What is the maximum number of times he can go to the driving range next month?
   ___ times

6. Ryan charges his neighbors $17.50 to wash their car. How many cars must he wash next summer if his goal is to earn at least $1500?
   ___ cars

7. Alejandro's water bill is $24.80 per month plus $2.20 per ccf (hundred cubic feet) of water. What is the maximum number of ccf Alejandro can use if he wants his bill to be no more than $60?
   ___ ccf

8. Look at the inequality $5m + 6 < 14 - 2m$. Which of the following is NOT true?

   a. If m is a negative number, then $14 - 2m$ will always be negative.
   b. If m is a positive number, then $5m + 6$ will always be positive.
   c. If m is a negative number, then $14 - 2m$ will always be positive.
   d. If m is a negative number, then $5m + 6$ could be a positive or negative number.

9. Solve the inequality: $x + 8 > -12$

   a. $x > 4$
   b. $x > -4$
   c. $x > 20$
   d. $x > -20$

10. Solve the inequality: $x - 15 < -3$

    a. $x < -18$
    b. $x < 12$
    c. $x < -12$
    d. $x < 18$

11. Solve the inequality: $x + 19 \leq 23$

    a. $x \leq 4$
    b. $x \leq 42$
    c. $x \leq -4$
    d. $x \geq 4$

2.11.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 3, determine whether each ordered pair is a solution to the inequality $y > x - 1$.

1. $(0, 1)$

a. Yes
b. No

2. $(-4, -1)$

a. Yes
b. No

3. $(4, 2)$

a. Yes
b. No

For questions 4 - 6, determine whether each ordered pair is a solution to the inequality $y > x - 3$.

4. $(0, 0)$

a. Yes
b. No

5. $(2, 1)$

a. Yes
b. No

6. $(-1, -5)$

a. Yes
b. No

For questions 7 and 8, determine whether each ordered pair is a solution to the inequality $2x + 3y > 2$.

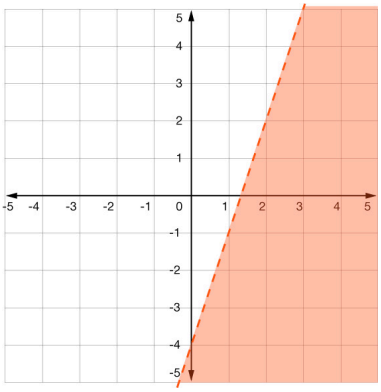
7. $(1, 1)$

a. Yes
b. No

8. $(4, -3)$

a. Yes
b. No

9. Which inequality represents the graph?



a. $y \geq 3x - 4$
b. $y > 3x - 4$

2.12.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, use the following scenario:

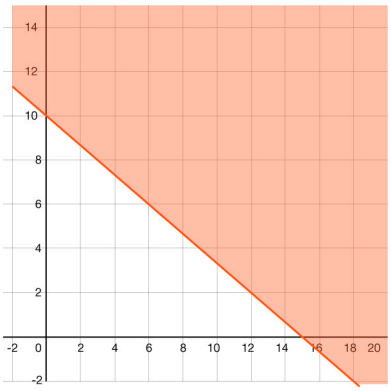
> Harrison works two part time jobs. One at a gas station that pays $11 an hour and the other is IT troubleshooting for $16.50 an hour. Between the two jobs, Harrison wants to earn at least $330 a week. How many hours does Harrison need to work at each job to earn at least $330?

1. Let x be the number of hours he works at the gas station and let y be the number of hours he works troubleshooting. Which inequality correctly models this situation?

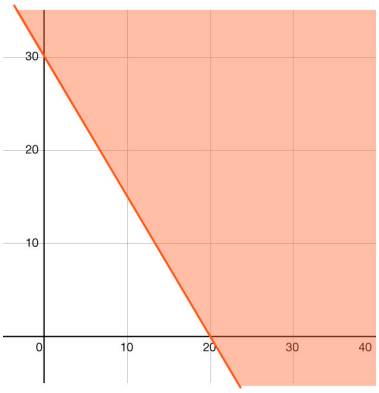
a. $16.5y + 11x > 330$
b. $11x + 16.5y \leq 330$
c. $16.5x + 11y \geq 330$
d. $11x + 16.5y \geq 330$

2. Which graph correctly represents the inequality from the previous question?

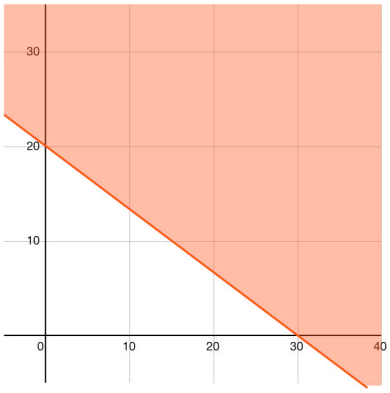
a.



b.



c.



2.13.6 Practice

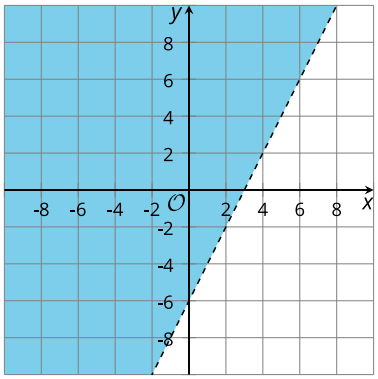
Complete the following questions to practice the skills you have learned in this lesson.

This year, students in the 9th grade are collecting dimes and quarters for a school fundraiser. They are trying to collect more money than the students who were in the 9th grade last year. The students in 9th grade last year collected $143.88.

1. Using d to represent the number of dimes collected and q to represent the number of quarters, which statement best represents this situation?

a. $0.25q + 0.1d > 143.88$
b. $0.25d + 0.1q > 143.88$
c. $0.25q + 0.1d < 143.88$
d. $0.25d + 0.1q \geq 143.88$

2. Which inequality is represented by the graph?



a. $4x + 2y < 12$
b. $4x + 2y > 12$
c. $4x - 2y < 12$
d. $4x - 2y > 12$

For questions 3 – 4, use the scenario below to answer the questions:

A farmer is creating a budget for planting soybeans and wheat. Planting soybeans costs $200 per acre and planting wheat costs $500 per acre. He wants to spend no more than $100,000 planting soybeans and wheat.

3. Let s represent the number of acres of soybeans and w represent the number of acres of wheat. Write an inequality to describe the situation.

a. $200s + 500w > 100,000$
b. $200s + 500w \leq 100,000$
c. $500s + 200w \leq 100,000$
d. $500s + 200w < 100,000$

4. Name one solution to the inequality and explain what it represents in that situation.

a. $(300, 90)$
b. $(170, 140)$
c. $(250, 100)$
d. $(200, 130)$

For questions 5 – 7, use the passage below to answer the questions:

Amira is ordering dried chili peppers and corn husks for her cooking class. Chili peppers cost $16.95 per pound and corn husks cost $6.49 per pound.

Amira spends less than $50 on d pounds of dried chili peppers and h pounds of corn husks.

2.14.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

In questions 1 - 5, determine whether each ordered pair is a solution to the system.

1. $\begin{cases} 3x + y > 5 \\ 2x - y \leq 10 \end{cases}$

a. (3, -3)

a. Yes
b. No

b. $(7, 1)$

a. Yes
b. No

2. $\begin{cases} 4x - y < 10 \\ -2x + 2y > -8 \end{cases}$

a. (5, -2)

a. Yes
b. No

b. $(-1, 3)$

a. Yes
b. No

3. $\begin{cases} y > \frac{2}{3}x - 5 \\ x + \frac{1}{2}y \leq 4 \end{cases}$

a. (6, −4)

a. Yes
b. No

b. $(3, 0)$

a. Yes
b. No

4. $\begin{cases} y < \frac{3}{2}x + 3 \\ \frac{3}{4}x - 2y < 5 \end{cases}$

a. (−4, −1)

a. Yes
b. No

b. $(8, 3)$

a. Yes
b. No

5. $\begin{cases} 6x - 5y < 20 \\ -2x + 7y > -8 \end{cases}$

2.15.5 Practice

For questions 1 – 4, use this scenario: Jada has p pennies and n nickels that add up to more than 40 cents. She has fewer than 20 coins altogether.

1. Which two inequalities could make up a system of inequalities that represents how many pennies and nickels that Jada could have? Select **two** inequalities.

a. $p + n < 20$
b. $p + n > 20$
c. $0.01p + 0.05n < 0.40$
d. $0.01p + 0.05n > 0.40$

For questions 2 – 4:

Determine if it is possible that Jada has each of the following combinations of coins. If so, explain or show how you know. If not, state which constraint—the amount of money or the number of coins—it does not meet.

2. 15 pennies and 5 nickels

a. Yes
b. No

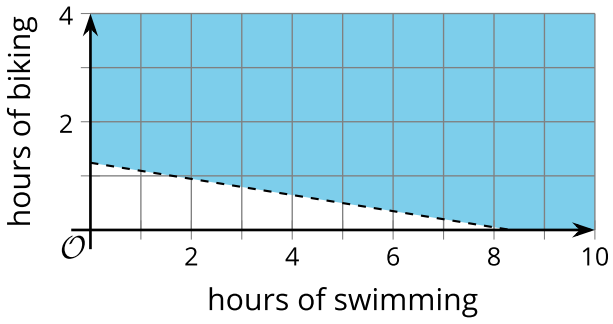
3. 16 pennies and 2 nickels

a. Yes
b. No

4. 10 pennies and 8 nickels

a. Yes
b. No

5. A triathlon athlete swims at an average rate of 2.4 miles per hour, and bikes at an average rate of 16.1 miles per hour. At the end of one training session, she has swam and biked more than 20 miles in total.
The inequality $2.4s + 16.1b > 20$ and this graph represent the relationship between the hours of swimming, s, the hours of biking, b, and the total distance the athlete could have traveled in miles.



Mai said, "I'm not sure the graph is right. For example, the point $(10, 3)$ is in the shaded region, but it's not realistic for an athlete to swim for 10 hours and bike for 3 hours in a training session! I think triathlon athletes generally train for no more than 2 hours a day."

Which inequality represents Mai's last statement?

a. $s + b = 2$
b. $s + b > 2$
c. $s + b < 2$
d. $s + b \leq 2$

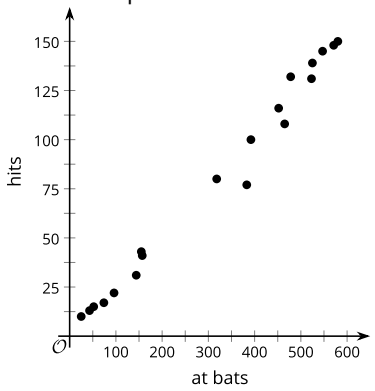
For questions 6 and 7, use the following situation and graph:

Elena is considering buying bracelets and necklaces as gifts for her friends. Bracelets, b, cost \$3, and necklaces, n, cost \$5. She can spend no more than \$30 on the gifts. Elena needs at least 7 gift items.
This graph represents the inequality $3b + 5n \leq 30$, which describes the cost constraint in this situation.

3.1.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The scatter plot shows the number of times a player came to bat and the number of hits they had.

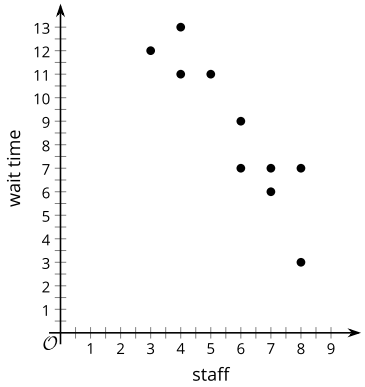


The scatter plot includes a point at $(318, 80)$. Describe the meaning of this point in this situation.

a. For every 80 at bats, there are 318 hits.
b. For every 318 at bats, there are 80 hits.

For questions 2–5, use the following to answer each question.

The scatter plot shows the number of minutes people had to wait for service at a restaurant and the number of waitstaff available at the time.



A line that models the data is given by the equation $y = -1.62x + 18$, where y represents the wait time, and x represents the number of waitstaff available.

2. The slope of the line is -1.62. What does this mean in this situation?

a. When there are 18 staff people, there is no wait time
b. When there are 0 staff people, the wait time is 18 minutes
c. For every wait time increase of 1 minute the staff people increase by 1.62
d. For every increase of 1 staff person, the wait time decreases 1.62 minutes

3. The slope of the line is -1.62. Is it realistic to conclude, for this slope, as x increases, y will decrease?

a. Yes
b. No

4. The y-intercept is $(0, 18)$. What does this mean in this situation?

a. When there are 18 staff people, there is no wait time
b. When there are 0 staff people, the wait time is 18 minutes
c. For every wait time increase of 1 minute the staff people increase by 1.62
d. For every increase of 1 staff person, the wait time decreases 1.62 minutes

3.2.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Use graphing technology to create a scatter plot and find the best fit line.

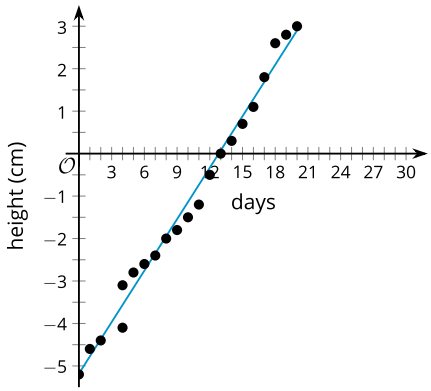
What does the best fit line estimate for the y-value when x is 100?

x	y
83	102
87	115
91	107
93	122
97	125
97	127
101	120
104	127

a. 120
b. 150
c. 125
d. 100

Access multimedia content (http://openstax.org/books/algebra-1/pages/3-2-6-practice)

2. A seed is planted in a glass pot, and its height is measured in centimeters every day.



The best fit line is given by the equation $y = 0.404x - 5.18$, where y represents the height of the plant above ground level, and x represents the number of days since it was planted.

What is the slope of the best fit line? What does the slope of the line mean in this situation?

a. The slope is 0.404, which means the plant grows 1 centimeter every 0.404 days.
b. The slope is -5.18, which means for every day, the plant shrinks -5.18 centimeters.
c. The slope is 0.404, which means for every day, the plant grows 0.404 centimeters.
d. The slope is 5.18, which means for every day, the plant grows 5.18 centimeters.

Use the following information to answer questions 3 - 5.

3.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Using the following information, find the residual for each of the coordinate pairs, (x, y) in 1 - 6.

A line of best fit for a data set is $y = 1.1x + 3.4$.

1. $(5, 8.8)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

2. $(2.5, 5.95)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

3. $(0, 3.72)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

4. $(1.5, 5.05)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

5. $(-3, 0)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

6. $(-5, -4.86)$

a. -2.76
b. 0
c. 0.32
d. -0.2
e. -0.1

7. In the scatter plot below, the length of fish and the age of the fish are shown. The residual for the fish that is 2 years old and 100 mm long is 8.06 mm.

3.4.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Select **three** values for *r* that indicate a positive slope for the line of best fit.

a. 1
b. -1
c. 0.5
d. -0.5
e. 0
f. 0.8
g. -0.8

2. Select **three** values for *r* that indicate a negative slope for the line of best fit.

a. 0
b. 1
c. -1
d. -0.87
e. 0.32
f. -0.24
g. 0.24

3. The correlation coefficient, *r*, is given for several different linear models. Order the correlation coefficients as follows: start with the *r*-value indicating the strongest linear relationship and end with the *r*-value indicating the weakest linear relationship.

$0.4, -0.85, 0.72, -0.31$

a. $-0.85, 0.72, 0.4, -0.31$
b. $-0.31, 0.4, 0.72, -0.85$
c. $-0.85, -0.31, 0.4, 0.72$
d. $0.72, 0.4, -0.31, -0.85$

4. The correlation coefficient, *r*, is given for several different linear models. Order the correlation coefficients as follows: start with the *r*-value indicating the strongest linear relationship and end with the *r*-value indicating the weakest linear relationship

$-0.65, 0.47, -1, 0.89$

a. $0.89, 0.47, -1, -0.65$
b. $-1, -0.65, 0.47, 0.89$
c. $0.47, -0.65, 0.89, -1$
d. $-1, 0.89, -0.65, 0.47$

5. Match the following scatter plots to the correct correlation coefficient from the following:

$r = 0.67$

$r = -0.33$

$r = 0.88$

$r = -0.79$

3.5.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The number of hours worked, x, and the total dollars earned, y, have a strong positive relationship.

Explain what it means to have a strong positive relationship in this situation.

a. As the number of hours worked increases, the total dollars earned remains unchanged.
b. As the number of hours worked decreases, the total dollars earned increases.
c. As the number of hours worked increases, the total dollars earned increases.
d. As the number of hours worked increases, the total dollars earned decreases.

2. The number of minutes on the phone and the customer satisfaction rating have a weak negative relationship.

Explain what it means to have a weak negative relationship in this context.

a. As the number of minutes the customer waits on the phone increases, the customer satisfaction rating increases.
b. As the number of minutes the customer waits on the phone decreases, the customer satisfaction rating decreases.
c. As the number of minutes the customer waits on the phone decreases, the customer satisfaction rating remains unchanged.
d. As the number of minutes the customer waits on the phone increases, the customer satisfaction rating decreases.

For questions 3 – 6, use a graphing calculator to answer the questions.

3. Find the equation of the line of best fit. What is the value of the correlation coefficient?

x	y
5	2
6	4.6
7.5	7.2
8	8.4
8.3	8.2
9	9.1
10.2	10.3
11.4	9.9
11.4	11
12	12.5

3.6 CAUSAL RELATIONSHIPS

3.6.0 Lesson Overview

In this lesson, you will compute and interpret the correlation coefficient for a bivariate, numerical data set.

When you finish this lesson, you will be able to:

- Describe the strength of a relationship between two variables.
- Analyze connections between two variables to determine whether or not there is a causal relationship.

Here are the **activities** that will help you reach those goals:

- 3.6.1: Describing the Relationship between Variables
- 3.6.2: Describing How Two Variables Are Related
  - 3.6.2: Self Check
  - 3.6.2: Additional Resources
- 3.6.3: Using the Term Causal Relationship
  - 3.6.3: Self Check
  - 3.6.3: Additional Resources
- 3.6.4: Determining the Type of Relationship

After that, you'll **practice and review**.

- 3.6.5: Practice
- 3.6.6: Lesson Summary

3.6.1 Describe the Relationship between Variables

Warm Up

Describe the strength and sign of the relationship you expect for each pair of variables. Be prepared to show your reasoning.

1. Used car price and original sale price of the car
2. Used car price and number of cup holders in the car
3. Used car price and number of oil changes the car has had
4. Used car price and number of miles the car has been driven

3.6.2 Describing How Two Variables Are Related

Activity

Each of the scatter plots shows a strong relationship. Write a sentence or two describing how you think the variables are related.

1. During the month of April, Elena keeps track of the number of inches of rain, or precipitation, recorded for the day and the percentage of people who come to school with rain jackets.

3.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Chloe creates a scatter plot showing the relationship between the number of steps she takes and her heart rate. The correlation coefficient of the line of best fit is 0.88.

Are they correlated? Be prepared to show your reasoning.

a. No, the correlation coefficient is more than 0.
b. Yes, the correlation coefficient is more than 0.
c. No, the correlation coefficient is close to 1.
d. Yes, the correlation coefficient is close to 1.

2. Kiran creates a scatter plot showing the relationship between the number of students attending drama club and the number of students attending poetry club each week. The correlation coefficient for the line of best fit is –0.36. Are they strongly correlated? Be prepared to show your reasoning.

a. Yes, the correlation coefficient is close to 0.
b. No, the correlation coefficient is negative.
c. Yes, the correlation coefficient is more than 0.
d. No, the correlation coefficient is closer to 0.

3. Which of the following is a causal relationship?

a. The number of minutes spent running and the number of calories burned.
b. The temperature outside and number of people wearing shorts.
c. The number of people in a family and the amount of the electric bill.
d. The age of a person and the number of words they know.

4. Which of the following is a causal relationship?

a. The number of children in a family and the number of pets in the home.
b. The amount of sugar a person eats and their level of happiness.
c. The number of students in a school and the state test scores.
d. The number of miles driven and the cost of gas for a trip.

4.1.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 4, use the following scenario.

The relationship between the amount of time a car is parked, in hours, and the cost of parking, in dollars, can be described with a function.

1. Identify the independent variable in this function.

a. Cost of parking
b. Amount of time car is parked

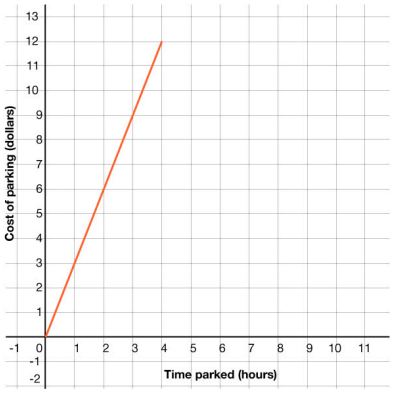
2. Identify the dependent variable in this function.

a. Cost of parking
b. Amount of time car is parked

3. Correctly describe the function with a sentence of the form "___ is a function of ___."

a. The cost of parking is a function of the amount of time a car is parked.
b. The amount of time a car is parked is a function of the cost of parking.

4. Suppose it costs \$3 per hour to park, with a maximum cost of \$12. Here is a graph of the situation.



What does the point $(2, 6)$ mean in this situation?

a. 6 hours of parking costs a total of \$2
b. 2 hours of parking costs a total of \$6
c. 6 hours of parking costs \$2 per hour
d. 2 hours of parking costs \$6 per hour

5. The distance a person walks, d, in kilometers, is a function of time, t, in minutes, since the walk began.

Select the **two statements** about the input variable of this function that are true.

a. Distance is the input.
b. Time of day is the input.
c. Time since the person starts walking is the input.
d. t represents the input.
e. d represents the input.
f. The input is not measured in any particular unit.
g. The input is measured in hours.
h. For each input, there are sometimes two outputs.

6. Determine if the following relation is a function.

4.2.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The height of water in a bathtub, w, is a function of time, t. Let P represent this function. Height is measured in inches and time in minutes.

What does each equation mean in this situation?

a. $P(0) = 0$

a. The height of the water is 10 inches after 4 minutes.
b. After 10 minutes, the height of the water is 4 inches.
c. The bathtub starts out with no water.
d. After 20 minutes, the bathtub is empty.

b. $P(4) = 10$

a. The height of the water is 10 inches after 4 minutes.
b. After 10 minutes, the height of the water is 4 inches.
c. The bathtub starts out with no water.
d. After 20 minutes, the bathtub is empty.

c. $P(10) = 4$

a. The height of the water is 10 inches after 4 minutes.
b. After 10 minutes, the height of the water is 4 inches.
c. The bathtub starts out with no water.
d. After 20 minutes, the bathtub is empty.

d. $P(20) = 0$

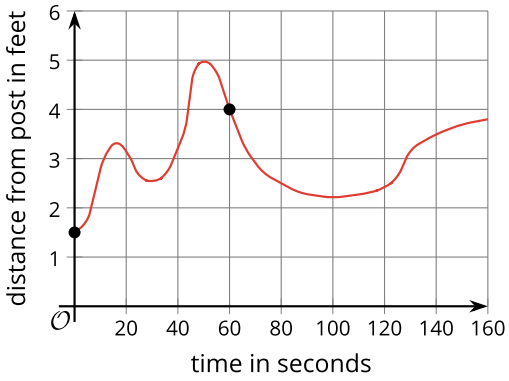
a. The height of the water is 10 inches after 4 minutes.
b. After 10 minutes, the height of the water is 4 inches.
c. The bathtub starts out with no water.
d. After 20 minutes, the bathtub is empty.

2. Function C takes time for its input and gives a student's Monday class for its output.

Use function notation to represent: A student has English at 10:00.

a. C(10:00)=Monday
b. C(Monday)=English at 10:00
c. C(10:00)=English
d. C(English)=10:00

3. Function f gives the distance of a dog from a post, in feet, as a function of time, in seconds, since its owner left.



a. Find the value of $f(20)$.

4.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Function f gives the temperature, in degrees Celsius, t hours after midnight.

Choose which equation or expression represents the statement in function notation.

a. The temperature was the same at 9 a.m. and at 4 p.m.

a. $f(9) = f(16)$
b. $f(9) = f(4)$

b. It was warmer at 9 a.m. than at 6 a.m.

a. $f(9) < f(6)$
b. $f(9) > f(6)$

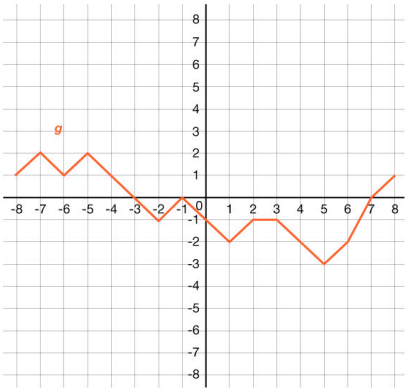
c. Some time after midnight, the temperature was 24 degrees Celsius.

a. $f(12) = 24$
b. $f(t) = 24$
c. $f(0) < 24$
d. $f(24) = t$

2. Select the **two** points that are on the graph of f if we know that $f(2) = -4$ and $f(5) = 3.4$.

a. (–4, 2)
b. (2, –4)
c. (3.4, 5)
d. (5, 3.4)
e. (2, 5)

3. Looking at the graph of the function g, below, which is greater, $g(-5)$ or $g(5)$?



a. $g(-5)$
b. $g(5)$

4. The function T tells how many people are living on floor n of an apartment building. What does $T(4) = 82$ mean?

a. There are 4 people living on floor 82.
b. There are 82 people living on floor 4.

5. The function, T, gives the temperature of a house, in degrees Fahrenheit, h hours after midnight. Choose which function notation represents the temperature at 6 a.m. was 70 degrees.

a. $70(T) = 6$
b. $6(T) = 70$
c. $T(6) = 70$

4.4.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Match each equation with a verbal description that represents the same function.

a. $f(x) = 2x + 4$

a. To get the output, multiply the input by 4, then add 2 to the result.
b. To get the output, multiply the input by 2, then add 4 to the result.
c. To get the output, add 2 to the input, then multiply the result by 4.
d. To get the output, add 4 to the input, then multiply the result by 2.

b. $g(x) = 2(x + 4)$

a. To get the output, multiply the input by 4, then add 2 to the result.
b. To get the output, multiply the input by 2, then add 4 to the result.
c. To get the output, add 2 to the input, then multiply the result by 4.
d. To get the output, add 4 to the input, then multiply the result by 2.

c. $h(x) = 4x + 2$

a. To get the output, multiply the input by 4, then add 2 to the result.
b. To get the output, multiply the input by 2, then add 4 to the result.
c. To get the output, add 2 to the input, then multiply the result by 4.
d. To get the output, add 4 to the input, then multiply the result by 2.

d. $k(x) = 4(x + 2)$

a. To get the output, multiply the input by 4, then add 2 to the result.
b. To get the output, multiply the input by 2, then add 4 to the result.
c. To get the output, add 2 to the input, then multiply the result by 4.
d. To get the output, add 2 to the input, then multiply the result by 2.

2. Functions A and B are defined by these equations.

$A(x) = 25 + 10x$
$B(x) = 80 - 15x$

Which function has a greater value when x is 2.5?

a. $A(x) = 25 + 10x$
b. $B(x) = 80 - 15x$

3. Which function is greater when $x = 4$?

a. $f(x) = x - 10$
b. $f(x) = -5x + 2$
c. $f(x) = -3x + 1$
d. $f(x) = 2x - 7$

4. Which function rule describes the table below:

x	2	3	4	5	6
y	3	5	7	9	11

a. $f(x) = x + 2$
b. $f(x) = 2x - 1$
c. $f(x) = 3x - 3$
d. $f(x) = x + 1$

4.5.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The total monthly cost of a cell phone plan from Company C costs \$10, plus \$15 per gigabyte for data used. The total monthly cost from Company D costs \$80 with unlimited data.

Rule C gives the monthly cost, in dollars, of using g gigabytes of data on Company C's plan. Rule D gives the monthly cost, in dollars, of using g gigabytes of data on Company D's plan.

a. Which sentence describes the meaning of the statement $C(2) = 40$?

a. It costs \$2 to use 40 gigabytes of data using plan D.
b. It costs \$2 to use 40 gigabytes of data using plan C.
c. It costs \$40 to use 2 gigabytes of data using plan D.
d. It costs \$40 to use 2 gigabytes of data using plan C.

b. Which is less, $C(4)$ or $D(4)$?

a. $C(4)$
b. $D(4)$

c. Which is less, $C(5)$ or $D(5)$?

a. $C(5)$
b. $D(5)$

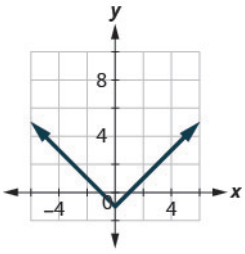
d. For what number g is $C(g) = 130$?

2. Function G takes a student's first name for its input and gives the number of letters in the first name for its output.

a. Find the value of $G(\text{Jada})$.

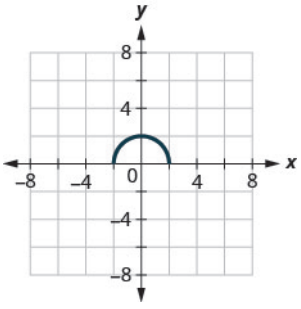
b. Find the value of $G(\text{Diego})$.

3. Is the following graph a function?



a. Yes
b. No

4. Is the following graph a function?

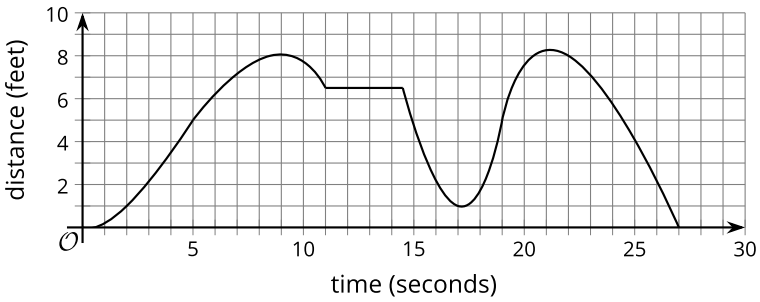


a. Yes
b. No

4.6.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. This graph represents Andre's distance from his bicycle as he walks in a park.



Decide whether the following statements are true or false.

a. The graph has multiple horizontal intercepts.

   a. true
   b. false

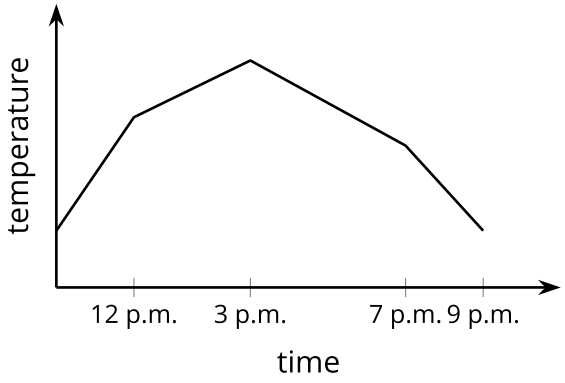
b. A horizontal intercept of the graph represents the time when Andre was with his bike.

   a. true
   b. false

c. A maximum of the graph is the point $(17, 1)$.

   a. true
   b. false

2. The graph represents the temperature, in degrees Fahrenheit, as a function of time.



At what hour in the afternoon does the maximum of the graph occur?

3. Match each feature of the situation with a corresponding statement in function notation.

4.7 FINDING SLOPE

4.7.0 Lesson Overview

In this lesson, you will work to find the slope of a line from two points, a graph, and a table. You will then use this knowledge to write equations of lines in three different forms: slope-intercept form, point-slope form, and standard form.

When you finish this lesson, you will be able to:

- Write the slope of a line.
- Write the equation of a line in different forms.

Here are the **activities** that will help you reach those goals:

- 4.7.1: Evaluating Fractions
- 4.7.2: Finding Slope From Tables, Graphs, and Points
  - 4.7.2: Self Check
  - 4.7.2: Additional Resources
- 4.7.3: Writing Linear Equations
  - 4.7.3: Self Check
  - 4.7.3: Additional Resources
- 4.7.4: Writing Equations From Two Points

After that, you'll **practice and review**.

- 4.7.5: Practice
- 4.7.6: Lesson Summary

4.7.1 Evaluating Fractions

Warm Up

Evaluate Mentally.

1. $\frac{102-96}{45-42}$

2. $\frac{-8-4}{6-2}$

3. $\frac{31-18}{5-10}$

4. $\frac{4-9}{12-18}$

4.7.2 Finding Slope From Tables, Graphs, and Points

Activity

Recall that the **slope** of a line is the rate of change of the linear function. On the graph, it is the steepness of the line.

1. Find the slope of the line that connects $(0, 0)$ and $(3, 2)$

2. Find the slope of the line that connects $(4, 2)$ and $(10, 7)$

3. Find the slope of the line that connects $(1, -2)$ and $(2, 5)$

4. Find the slope of the line that connects $(-3, 4)$ and $(-5, -2)$

5. Find the slope of the line that connects $(8, 3)$ and $(10, -9)$

For questions 6 – 9, use the graph to find the slope of the line that goes through the two points.

4.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Evaluate $\frac{140-20}{9-12}$ mentally.

2. Find the slope of the line that goes through the points $(1, -3)$ and $(2, 4)$.

3. Write the equation of the line in slope-intercept form that goes through $(0, 3)$ and has a slope of $\frac{1}{4}$.

a. $4x - y = -3$
b. $y = 3x + \frac{1}{4}$
c. $y = \frac{1}{4}x + 3$
d. $y = -\frac{1}{4}x + 3$

4. Write the equation of the line that passes through $(2, -1)$ and has a slope of $-\frac{1}{2}$ in point-slope form.

a. $2x + y = 0$
b. $y = -\frac{1}{2}x$
c. $y + 1 = -\frac{1}{2}(x - 2)$
d. $y = -\frac{1}{2}x - 1$

5. **Select two** equations that are correctly in standard form.

a. $6x - 3y = 3$
b. $-4x + 5y = 9$
c. $\frac{1}{2}x + 3y = 8$
d. $5x + 2y = 9$
e. $5y + 2x = 10$

6. Write $y = 4x - 2$ in standard form.

7. Write the equation $y - 4 = -2(x - 5)$ in slope-intercept form.

8. Find the slope of the line between the points $(3, -1)$ and $(2, -4)$.

9. Write the equation $y - 2 = 3(x + 5)$ in standard form.

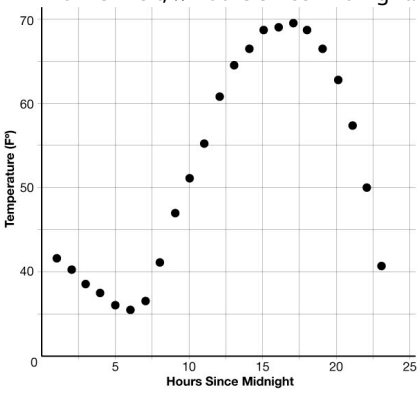
10. Evaluate $\frac{36-48}{10-14}$ mentally.

4.8.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following information to answer questions 1 - 5.

The temperature was recorded at several times during the day. Function T gives the temperature in degrees Fahrenheit, n hours since midnight. A graph for this function is provided.



1. For the interval from $n = 1$ to $n = 5$, determine if the average rate of change of the temperature is positive, negative, or zero.

   a. positive
   b. negative
   c. zero

2. For the interval from $n = 5$ to $n = 7$, determine if the average rate of change of the temperature is positive, negative, or zero.

   a. positive
   b. negative
   c. zero

3. For the interval from $n = 10$ to $n = 20$, determine if the average rate of change of the temperature is positive, negative, or zero.

   a. positive
   b. negative
   c. zero

4. For the interval from $n = 15$ to $n = 18$, determine if the average rate of change of the temperature is positive, negative, or zero.

   a. positive
   b. negative
   c. zero

5. For the interval from $n = 20$ to $n = 24$, determine if the average rate of change of the temperature is positive, negative, or zero.

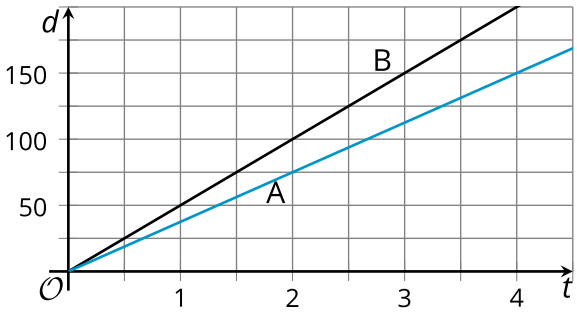
   a. positive
   b. negative
   c. zero

The graph below shows the total distance, in feet, walked by a person as a function of time, in seconds. Use this information to answer questions 6 - 7.

4.9.6 Practice

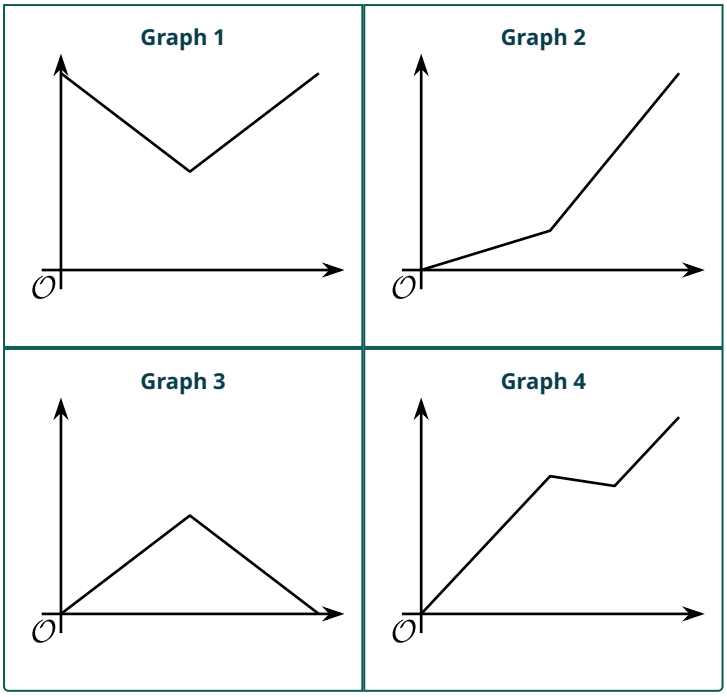
Complete the following questions to practice the skills you have learned in this lesson.

1. The graphs show the distance, d, traveled by two cars, A and B, over time, t. Distance is measured in miles, and time is measured in hours. Which car traveled slower?



a. Car A
b. Car B

For questions 2 – 5, use the graphs below to match each description of a situation in which the volume of water in a tank is a function of time to its corresponding graph.



2. An empty 20-gallon water tank is filled at a constant rate for 3 minutes until it is half full. Then, it is emptied at a constant rate for 3 minutes.

a. Graph 1
b. Graph 2
c. Graph 3
d. Graph 4

3. A full 10-gallon water tank is drained for 30 seconds, until it is half full. Afterward, it gets refilled.

a. Graph 1

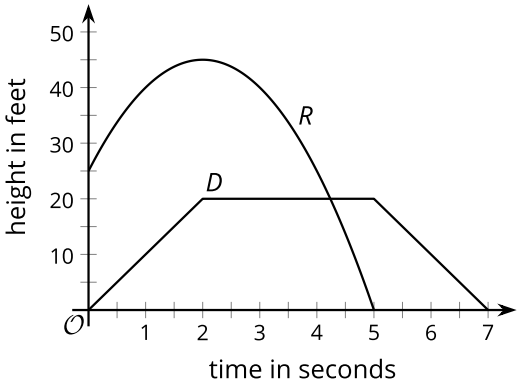
4.10.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following scenario and graph to answer question 1.

Functions R and D give the height, in feet, of a toy rocket and a drone, t seconds after they are released.

Here are the graphs of R (for the rocket) and D (for the drone).



1. Decide whether the following statements are true or false.

a. $R(3) > D(3)$

a. true
b. false

b. $R(0) = D(0)$

a. true
b. false

c. $R(4.5) = D(4.5)$

a. true
b. false

d. $R(0) = D(0) + 25$

a. true
b. false

e. $D(7) > R(5)$

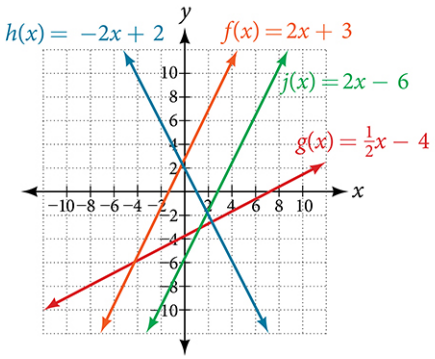
a. true
b. false

2. $A(t)$ is the average high temperature in Aspen, Colorado, t months after the start of the year. $M(t)$ is the average high temperature in Minneapolis, Minnesota, t months after the start of the year. Temperature is measured in degrees Fahrenheit.

4.11.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 3, use the graph below.



1. Tell the shift from $j(x) = 2x - 6$ to $f(x) = 2x + 3$.

a. Up 8
b. Up 9
c. Down 8
d. Down 3

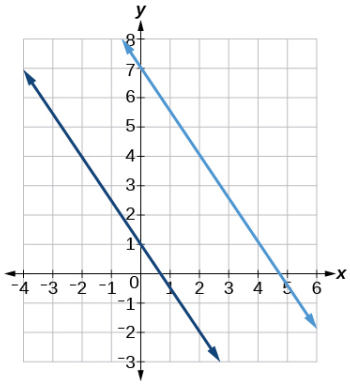
2. Tell the transformations from $f(x) = 2x + 3$ to $h(x) = -2x + 2$.

a. Vertical stretch, left 1
b. Vertical stretch, right 1
c. Vertical reflection, down 1
d. Vertical reflection, up 1

3. Describe the transformations made from the parent linear function to achieve $g(x) = \frac{1}{2}x - 4$. **Select two** that apply:

a. Vertical stretch of $\frac{1}{2}$
b. Vertical compression of $\frac{1}{2}$
c. Up 4
d. Down 4
e. Right 4
f. Left 4

4. What are the equations of the two lines shown?



Select two that apply:

a. $y = x + 4$
b. $y = x - 4$

4.12.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The cost for an upcoming field trip is $15 per student. The cost of the field trip C, in dollars, is a function of the number of students x. **Select two** possible outputs for the function defined by $C(x) = 15x$.

a. 20
b. 30
c. 50
d. 90
e. 100

2. A rectangle has an area of $24cm^2$. Function f gives the length of the rectangle, in centimeters, when the width is w cm. Determine which values, in centimeters, are possible inputs of the function. **Select three** that apply.

a. 3
b. 0.5
c. 48
d. –6
e. 0

3. **Select three** possible input-output pairs for the function $y = x^3$.

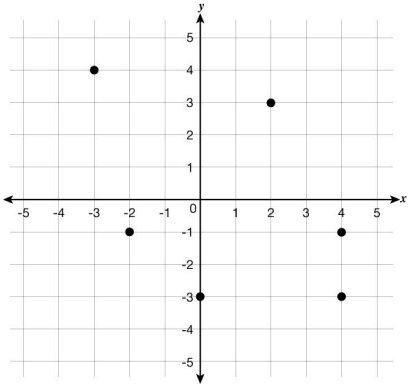
a. $(-1, -1)$
b. $(-2, 8)$
c. $(3, 9)$
d. $\left(\frac{1}{2}, \frac{1}{8}\right)$
e. $(4, 64)$
f. $(1, -1)$

4. Two functions are defined by the equations $f(x) = 5 - 0.2x$ and $g(x) = 0.2(x + 5)$.

Select two possible statements that are true about the functions.

a. $f(3) > 0$
b. $f(3) > 5$
c. $g(-1) = 0.8$
d. $f(0) = g(0)$

5. Is the relation graphed below a function?



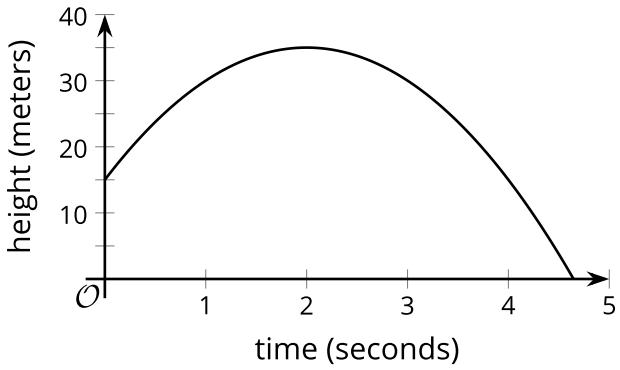
a. Yes
b. No

6. Is the relation graphed below a function?

4.13.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The graph of function H shows the height, in meters, of a rocket t seconds after it was launched.



a. Find $H(0)$ in meters.

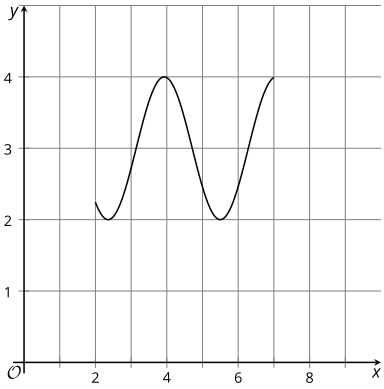
b. Which of the following describes the domain of this function?

a. $15 \leq x \leq 35$
b. $0 \leq x \leq 2$
c. $0 \leq x \leq 4.5$
d. $0 \leq x \leq 35$

c. Which of the following describes the range of this function?

a. $15 \leq y \leq 35$
b. $0 \leq y \leq 2$
c. $0 \leq y \leq 4.5$
d. $0 \leq y \leq 35$

2. Which of the following describes the domain of the function graphed below?



a. $2 \leq x \leq 4$
b. $2 \leq x \leq 7$
c. $0 \leq x \leq 4$
d. $0 \leq x \leq 9$

3. Which of the following describes the range of the graph below?

4.14 SEQUENCES

4.14.0 Lesson Overview

In this lesson, you will learn how to work with sequences and describe them recursively in an informal way. You will see how sequences are used to accomplish recurring tasks.

When you finish this lesson, you will be able to:

- Give examples of a sequence.
- Explain the meaning of the term “sequence.”
- Explain the meaning of the term “term,” as it is used in a mathematical sequence.

Here are the **activities** that will help you reach those goals:

- 4.14.1: Describe the Pattern
- 4.14.2: What Is a Sequence?
  - 4.14.2: Self Check
  - 4.14.2: Additional Resources
- 4.14.3: The Connection Between Sequences and Terms
  - 4.14.3: Self Check
  - 4.14.3: Additional Resources
- 4.14.4: Rules for a Sequence

After that, you’ll **practice and review**.

- 4.14.5: Practice
- 4.14.6: Lesson Summary

4.14.1 Describe the Pattern

Warm Up

Here is a rule for making a list of numbers: **Each number is 1 less than twice the previous number.**

Pick a number to start with. Then follow the rule to build a list of 5 numbers.

4.14.2 What Is a Sequence?

Activity

Access multimedia content (http://openstax.org/books/algebra-1/pages/4-14-2-what-is-a-sequence)
In the Tower of Hanoi puzzle, a set of discs sits on a peg, while there are 2 other empty pegs.

A move in the Tower of Hanoi puzzle involves taking a disc and moving it to another peg. There are two rules:

- Only move 1 disc at a time.
- Never put a larger disc on top of a smaller one.

You complete the puzzle by building the complete tower on a different peg from the starting peg.

1. What is the smallest number of moves in which you are able to complete the puzzle with 3 discs?

2. What is the smallest number of moves in which you are able to complete the puzzle with 4 discs?

3. Jada says she used the solution for 3 discs to help her solve the puzzle for 4 discs. Describe how this might happen.

4. How many moves do you think it will take to complete a puzzle with 5 discs? Explain.

5. How many moves do you think it will take to complete a puzzle with 7 discs?

In this activity, you have started to find a **sequence** of **terms**.

4.14.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here is a rule to make a list of numbers: Each number is the sum of the previous two numbers. Start with the numbers 0 and 1, and then follow the rule to build a sequence of 10 numbers.

Enter the 10th number in the sequence.

2. A sequence of numbers follows the rule: Multiply the previous number by –2 and add 3. The 4th term in the sequence is –7.

a. Give the 8th term in the sequence.

b. Give the term that came before –7 in the sequence.

3. Find the 4th term if the first term is 2 and each term is 3 more than the previous number

4. Find the 8th term of the sequence: 4, 7, 12, 19, 28, . . .

5. Find the 8th term of the sequence: –8, –6, –3, 1, 6, . . .

6. Find the next term in the sequence: –3, 6, –12, 24, . . .

7. Starting with 2, each number is three times the previous number.

What is the 4th term in the sequence?

4.15 INTRODUCING GEOMETRIC SEQUENCES

4.15.0 Lesson Overview

In this lesson, you will begin to understand what makes a sequence a geometric sequence and how that connects to exponential functions. Using graphs, you will explore how growing and shrinking patterns represent geometric sequences.

When you finish this lesson, you will be able to:

- Find missing terms in a geometric sequence.

Here are the **activities** that will help you reach those goals:

- 4.15.1: What Are the Patterns?
- 4.15.2: Using Tables and Graphs to Represent Geometric Sequences
  - 4.15.2: Self Check
  - 4.15.2: Additional Resources
- 4.15.3: Complete the Sequence
  - 4.15.3: Self Check
  - 4.15.3: Additional Resources
- 4.15.4: Connecting the Common Ratio and the Term in a Sequence

After that, you'll **practice and review**.

- 4.15.5: Practice
- 4.15.6: Lesson Summary

4.15.1 What Are the Patterns?

Warm Up

Examine the patterns of numbers given below.

- 40, 120, 360, 1,080, 3,240
- 2, 8, 32, 128, 512
- 1,000, 500, 250, 125, 62.5
- 256, 192, 144, 108, 81

1. What do you notice?

2. What do you wonder?

4.15.2 Using Tables and Graphs to Represent Geometric Sequences

Activity

Clare takes a piece of paper, cuts it in half, and then stacks the pieces. She takes the stack of two pieces and cuts it in half again to form four pieces, stacking them. She keeps repeating the process.

1. The original piece of paper has length 8 inches and width 10 inches. Complete the table.

number of cuts	number of pieces	area in square inches of each piece
0		
1		
2		

4.15.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here are the first two terms of a geometric sequence: 2, 4. What is the 5th term in the sequence?

2. What are the common ratios of each geometric sequence?

a. 1, 1, 1, 1

b. 256, 128, 64

c. 18, 54, 162

d. 0.8, 0.08, 0.008

e. 0.008, 0.08, 0.8

3. A person owes $1,000 on a credit card that charges an interest rate of 2% per month.

Complete this table showing the credit card balance each month if they do not make any payments. Enter each answer to the hundredths place.

Month	Total Bill In Dollars
1	1,000
2	1,020
3	1,040.40
4	a.
5	b.
6	c.
7	d.
8	e.

Table 4.15.0

a. Month 4
Round to the hundredths place.

b. Month 5
Round to the hundredths place.

c. Month 6
Round to the hundredths place.

d. Month 7
Round to the hundredths place.

e. Month 8
Round to the hundredths place.

4. A Sierpinski triangle can be created by starting with an equilateral triangle, breaking the triangle into 4 congruent equilateral triangles, and then removing the middle triangle. Starting from a single black equilateral triangle with

4.16 INTRODUCING ARITHMETIC SEQUENCES

4.16.0 Lesson Overview

In this lesson, you will learn how to explain the difference between an arithmetic and a geometric sequence. You will determine arithmetic sequences that will help you solve real-world problems, such as how much fuel is left in your vehicle or how fast the money in your bank account is increasing.

When you finish this lesson, you will be able to:

- Explain what it means for a sequence to be arithmetic or geometric.

Here are the **activities** that will help you reach those goals:

- 4.16.1: Interpreting Function Notation
- 4.16.2: What Is an Arithmetic Sequence?
  - 4.16.2: Self Check
  - 4.16.2: Additional Resources
- 4.16.3: A Sequence Is a Type of Function
  - 4.16.3: Self Check
  - 4.16.3: Additional Resources
- 4.16.4: Identify and Generate Sequences

After that, you'll **practice and review**.

- 4.16.5: Practice
- 4.16.6: Lesson Summary

4.16.1 Interpreting Function Notation

Warm Up

Consider the function f given by $f(n) = 3n - 7$. This function takes an input, multiplies it by 3, and then subtracts 7.

Evaluate.

1. $f(10)$

2. $f(10) - 1$

3. $f(10 - 1)$

4. $f(5) - f(4)$

4.16.2 What Is an Arithmetic Sequence?

Activity

Here are the values of the first 5 terms of 3 sequences:

1. 30, 40, 50, 60, 70, . . .
2. 0, 5, 15, 30, 50, . . .
3. 1, 2, 4, 8, 16, . . .

1. For each sequence, describe a way to produce a new term from the previous term.

a. 30, 40, 50, 60, 70, . . .

b. 0, 5, 15, 30, 50, . . .

c. 1, 2, 4, 8, 16, . . .

2. If the patterns you described continue, which sequence has the second-greatest value for the 10th term?

3. Which of these could be geometric sequences? Explain how you know.

4.16.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here are the first two terms of some different arithmetic sequences. What is the 5th term of each sequence?

a. –2, 4 ...

b. 11, 111 ...

c. 5, 7.5 ...

d. 5, –4 ...

2. For each sequence, decide whether it could be arithmetic, geometric, or neither.

a. 200, 40, 8, . . .

   a. Neither
   b. Arithmetic
   c. Geometric

b. 2, 4, 16, . . .

   a. Neither
   b. Arithmetic
   c. Geometric

c. 10, 20, 30, . . .

   a. Neither
   b. Arithmetic
   c. Geometric

d. 100, 20, 4, . . .

   a. Neither
   b. Arithmetic
   c. Geometric

e. 6, 12, 18, . . .

   a. Neither
   b. Arithmetic
   c. Geometric

3. For each arithmetic sequence with missing terms, find the common difference that might help determine the missing terms.

a. –3, –2, ___, ___, 1

b. ___, 13, 25, ___, ___

c. 1, 0.25, ___, –1.25, ___

d. 92, ___, ___ ,___, 80

4. Find the common difference for each of the following arithmetic sequences.
HINT: It may be helpful to complete the sequence with the missing terms.

a. ___, 6, 10, ___, 18

b. –1, ___, –11, –16, ___

4.17.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 3, match each sequence with one of the definitions. Note that only the part of the definition showing the relationship between the current term and the previous term is given so as not to give away the solutions.

1. 6, 12, 18, 24

a. $d(n) = d(n-1) + 20$
b. $c(n) = c(n-1) + 6$
c. $b(n) = \frac{1}{2}b(n-1)$
d. $a(n) = 7 \cdot a(n-1)$

2. 2, 14, 98, 686

a. $d(n) = d(n-1) + 20$
b. $c(n) = c(n-1) + 6$
c. $b(n) = \frac{1}{2}b(n-1)$
d. $a(\mathrm{n}) = 7 \cdot a(n-1)$

3. 160, 80, 40, 20

a. $d(n) = d(n-1) + 20$
b. $c(n) = c(n-1) + 6$
c. $b(n) = \frac{1}{2}b(n-1)$
d. $a(\mathrm{n}) = 7 \cdot a(n-1)$

For questions 4 – 5, select the first five terms of each sequence. Determine whether each sequence is arithmetic, geometric, or neither.

4. $a(1) = 7$, $a(n) = a(n-1) - 3$ for $n \geq 2$.

a. 7, 4, 1, –2, –5, arithmetic
b. 7, –21, 63, –189, arithmetic
c. 7, –21, 63, –189, geometric
d. 7, 4, 1, –2, –5, geometric

5. $c(1) = 3$, $c(n) = 10 \cdot c(n-1)$ for $n \geq 2$.

a. 3, 13, 23, 33, 43, arithmetic
b. 3, 13, 23, 33, 43, geometric
c. 3, 30, 300, 3000, 30000, geometric
d. 3, 30, 300, 3000, 30000, arithmetic

For questions 6 – 8, match each sequence with one of the recursive definitions. Note that only the part of the definition showing the relationship between the current term and the previous term is given so as not to give away the solutions.

6. 3, 15, 75, 375

a. $d(n) = d(n-1) + n - 1$
b. $c(n) = 5 \cdot c(n-1)$
c. $b(n) = b(n-1) - 4$
d. $a(n) = \frac{1}{3} \cdot a(n-1)$

7. 18, 6, 2, $\frac{2}{3}$

a. $d(n) = d(n-1) + n - 1$
b. $c(n) = 5 \cdot c(n-1)$
c. $b(n) = b(n-1) - 4$
d. $a(n) = \frac{1}{3} \cdot a(n-1)$

8. 17, 13, 9, 5

a. $d(n) = d(n-1) + n - 1$

4.18 THE Nth TERM OF A SEQUENCE

4.18.0 Lesson Overview

In this lesson, you will learn how a function can be represented by two different equations. You will find any term of an arithmetic sequence directly, by using the nth term formula to find any term number n.

When you finish this lesson, you will be able to:

- Describe the nth term.
- Find terms of sequences directly.
- Explain why different equations can represent the same sequence.

Here are the **activities** that will help you reach those goals:

- 4.18.1 Repeated Operations
- 4.18.2 Identifying Domain for a Function
- 4.18.3 Define an Arithmetic Sequence by the nth Term
  - 4.18.3: Self Check
  - 4.18.3: Additional Resources
- 4.18.4 Define a Geometric Sequence by the nth Term
  - 4.18.4: Self Check
  - 4.18.4: Additional Resources
- 4.18.5 Interchanging between Formulas
  - 4.18.5: Self Check
  - 4.18.5: Additional Resources
- 4.18.6 Different Types of Equations

After that, you'll **practice and review**.

- 4.18.7 Practice
- 4.18.8 Lesson Summary

4.18.1 Repeated Operations

Warm Up

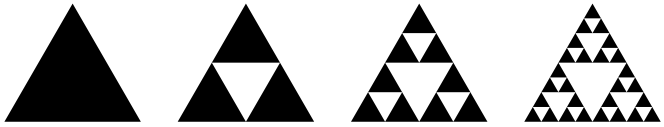
Which one doesn't belong?

1. $5+2+2+2+2+2+2$
2. $5+6 \cdot 2$
3. $5 \cdot 2^6$
4. $5 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

4.18.2 Identifying Domain for a Function

Activity

A Sierpinski triangle can be created by starting with an equilateral triangle, breaking the triangle into 4 congruent equilateral triangles, and then removing the middle triangle. Starting from a single black equilateral triangle:



1. Let S be the number of black triangles in Step n. Define $S(n)$ recursively.

4.18.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Find the value of each specified term of the arithmetic sequences below.

a. Find the 21st term of a sequence where the first term is 3 and the common difference is 8.

b. Find the 30th term of a sequence where the first term is -14 and the common difference is 5.

c. Find the 16th term of a sequence where the first term is 11 and the common difference is -6.

2. Find the value of each specified term of the geometric sequences below.

a. Find the 11th term given the first term is 8 and the common ratio is 3.

b. Find the 10th term given the first term is -6 and the common ratio is -2.

3. A sequence is defined recursively by $f_1 = -6$, $f_n = f_{n-1} - 2$, for $n \geq 2$. Which of the following gives the nth term for $n \geq 0$?

a. $a_n = -6 + (n)(-2)$
b. $a_n = -6 + (n-1) + (-2)$
c. $a_n = -6 + (n-1)(-2)$
d. $a_n = -2 + (n-1)(-6)$

4. A sequence is defined recursively by $f_1 = -20$, $f_n = f_{(n-1)} + 5$ for $n \geq 2$. Which of the following is the nth term?

a. $a_n = 5 + (n-1)(-20)$
b. $a_n = -20 + (n-1)(5)$
c. $a_n = -20 + (n-1)(-2)$
d. $a_n = -20 + (n-1) + 5$

5. A sequence is defined recursively by $f_1 = 3$, $f_n = 2 \cdot f_{n-1}$, $n \geq 2$. Which of the following gives the nth term for $n \geq 0$?

a. $f_n = 3 + 2^n$
b. $f_n = 3 + 2n$
c. $f_n = 3^n$
d. $f_n = 3 \cdot 2^{n-1}$

6. A sequence of numbers is given by $a_2 = 1$, $a_3 = 5$, $a_4 = 9$, $a_5 = 13$, $a_6 = 17$. Which expression can be used to find the nth term of the sequence?

a. $3n - 4$
b. $7n - 4$
c. $4n - 7$
d. $4n - 3$

7. A sequence of numbers is given by $\frac{1}{27}$, $\frac{1}{9}$, $\frac{1}{3}$, 1, ... What is the equation that can be used to find the nth term of the sequence?

a. $a_n = \frac{1}{27} \div 3(n-1)$
b. $a_n = \frac{1}{27}(3)^{n-1}$
c. $a_n = \frac{1}{27}(\frac{1}{3})^{n-1}$
d. $a_n = \frac{1}{27} - \frac{1}{3}(n-1)$

8. A sequence is given by the set of numbers $5, 1, -3, -7$ ... What is the equation that can be used to find the nth term of the sequence?

Interpret Linear Functions

A linear function is a function that represents a line on the coordinate plane. A line can be written in slope-intercept form, $y = mx + b$, where m is the slope and b is the y-intercept. The dependent variable is y, and the independent variable is x. The y can be replaced by $f(x)$, so the function can be written as $f(x) = mx + b$.

Andrei wants to fill a glass tank with marbles, each with the same volume, and then fill the remaining space with water. The function $V(n) = 32 - 0.05n$ represents the volume of water, in liters, Andrei uses if he uses n marbles.

You can rewrite the function in slope-intercept form: $V(n) = -0.05n + 32$. The slope is –0.05, and the y-intercept is 32.

1. What is the volume of the tank?

   $V(n) = -0.05n + 32$ represents the volume left in the tank after n marbles were added. The volume of the tank is the initial volume before any of the marbles have been added, or $V(0)$. $V(0) = -0.05(0) + 32 = 32$. The volume of the tank is 32 liters.

2. What is the volume of each marble?

   The volume of water needed to fill the tank decreases by 0.05 liters each time Andrei puts in a marble, n. This means that the volume of each marble is 0.05 liters.

› **TRY IT** **Interpret Linear Functions**

Rachel is a stunt driver, and she is escaping from a building that is about to explode. The function $D(t) = -38t + 220$ represents the remaining distance, D, in meters after t seconds.

How long is the distance Rachel has to drive?

Check Your Understanding

The amount of money that farmers in Massachusetts paid to maintain their crops between 1991 and 2008 is modeled by the function $P = 3.53t + 100$, where P is the amount of money the farmers paid, in millions of dollars, and t is the year (with 1991 as $t = 0$). What does the 3.53 mean in the equation?

a. The costs for maintaining crops increased by $3.53 million each year between 1991 and 2008.

b. The costs for maintaining crops increased a total of $3.53 million between 1991 and 2008.

c. The costs for maintaining crops was $3.53 million in 1991.

d. The costs for maintaining crops was $3.53 million in 2008.

Videos: Interpreting Linear Functions

Khan Academy: Interpret Linear Functions

Watch the videos to see how to interpret a linear function.

Linear equation word problems (https://www.youtube.com/embed/w6R8rywmgek)

Interpreting linear functions — Basic example | Math | SAT | Khan Academy (https://www.youtube.com/embed/hOsBsHA6okM)

Evaluate Exponential Expressions: Mini-Lesson Review

? **MINI LESSON QUESTION** **Question #3: Evaluate Exponential Expressions**

Evaluate $2(4)^3$.

a. 24

5.1.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Simplify each expression using the appropriate property from the lesson.

1. $f^2 \cdot f^6$

a. f^8
b. f^{12}
c. f^4
d. f^{36}

2. $\frac{(-y)^6}{(-y)^3}$

a. $(-y)^{18}$
b. $(-y)^9$
c. $(-y)^2$
d. $(-y)^3$

3. $(-4)^0$

a. -4
b. 1
c. 4
d. 0

4. 2^{-3}

a. $-\frac{1}{8}$
b. $\frac{1}{8}$
c. -6
d. 8

5. $\frac{1}{(-t)^{-5}}$

a. $(-t)^5$
b. $\frac{1}{t^5}$
c. t^5
d. t^4

6. $(-\frac{m}{n})^{-3}$

a. $-\frac{m^3}{n^3}$
b. $-\frac{n^3}{m^3}$
c. $\frac{m^3}{n^3}$
d. $\frac{n^3}{m^3}$

7. $(d^3)^5$

a. d^{-2}
b. d^5
c. d^8
d. d^{15}

5.2 RATIONAL EXPONENTS

5.2.0 Lesson Overview

In this lesson, you will learn how to write terms that include rational exponents as radicals and vice versa.

When you finish this lesson, you will be able to:

- Simplify radical expressions with even and odd indexes.
- Rewrite expressions with rational exponents as roots.
- Use properties for exponents with rational exponents.

Here are the **activities** that will help you reach those goals:

- 5.2.1: Perfect Squares Revisited
- 5.2.2: The nth Root
  - 5.2.2: Self Check
  - 5.2.2: Additional Resources
- 5.2.3: Radicals and Rational Exponents
  - 5.2.3: Self Check
  - 5.2.3: Additional Resources
- 5.2.4: Properties of Exponents with Rationals

After that, you'll **practice and review**.

- 5.2.5: Practice
- 5.2.6: Lesson Summary

5.2.1 Perfect Squares Revisited

Warm Up

1. What is the value of 4×4?
2. What is the value of 4^2?
3. Using what you know about roots, what is the value of $\sqrt{16}$?
4. Talk to a partner. How can you prove $\sqrt[2]{4^2} = 4$?
5. Talk to a partner. What do you think the answer to $\sqrt[5]{4^5}$ would be?

5.2.2 The nth Root

Activity

Work with a partner to discuss and answer the following questions.

1. Complete the table when $n = -2$ on your paper and check your answers.

Expression	Process	Simplified Result
n	(-2)	-2
n^2	$(-2)^2 = (-2)(-2)$	4
n^3		

5.2.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Simplify $\sqrt[4]{81}$.

2. Simplify $\sqrt[5]{-32}$.

3. Which of the following is equivalent to $\sqrt[4]{x^3}$?

a. $\frac{3}{4}x$
b. $4x^3$
c. $x^{\frac{4}{3}}$
d. $x^{\frac{3}{4}}$

4. Which of the following is equivalent to $\sqrt[9]{2^4}$?

a. $2^{\frac{4}{9}}$
b. $\frac{2^4}{9}$
c. $\frac{4}{9}(2)$
d. $2^{\frac{9}{4}}$

5. Simplify $k^3 \cdot k^{\frac{1}{3}}$.

a. $k^{\frac{10}{3}}$
b. $3k^{\frac{1}{3}}$
c. $\frac{1}{3}k^3$
d. k

6. Simplify $(m^{\frac{3}{4}})^{-2}$.

a. $-2m^{\frac{3}{4}}$
b. $m^{\frac{3}{2}}$
c. $\frac{1}{m^{\frac{3}{2}}}$
d. $m^{\frac{6}{8}}$

7. True or false:
$\sqrt[4]{-16} = -2$

a. true
b. false

8. True or false:
$\sqrt[3]{-64} = -4$

a. true
b. false

9. Which of the following expressions are equivalent to perfect squares? **Select the three** that apply.

a. 25

5.3.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

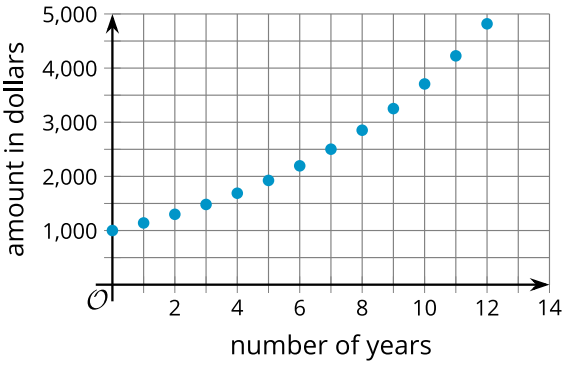
1. Which expression equals 2^7?

a. $2+7$
b. $2 \cdot 7$
c. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$
d. $2+2+2+2+2+2+2$

For questions 2 – 3, use the following information :

The graph shows the yearly balance, in dollars, of investment account A. The table shows the yearly balance, in dollars, of investment account B.

Investment account A



Investment account B

Number of years	Amount in dollars
0	$2,000
1	$2,150
2	$2,300
3	$2,450
4	$2,600

2. What type of functions do the graph and the table represent?

a. Investment A is a/n:

a. linear function
b. exponential function

b. Investment B is a/n:

a. linear function
b. exponential function

3. Which of the following statements are true? **Select two** that apply.

a. Investment A has a higher initial balance than investment B.
b. Investment A has a common ratio.
c. Investment B has a common ratio.
d. Investment A is growing by the same number of dollars each year.
e. Investment B is growing by the same number of dollars each year.
f. After 12 years, investment B will result in more money.

4. Investment account 1 starts with a balance of $200 and doubles every year. Investment account 2 starts with $1,000 and increases by $100 each year. How long does it take for each account to double?

a. How many years will it take investment account 1 to double?

a. 15
b. 10

5.4 REPRESENTING EXPONENTIAL GROWTH

5.4.0 Lesson Overview

In this lesson, you will learn about equations that represent exponential growth and the **growth factor** or **constant ratio** in an exponential function. You will apply this knowledge in a real-world activity about the growth of bacteria.

When you finish this lesson, you will be able to:

- Explain the connections between an equation and a graph that represents exponential growth.
- Write and interpret an equation that represents exponential growth.

Here are the **activities** that will help you reach those goals:

- 5.4.1: Exponent Rules
- 5.4.2: Zero Exponent Rule
  - 5.4.2: Self Check
  - 5.4.2: Additional Resources
- 5.4.3: Exponential Change: The Growth Factor
  - 5.4.3: Self Check
  - 5.4.3: Additional Resources
- 5.4.4: Graphing Exponential Expressions
  - 5.4.4: Self Check
  - 5.4.4: Additional Resources
- 5.4.5: Interpret the Growth Factor of the Equation

After that, you'll **practice and review**.

- 5.4.6: Practice
- 5.4.7: Lesson Summary

5.4.1 Exponent Rules

Warm Up

Find the value of x that makes the equation true.

1. $2^3 \cdot 2^4 = 2^x$

2. $2^5 \cdot 2 = 2^x$

3. $2^{10} \div 2^7 = 2^x$

4. $2^9 \div 2 = 2^x$

5.4.2 Zero Exponent Rule

Activity

1. Complete the table by entering the values for a, b, and c below the table. Take advantage of any patterns you notice.

x	4	3	2	1	0
3^x	81	27	a. _____	b. _____	c. _____

a. _____

b. _____

c. _____

2. Here are some equations. Find the solution to each equation using what you know about exponent rules. Be

5.4.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Which expression is equal to $4^0 \cdot 4^2$?

a. 64
b. 16
c. 1
d. 0

2. **Select three** expressions that are equivalent to 3^8.

a. 8^3
b. $\frac{3^{10}}{3^2}$
c. $3 \cdot 8$
d. $\left(3^4\right)^2$
e. $(3 \cdot 3)^4$
f. $\frac{1}{3^8}$

For 3 – 8 use the following information and table:

An investment bond is initially bought for $250. It doubles in value every decade. For each of the missing spaces in the table, complete the table.

decades since bond is bought	dollar value of bond
0	3. _____
1	4. _____
2	5. _____
3	6. _____

3. The dollar value of the bond, 0 decades

a. $750
b. $500
c. $250
d. $0

4. The dollar value of the bond, 1 decade

a. $1000
b. $750
c. $500
d. $250
e. $0

5. The dollar value of the bond, 2 decades

a. $1500

5.5 REPRESENTING EXPONENTIAL DECAY

5.5.0 Lesson Overview

In this lesson, you will work more with growth factors. You will continue to write equations and interpret graphs as you learn about exponential decay.

When you finish this lesson, you will be able to:

- Use only multiplication to represent "decreasing a quantity by a fraction of itself."
- Write an expression or equation to represent a quantity that decays exponentially.
- Know the meanings of "exponential growth" and "exponential decay."
- Explain the meanings of a and b in an equation that represents exponential decay and is written as $y = a \cdot b^x$.
- Find a growth factor from a graph and write an equation to represent exponential decay.
- Graph equations that represent quantities that change by a growth factor between 0 and 1.

Here are the **activities** that will help you reach those goals:

- 5.5.1: Identifying Linear and Exponential Quantities
- 5.5.2: Exponential Decay
  - 5.5.2: Self Check
  - 5.5.2: Additional Resources
- 5.5.3: Using Graphs to Represent Exponential Decay
  - 5.5.3: Self Check
  - 5.5.3: Additional Resources
- 5.5.4: Justifying Exponential Models
  - 5.5.4: Self Check
  - 5.5.4: Additional Resources
- 5.5.5: Writing Equations to Represent Exponential Decay

After that, you'll **practice and review**.

- 5.5.6: Practice
- 5.5.7: Lesson Summary

5.5.1 Identifying Linear and Exponential Quantities

Warm Up

Use the patterns you notice to complete the tables. Show your reasoning. Complete your work on paper.

1.

x	0	1	2	3	4	25
y	2.5	10	17.5	25		

Table A

2.

x	0	1	2	3	4	25
y	2.5	10	40	160		

Table B

5.5.2 Exponential Decay

Activity

Every year after a new car is purchased, it loses $\frac{1}{3}$ of its value. Another way to think about this is that the car retains $\frac{2}{3}$ of its value each year. Let's say that a new car's value is $18,000.

1. A buyer worries that the car will be worth nothing in three years. Do you agree? Be prepared to show your reasoning.

5.5.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. A new bicycle sells for \$300. It is on sale for $\frac{1}{4}$ off the regular price. **Select three** expressions that represent the sale price of the bicycle in dollars.

a. $300 \cdot \frac{1}{4}$
b. $300 \cdot \frac{3}{4}$
c. $300 \cdot (1 - \frac{1}{4})$
d. $300 - \frac{1}{4}$
e. $300 - \frac{1}{4} \cdot 300$

For questions 2–4, use the following information:

A computer costs \$800. It loses $\frac{1}{4}$ of its value every year after it is purchased.

2. Answer the following questions that would complete the table to show the value of the computer at the listed times.

Time (Years)	Value of Computer (Dollars)
0	
1	
2	
3	
t	

a. The value of the computer, in dollars, when it is purchased.

b. The value of the computer, in dollars, after 1 year.

c. The value of the computer, in dollars, after 2 years.

3. Which of the following expressions would represent the value of the computer, in dollars, after t years?

a. $v = 800 \cdot (\frac{3}{4})^t$
b. $v = 800 \cdot (\frac{1}{4})^t$
c. $v = 800 \cdot \frac{3}{4} \cdot t$
d. $v = 800 \cdot \frac{1}{4} \cdot t$

4. Use your equation from question 3 to find v when t is 5. What does this value of v mean?

a. 253.13; After 5 years, the computer's value is \$253.13.
b. 3,000; After 5 years, the computer's value is \$3,000.
c. 142.38; After 5 years, the computer's value is \$142.38.
d. 189.84; After 5 years, the computer's value is \$189.84.

5. A piece of paper is folded into thirds multiple times. The area, A, of the piece of paper in square inches, after n folds, is $A = 90 \cdot (\frac{1}{3})^n$.

a. What is the value of A when $n = 0$?

b. How many folds are needed before the area is less than 1 square inch?

5.6.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following scenario to answer questions 1 – 3.

- In a newly formed swamp, it is estimated that there are 500 mosquitoes. The population is estimated to double each day.

1. Let m represent the total population of mosquitoes and d represent the number of days that have passed since the first estimate. What is the equation that represents the growth of the mosquito population?

a. $m = 500 \cdot d^2$
b. $m = 500 \cdot 2^d$
c. $m = 2 \cdot 500^d$
d. $m = d \cdot 500^2$

2. What does $d = -2$ represent in this context?

a. This represents 2 days after the first estimate was taken.
b. This represents the day the first estimate was taken.
c. This represents the day before the first estimate was taken.
d. This represents 2 days before the first estimate was taken.

3. What is the estimated number of mosquitoes that were in the new swamp 1 day before the estimate was taken?

a. 125
b. 1000
c. 250
d. 500

Use the following scenario to answer questions 4 – 6.

A desert pond is slowly drying up. The average volume of the pond is shown in the table. The first year it was measured, the average volume was determined to be 40,000 liters.

y, years	v, volume (liters)
–4	
–2	
0	40,000
1	20,000
3	5,000

4. Which equation represents the pond's water loss?

a. $v = \frac{1}{2} \cdot (40,000)^y$
b. $v = 20,000 \cdot \left(\frac{1}{4}\right)^y$
c. $v = 40,000 \cdot \left(\frac{1}{2}\right)^y$
d. $y = 40,000 \cdot \left(\frac{1}{2}\right)^v$

5. What does $y = -2$ represent in this context?

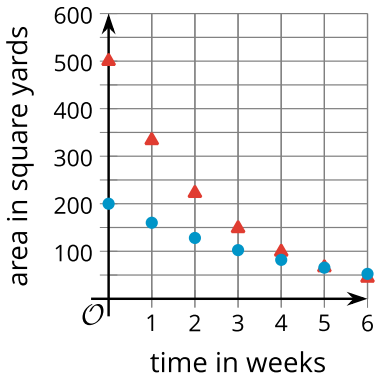
a. the time 2 years before the first time the volume was measured
b. the time 2 years after the first time the volume was measured
c. the first time the volume was measured

5.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 2, use the following scenario:

The two graphs show models characterized by exponential decay representing the area covered by two different algae blooms, in square yards, w weeks after different chemicals were applied.



1. Which algae bloom covered a larger area when the chemicals were applied?

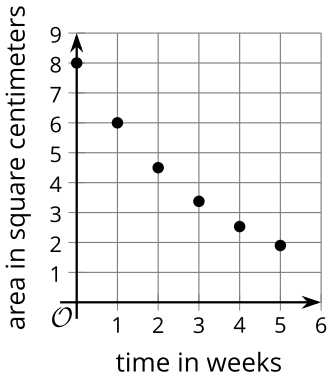
a. The algae bloom represented by the circles because the curve is always below the other one on the graph.
b. Both algae blooms covered the same area after 5 weeks.
c. The algae bloom represented by the circles because the curve has a lower y-intercept.
d. The algae bloom represented by the triangles because the curve has a higher y-intercept.

2. Which algae population is decreasing more rapidly? Explain how you know.

a. The algae bloom represented by the triangles because its graph is steeper.
b. The algae bloom represented by the circles because the curve is always below the other one on the graph.
c. Both algae blooms are decreasing at the same rate.
d. The algae bloom represented by the triangles because the curve has a higher y-intercept.

For questions 3 – 5, use the following scenario:

A medicine is applied to a burn on a patient's arm. The area of the burn in square centimeters decreases exponentially and is shown in the graph.



3. What fraction of the burn area remains each week?

a. $\frac{2}{3}$ or 0.66
b. $\frac{1}{4}$ or 0.25
c. $\frac{3}{4}$ or 0.75

5.8.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. For an experiment, a scientist designs a can, 20 cm in height, that holds water. A tube is installed at the bottom of the can allowing water to drain out. At the beginning of the experiment, the can is full. Every minute after the start of the experiment, $\frac{2}{3}$ of the water is drained.

The height of the water, h, in cm, is a function f of time t in minutes since the beginning of the experiment, $h = f(t)$. Which is an expression for $f(t)$?

a. $f(t) = 20 + \frac{2}{3}t$
b. $f(t) = 20(\frac{2}{3})^t$
c. $f(t) = 20(\frac{1}{3})^t$
d. $f(t) = 20 + \frac{1}{3}t$

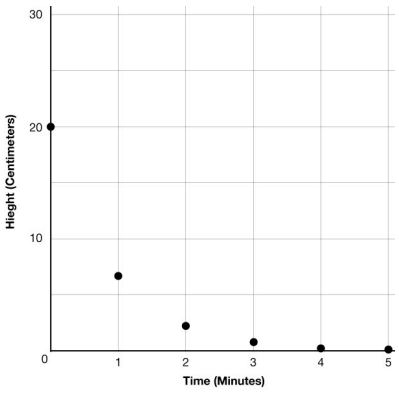
2. A bacteria population is 10,000. It triples each day.
The bacteria population, b, is a function of the number of days, d. Which equation relates b and d?

a. $b = 10000 + 3^d$
b. $b = 10000(\frac{1}{3})^d$
c. $b = 10000 \cdot 3d$
d. $b = 10000 \cdot 3^d$

3. The area, a, covered by a city is 20 square miles. The area grows by a factor of 1.1 each year, t, since it was 20 square miles. Which equation expresses a in terms of t?

a. $a = 20 - 1.1^t$
b. $a = 20 \cdot 1.1^t$
c. $a = 20 \cdot 1.1t$
d. $a = 20 + 1.1^t$

4. The graph below models the water draining out of a tube during an experiment represented by the function, $f(t) = 20(\frac{1}{3})^t$. What does the value $(2, \frac{20}{9})$ mean in the context of the problem?



a. After $\frac{20}{9}$ minutes, the height of the water is 2 centimeters less than at the start.
b. After 2 minutes, the height of the water is $\frac{20}{9}$ centimeters.
c. After $\frac{20}{9}$ minutes, the height of the water is 2 centimeters.
d. After 2 minutes, the height of the water is $\frac{20}{9}$ centimeters less than at the start.

5. A scientist measures the height, h, of a tree each month, and m is the number of months since the scientist first measured the height of the tree. Is the height, h, a function of the month, m?

5.9.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

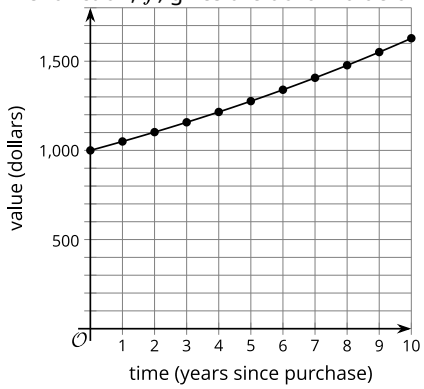
1. The number of people with the flu during an epidemic is a function, f, of the number of days, d, since the epidemic began. The equation $f(d) = 50 \cdot (\frac{3}{2})^d$ defines f.

a. How many people had the flu at the beginning of the epidemic?

b. What does $f(1)$ mean in this situation?

a. The number of people infected after 1 year.
b. The number of people infected grows by a factor of 1.
c. The number of people infected 1 day after the epidemic began.
d. There was 1 year of the epidemic.

2. The function, f, gives the dollar value of a bond t years after the bond was purchased. The graph of f is shown.



a. What is $f(0)$?

b. When is $f(t) = 1500$? Estimate to the nearest whole number.

3. A function f gives the number of stray cats in a town t years since the town started an animal control program. The program includes both sterilizing stray cats and finding homes to adopt them. An equation representing f is $f(t) = 243(\frac{1}{3})^t$.

a. What is the value of $f(t)$ when t is 0?

b. What is the value of $f(t)$ when t is 1?

4. Function g gives the amount of a chemical in a person's body, in milligrams, t hours since the patient took the drug. The equation $g(t) = 600 \cdot (\frac{3}{5})^t$ defines this function.

a. Which is the best choice for the domain of g?

a. $0 \leq t \leq 600$
b. $t \leq 600$
c. $t \geq 600$
d. $t \geq 0$

b. Which is the best choice for the range of g?

a. $0 < g \leq 600$
b. $g \leq 600$
c. $g \geq 600$
d. $g > 0$

5.10.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. A store receives 2,000 decks of popular trading cards. The number of decks of cards is a function, d, of the number of days, t, since the shipment arrived. Here is a table showing some values of d.

t	$d(t)$
0	2,000
5	1,283
10	823
15	528
20	338

Calculate the average rate of change for the following intervals:

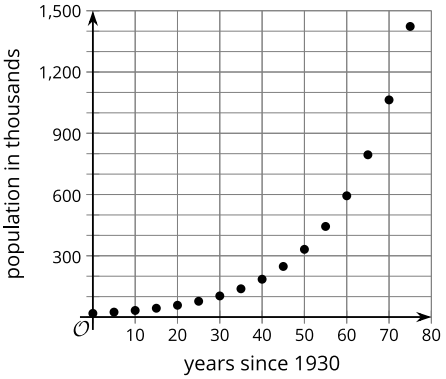
a. day 0 to day 5

b. day 15 to day 20

2. A study was conducted to analyze the effects on deer population in a particular area. Let f be an exponential function that gives the population of deer t years after the study began.
If $f(t) = a \cdot b^t$ and the population is increasing, **select three** statements that must be true.;

a. $b > 1$
b. $b < 1$
c. The average rate of change from year 0 to year 5 is less than the average rate of change from year 10 to year 15.
d. The average rate of change from year 0 to year 5 is greater than the average rate of change from year 10 to year 15.
e. $a > 0$

3. Function f models the population, in thousands, of a city t years after 1930.



a. The average rate of change of f from $t = 0$ to $t = 80$ is approximately 1,400 thousand people per year. Is this value a good way to describe the population change of the city over that time period?

a. Yes

5.11.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, use the following:

- Function h describes the height of a ball, in inches, after n by the equation $h(n) = 120 \cdot (\frac{4}{5})^n$.

1. What is $h(3)$? Round to the nearest integer. Think about what it represents in this situation.

2. Could $h(n)$ be 150?

a. Yes
b. No

3. Which ball loses its height more quickly, this ball, modeled by $h(n) = 120 \cdot (\frac{4}{5})^n$, or a tennis ball whose height in inches after n bounces is modeled by the function f where $f(n) = 50 \cdot (\frac{5}{9})^n$?

a. the ball modeled by $h(n)$
b. the ball modeled by $f(n)$

4. How many bounces would it take before the ball, modeled by $h(n)$, bounces less than 12 inches from the surface?

5. After a second bounce, a ball reached a height of 70 cm. The rebound factor for the ball was 0.8. From approximately what height, in cm, was the ball dropped?

a. 109
b. 21
c. 56
d. 560

6. Using the table below, find the height, in cm, at which the ball was dropped.

Number of Bounces	1	2	3	4
Height in cm	90	54	32	19

5.12.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 5, use the the three exponential functions defined below:

- $f(x) = 100 \cdot 3^x$
- $g(x) = 100 \cdot (3.5)^x$
- $h(x) = 100 \cdot 4^x$

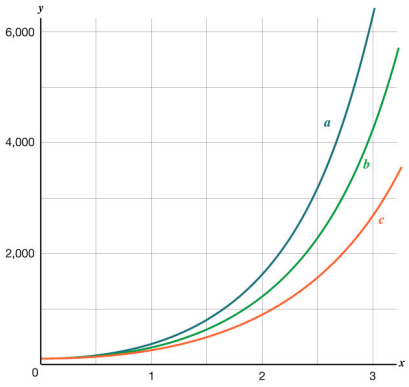
1. Which of these functions grows the least quickly?

a. $h(x)$
b. $g(x)$
c. $f(x)$

2. Which function grows the most quickly?

a. $h(x)$
b. $g(x)$
c. $f(x)$

The three given graphs represent the functions f, g, and h defined above.



3. Which graph corresponds to $f(x)$?

a. Graph A
b. Graph B
c. Graph C

4. Which graph corresponds to $g(x)$?

a. Graph A
b. Graph B
c. Graph C

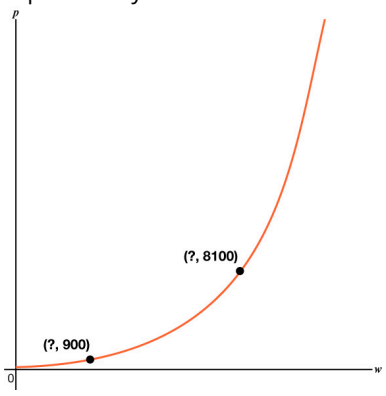
5. Which graph corresponds to $h(x)$?

a. Graph A
b. Graph B
c. Graph C

5.13.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here is a graph of p, an insect population, w weeks after it was first measured. The population grows exponentially.



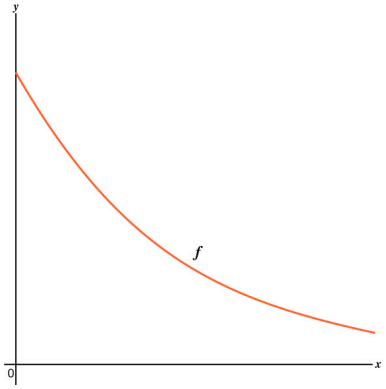
a. What is the weekly factor of growth for the insect population?

b. What was the population when it was first measured?

c. Write an equation relating p and w.

a. $p = -300 \cdot 3^w$
b. $p = \frac{300}{3^w}$
c. $p = 300 \cdot 3^w$
d. $p = 300 + 3^w$

2. Here is a graph of the function f defined by $f(x) = a \cdot b^x$.



Select three possible values of b.

a. 0
b. $\frac{1}{10}$
c. $\frac{1}{2}$
d. $\frac{9}{10}$
e. 1
f. 1.3
g. $\frac{18}{5}$

3. The function f is given by $f(x) = 50 \cdot (\frac{1}{2})^x$, and the function g is given by $g(x) = 50 \cdot (\frac{1}{3})^x$
Here are graphs of f and g.

5.14.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Functions a, b, c, d, e, and f are given below. Which functions are linear? **Select the three** answers that apply.

a. $a(x) = 3x$
b. $b(x) = 3^x$
c. $c(x) = x^3$
d. $d(x) = 9 + 3x$
e. $e(x) = 9 \cdot 3^x$
f. $f(x) = 9 \cdot 3x$

2. Functions a, b, c, d, e, and f are given below. Which functions are exponential? **Select the two** answers that apply.

a. $a(x) = 3x$
b. $b(x) = 3^x$
c. $c(x) = x^3$
d. $d(x) = 9 + 3x$
e. $e(x) = 9 \cdot 3^x$
f. $f(x) = 9 \cdot 3x$

3. Here are 4 equations defining 4 different functions, a, b, c, and d. List them in order of increasing rate of change. That is, start with the one that grows the slowest and end with the one that grows the quickest.

- $a(x) = 5x + 3$
- $b(x) = 3x + 5$
- $c(x) = x + 4$
- $d(x) = 1 + 4x$

a. Option 1: $a(x)$, $b(x)$, $c(x)$, $d(x)$
b. Option 2: $d(x)$, $c(x)$, $b(x)$, $a(x)$
c. Option 3: $a(x)$,$d(x)$, $b(x)$, $a(x)$
d. Option 4: $c(x)$, $b(x)$, $d(x)$, $a(x)$

Use the given information for problems 4 – 6. Technology is required. Function f is defined by $f(x) = 3x + 5$, and function g is defined by $g(x) = (1.1)^x$.

	$f(x)$	$g(x)$
1		
5		
10		
20		

4. Which values will complete the table for $f(x)$?

a. 8, 15, 30, 60
b. 3, 15, 30, 60
c. 8, 20, 35, 65
d. 8, 11, 14, 17

5. Which of the following sets of values will complete the table for $g(x)$?

a. 1.1, 1.21, 1.33, 1.46
b. 1.1, 1.6, 2.59, 6.73

5.15.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

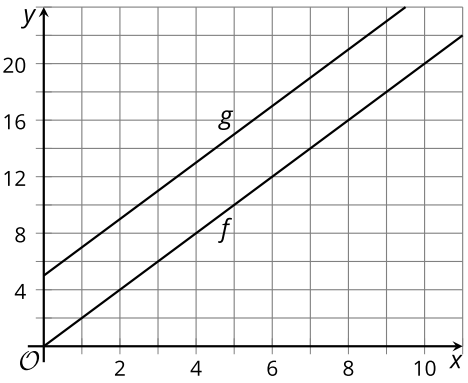
1. Whenever the input of a function f increases by 1, the output increases by 5. Which of these equations could define f?

a. $f(x) = x^5$
b. $f(x) = 5^x$
c. $f(x) = 5x + 3$
d. $f(x) = 3x + 5$

2. The function f is defined by $f(x) = 2^x$. Which of the following statements is true about the values of f? **Select the two statements** that apply.

a. When the input x increases by 1, the value of f increases by 2.
b. When the input x increases by 1, the value of f increases by a factor of 2.
c. When the input x increases by 3, the value of f increases by 8.
d. When the input x increases by 3, the value of f increases by a factor of 8.
e. When the input x increases by 4, the value of f increases by a factor of 4.

Use the graph below for questions 3 and 4.
The two lines on the coordinate plane are graphs of functions f and g.



3. Use the graph to explain why the value of f increases by 2 each time the input x increases by 1.

a. The y-axis counts by 2's so f must increase 2 each time the input increases 1.
b. There are two graphs so f increases by 2 each time the input x increases by 1.
c. $(0, 0)$ and $(1, 2)$ are two points on the graph of f. Between the two points, y increases by 2 and x increases by 1.
d. f does not increase by 2 each time x increases by 1.

4. Use the graph to explain why the value of g increases by 2 each time the input x increases by 1. **Select two** that apply.

a. f and g never intersect and stay an equal distance apart. f increases by 2 every time x increases by 1, so g must do so as well.
b. You can't use the graph because the points are not labeled.
c. There are two graphs so g increases by 2 each time the input x increases by 1.
d. $(0, 5)$and $(1, 7)$ are points on the graph of g. Between the two points, y increases by 2 and x increases by 1.
e. $(0, 0)$ and $(1, 2)$ are two points on the graph of g. Between the two points, y increases by 2 and x increases by 1.

5. The function h is given by $h(x) = 5^x$. Find the quotient $\frac{h(x+2)}{h(x)}$.

6. The function h is given by $h(x) = 5^x$. Find the quotient $\frac{h(x+3)}{h(x)}$.

a. 125

6.1.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 5, simplify each expression completely.

1. $14ab^2 - 8ab^2$

a. $8ab^2$
b. $14ab^2$
c. $22ab^2$
d. $6ab^2$

2. $(8k^2 - 3k + 14) + (8k + 17)$

3. $(7b^2 - 2b - 12) - (4b^2 + 9b - 7)$

a. $11b^2 - 11b - 5$
b. $3b^2 - 11b - 5$
c. $3b^2 + 7b - 19$
d. $11b^2 + 7b - 19$

4. $(7mn^3 - 2m^2n - 12mn) - (3mn^3 - 4mn)$

a. $4mn^3 - 10m^2n$
b. $4mn^3 - 2m^2n - 16mn$
c. $4mn^3 - 2m^2n - 8mn$
d. $10mn^3 - 2m^2n - 16mn$

5. $(r^3 - r^2s) - 2(rs^2 + s^3) + (r^2s + 2rs^2)$

a. $r^3 + rs^2 - 2s^3$
b. $r^3 - 2s^3$
c. $r^3 - rs^2 - 2s^3$
d. $r^3 + 2s^3$

Use the following polynomial functions to answer questions 6 – 10.

- $p(x) = 12x^2 + 3x - 4$
- $d(x) = 3x^2 - 11x - 6$

6. $p(4)$

a. 48
b. 192
c. 200
d. –2

7. $d(-3)$

a. –12
b. 54
c. 113
d. 95

8. $p(x) + d(x)$

9. $p(x) - d(x)$

10. $p(5) - d(-6)$

a. 479
b. 168
c. 311
d. 143

6.2.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 8, simplify each expression completely.

1. $(5h^2)(-3h^2)$

a. $-15h^8$
b. $-15h$
c. $-15h^4$
d. $-15h^2$

2. $(-\frac{4}{5}d^2s^2)(6ds^2)$

a. $-\frac{24}{30}d^2s^4$
b. $-\frac{24}{5}d^3s^4$
c. $(-\frac{24}{30}d^3s^4)$
d. $(-\frac{24}{5}d^2s^4)$

3. $(n-4)(3n+7)$

a. $3n^2+5n-28$
b. $3n^2-5n-28$
c. $3n^2-12n+28$
d. $3n^2-19n+28$

4. $(3x^2+3)(7x-5)$

5. $(m-6)(2m^2-7m+8)$

a. $2m^3-19m^2+50m-48$
b. $2m^3+19m^2+34m-48$
c. $-6m^2-42m-48$
d. $2m^3-7m^2+8m$

6. $(c-4)(c^2+5c-9)$

a. $-4c^3+c^2-29c+36$
b. $-4c^2-20c+36$
c. $c^3+c^2-29c+36$
d. c^3+5c^2-9c

7. $(5x-3)^2$

8. $(11z+7)(11z-7)$

a. $121z^2+49$
b. $121z^2+144z-49$
c. $121z^2-49$
d. $121z^2-144z-49$

For questions 9–10, use the following polynomial functions.

- $f(x)=5x+4v$
- $g(x)=5x-4v$

9. $f(x)\cdot g(x)$

a. $25x^2-16v^2$
b. $25x^2+40xv+16v^2$
c. $25x^2-40xv+16v^2$
d. $25x^2+16v^2$

6.3.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 2, find the quotients using division.

1. $\frac{45x^5y^9}{-60x^8y^6}$

a. $-\frac{3}{4}x^3y^3$
b. $-\frac{3y^3}{4x^3}$
c. $-\frac{4y^3}{3x^3}$
d. $-\frac{3x^3}{4y^3}$

2. $(48y - 24x) \div (2y - x)$

a. -24
b. $24y$
c. $-24x$
d. 24

For questions 3 and 4, divide using long division.

3. $(m^3 + 1000) \div (m + 10)$

a. $m^2 - 10m + 100$
b. $m^2 + 10m - 100$
c. $m^2 - 100$
d. $m^2 + 10m$

4. $(125y^3 - 64) \div (5y + 4)$

a. $25y^2 - 20y + 16$
b. $25y^2 + 20y + 16$
c. $25y^2 - 20y + 16 - \frac{64}{5y+4}$
d. $25y^2 - 20y + 16 - \frac{128}{5y+4}$

For questions 5 - 6, divide using synthetic division.

5. $(b^3 - 3b^2 - 4b + 12) \div (b + 2)$

a. $b^2 + 5b - 6$; remainder 0
b. $b^2 - 5b$; remainder 6
c. $b^2 - 5b + 6$; remainder 0
d. $b^2 + 5b$; remainder 6

6. $(3x^4 - 11x^3 + 2x^2 + 10x + 6) \div (x - 3)$

a. $3x^3 - 2x^2 - 4x$; remainder -2
b. $3x^3 - 2x^2 - 4x - 2$; remainder 0
c. $3x^3 - 20x^2 - 62x - 180$; remainder 0
d. $3x^3 + 2x^2 + 4x + 2$; remainder 0

For questions 7 and 8, use the following polynomial functions.

- $f(x) = x^2 - 13x + 36$
- $g(x) = x - 4$

7. Find $(\frac{f}{g})(x)$.

6.4 GREATEST COMMON FACTOR AND FACTOR BY GROUPING

6.4.0 Lesson Overview

In this lesson, you will use your skills finding common factors in different terms to find the greatest common factor (GCF) of two or more expressions. You will then use the GCF to factor these expressions. In some cases, you will need to group terms together so that the expressions can be factored. These skills will allow you to rewrite expressions into a simpler, fully factored form.

When you finish this lesson, you will be able to:

- Find the greatest common factor of two or more expressions.
- Factor the greatest common factor from a polynomial.
- Factor by grouping.

Here are the **activities** that will help you reach those goals:

- 6.4.1: Finding Factors
- 6.4.2: Finding the GCF of Two or More Expressions
  - 6.4.2: Self Check
  - 6.4.2: Additional Resources
- 6.4.3: Factoring the GCF from Polynomials
  - 6.4.3: Self Check
  - 6.4.3: Additional Resources
- 6.4.4: Factoring Polynomials by Grouping
  - 6.4.4: Self Check
  - 6.4.4: Additional Resources
- 6.4.5: Can You Factor the Polynomial?

After that, you'll **practice and review**.

- 6.4.6: Practice
- 6.4.7: Lesson Summary

6.4.1 Finding Factors

Warm Up

Finding the factors of a number, monomial, or polynomial is an important step used in the activities of this lesson.

1. Factor 56 into prime numbers to find all of its factors.

2. Find all of the factors of $42m^3y^2$.

3. Find all of the common factors between $6n^2p^2$ and $9ns^2$.

4. Find the least common multiple (LCM) of $3x^2$ and $8x^4$.

6.4.2 Finding the GCF of Two or More Expressions

Activity

For the first part of the activity, you will work with a partner. Your teacher will hand out two pieces of paper, one to each of you. It is important for the activity that you do not show your partner your paper.

1. When you receive your paper, look at your expression. Write down all of the factors in your expression.

For example, if your expression was $100x^3y^2$, you would write down $2{\cdot}2{\cdot}5{\cdot}5{\cdot}x{\cdot}x{\cdot}x{\cdot}y{\cdot}y$. Do not share these factors with your partner yet.

2. When you and your partner have finished finding the factors of your expression, you will try to determine the GCF of

6.4.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 3, find the GCF of each set of expressions.

1. $12m^2n^3, 30m^5n^3$

a. $6m^2n^3$
b. $2m^5n^3$
c. $3m^2n^3$
d. $6m^3n^3$

2. $28x^2y^4, 42x^4y^4$

3. $27p^2q^3, 45p^3q^4, 9p^4q^3$

a. $9p^3q^3$
b. $9p^2q^3$
c. $27p^2q^3$
d. $9pq^3$

For questions 4 – 6, factor the expression.

4. $15x^3y - 3x^2y^2 + 6xy^3$

5. $-7a^3 + 21a^2 - 14a$

a. $-7a(a^2 - 3a + 2)$
b. $-7a^2(a - 1)$
c. $7a(a^2 - 3a + 2)$
d. $-7(a^2 - 3a + 2)$

6. $8n(n - 4) + 5(n - 4)$

a. $(8n - 4)(n + 5)$
b. $(n - 4)(8n + 5)$
c. $(n - 4)(8n - 5)$
d. $(n + 4)(8n + 5)$

For questions 7–10, factor by grouping

7. $u^2 - u + 6u - 6$

a. $(u - 1)(u + 6)$
b. $(u + 1)(u - 6)$
c. $(u - 1)(u - 6)$
d. $(u + 1)(u + 6)$

8. $16q^2 + 20q - 28q - 35$

a. $(4q - 5)(4q - 7)$
b. $(4q + 5)(4q - 7)$
c. $(4q - 5)(4q + 7)$
d. $(4q + 5)(4q + 7)$

9. $r^2 - 3r - r + 3$

a. $(r + 3)(r - 1)$
b. $(r - 3)(r + 1)$
c. $(r + 3)(r + 1)$
d. $(r - 3)(r - 1)$

10. $6y^2 + 7y + 24y + 28$

6.5.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 3, factor.

1. $u^2 - 9u + 18$

a. $(u + 3)(u - 6)$
b. $(u + 3)(u + 6)$
c. $(u - 3)(u - 6)$
d. $(u + 9)(u + 2)$

2. $8 - 6x + x^2$

a. $(x + 4)(x - 2)$
b. $(x - 4)(x - 2)$
c. $(x - 8)(x + 1)$
d. $(x + 4)(x + 2)$

3. $a^2 + 5ab - 24b^2$

a. $(a + 8b)(a - 3b)$
b. $(a + 8b)(a + 3b)$
c. $(a - 8b)(a - 3b)$
d. $(a - 8b)(a + 3b)$

For questions 4 – 5, factor using the trial and error method.

4. $4b^2 + 5b + 1$

a. $(b + 1)(4b + 1)$
b. $(5b + 1)(b + 1)$
c. $(5b + 1)(b - 1)$
d. $(b - 1)(4b - 1)$

5. $30x^2 - 53xy - 21y^2$

a. $(3x - y)(10x - 21y)$
b. $(5x + y)(6x - 21y)$
c. $(15x + y)(2x - 21y)$
d. $(3x + y)(10x - 21y)$

6. Factor completely using the "ac" method: $4y^2 + 8y + 3$.

a. $(2y + 1)(2y - 3)$
b. $(2y - 1)(2y + 3)$
c. $(2y + 1)(2y + 3)$
d. $(2y - 1)(2y - 3)$

7. Factor completely using the "ac" method: $18w^2 - 39w + 18$.

a. $3(3w + 2)(2w - 3)$
b. $3(3w + 2)(2w + 3)$
c. $3(3w - 2)(2w - 3)$
d. $(9w - 2)(2w - 3)$

8. Factor by substitution: $h^4 + 4h^2 - 12$.

a. $(h^2 - 2)(h^2 + 6)$
b. $(h - 2)(h + 6)$
c. $(h + 2)(h - 6)$
d. $(h^2 + 2)(h^2 - 6)$

6.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 5, factor using perfect square trinomials.

1. $9y^2 + 24y + 16$

a. $(4y + 3)^2$
b. $(3y + 4)^2$
c. $(3y - 4)^2$
d. $(4y - 3)^2$

2. $16z^2 - 72z + 81$

a. $(4z + 9)^2$
b. $(9z + 4)^2$
c. $(9z - 4)^2$
d. $(4z - 9)^2$

3. $27p^2q + 90pq + 75q$

a. $3q(3p - 5)^2$
b. $3q(3p + 5)^2$
c. $9q(p + 5)^2$
d. $(27p + 5)^2$

4. $64z^2 - 16z + 1$

a. $(8z - 1)^2$
b. $(z + 8)^2$
c. $(8z + 1)^2$
d. $(z - 8)^2$

5. $64x^2y - 96xy + 36y$

a. $4y(4x + 3)^2$
b. $4y(3x - 4)^2$
c. $2y(8x - 6)^2$
d. $4y(4x - 3)^2$

For questions 6 – 10, factor using the difference of squares.

6. $25v^2 - 1$

a. $(v - 5)(v + 5)$
b. $(v - 5)(5v + 1)$
c. $(5v - 1)(5v + 1)$
d. $(5v - 1)(v + 5)$

7. $4 - 49x^2$

a. $(7 - 2x)(7 + 2x)$
b. $(2 - 7x)(2 - 7x)$
c. $(2 - 7x)(2 + 7x)$
d. $(2 - 7x)(7 + 2x)$

8. $169c^2 - 36d^2$

a. $(13c - 6d)(13c - 6d)$
b. $(13d - 6c)(13d + 6c)$
c. $(13c + 6d)^2$
d. $(13c - 6d)(13c + 6d)$

9. $121x^2 - 144y^2$

Problem Animation: Using Multiple Processes to Factor a Polynomial (https://openstax.org/r/factor-polynomial)

2. $75m^3 + 12m$

3. $16x^2 + 24xy - 4x - 6y$

4. $2n^2 + 13n - 7$

5. $121r^2 - s^2$

6. $25w^2 - 60w + 36$

7. $(3x + 1)^2 - 6(3x + 1) + 9$

8. $16x^2 - 24xy + 9y^2 - 64$

Self Check

Which strategy would you use to factor $9x^2 - 24xy + 16y^2$? What is the factored result?

a. trinomial square pattern; $(3x - 4y)^2$
b. difference of squares; $(4x - 3y)^2$
c. grouping; $(3x - 4y)(3x + 4y)$
d. trinomial square pattern; $(3x + 4y)^2$

Additional Resources

Implementing General Strategies for Factoring

Let's practice recognizing and implementing these strategies as we factor polynomials.

Remember, a polynomial is completely factored if, other than monomials, its factors are prime!

EXAMPLE 1

Find a strategy that fits and factor completely: $7x^3 - 21x^2 - 70x$.

Step 1 - Is there a GCF?

$7x^3 - 21x^2 - 70x$

Yes, $7x$.

Step 2 - Factor out the GCF.

$7x(x^2 - 3x - 10)$

Step 3 - In the parentheses, is it a binomial, a trinomial, or are there more than three terms?

Trinomial with leading coefficient 1.

Step 4 - "Undo" FOIL.

$7x(x \quad)(x \quad)$

$7x(x + 2)(x - 5)$

Step 5 - Is the expression factored completely?

Yes. Neither binomial can be factored.

Step 6 - Check your answer by multiplying.

$7x(x + 2)(x - 5)$

$7x(x^2 - 5x + 2x - 10)$

$7x(x^2 - 3x - 10)$

$7x^3 - 21x^2 - 70x$✓

6.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 10, factor each of the expressions completely.

1. $36x^2y + 15xy - 6y$

a. $3y(3x - 2)(4x + 1)$
b. $3y(3x - 2)(4x - 1)$
c. $(3x + 2)(4x - 1)$
d. $3y(3x + 2)(4x - 1)$

2. $49b^2 - 36a^2$

a. $(6b - 7a)(6b + 7a)$
b. $(7b - 6a)^2$
c. $(7b - 6a)(7b + 6a)$
d. $(7b + 6a)^2$

3. $49b^2 - 112b + 64$

a. $(7b + 8)(7b - 8)$
b. $(7b + 8)^2$
c. $(8b + 7)(8b - 7)$
d. $(7b - 8)2$

4. $7b^2 + 7b - 42$

a. $7(b + 3)(b - 2)$
b. $7(b - 3)(b - 2)$
c. $7(b - 3)(b + 2)$
d. $7(b + 3)(b + 2)$

5. $15pq - 15p + 12q - 12$

a. $3(5p + 4)(q - 1)$
b. $3(5p + 4)(q + 1)$
c. $3(4p + 5)(q - 1)$
d. $3(5p - 4)(q - 1)$

6. $4x^2 + 40x + 84$

a. $4(x + 3)(x - 7)$
b. $(4x + 3)(x + 7)$
c. $4(x - 3)(x - 7)$
d. $4(x + 3)(x + 7)$

7. $m^2 + 14mn + 49n^2$

a. $(7m + n)^2$
b. $(m + 7n)^2$
c. $(m + 7n)(m - 7n)$
d. $(7m + n)(7m - n)$

8. $(4x - 5)^2 - 7(4x - 5) + 12$

a. $(4x - 4)(4x - 3)$
b. $(x - 4)(x - 3)$
c. $(4x - 9)(4x - 8)$
d. $(4x - 4)(4x + 3)$

9. $8m^2 - 32$

a. $4(2m - 2)(2m + 2)$
b. $8(m + 2)^2$

7.1.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here are a few pairs of positive numbers whose sum is 50.

Row	First number	Second number	Product
a	1	49	
b	2	48	
c	10	40	

a. Find the product in Row a.

b. Find the product in Row b.

c. Find the product in Row c.

Use the following table to answer questions 2 – 5.

Here are some lengths and widths of a rectangle whose perimeter is 20 meters.

Row	Length (meters)	Width (meters)	Area (square meters)
a	1	9	
b	3	7	
c	5		
d	7		
e	9		

2. Complete the **Width** column of the table.

a. Width in Row c

b. Width in Row d

c. Width in Row e

3. Complete the **Area** column of the table. What do you notice about the areas?

a. Area in Row a

b. Area in Row b

c. Area in Row c

d. Area in Row d

e. Area in Row e

4. Without calculating, predict whether the area of the rectangle will be greater than or less than 25 square meters if the length is 5.25 meters.

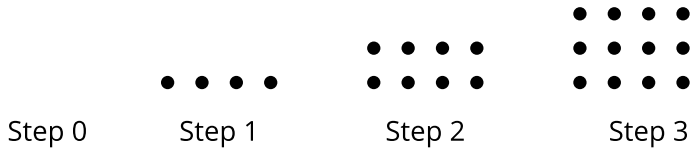
a. greater than
b. less than

7.2.5 Practice

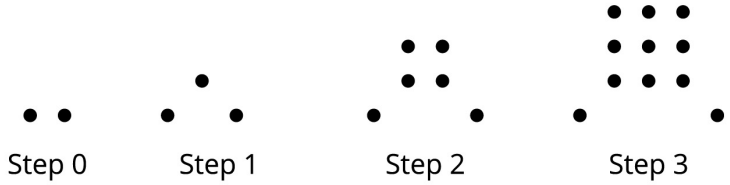
Complete the following questions to practice the skills you have learned in this lesson.

1. Here are two patterns of dots.

Pattern A



Pattern B



a. How many dots will there be in Step 4 of Pattern A?

b. How many dots will there be in Step 4 of Pattern B?

c. Which pattern shows a quadratic relationship between the step number and the number of dots?

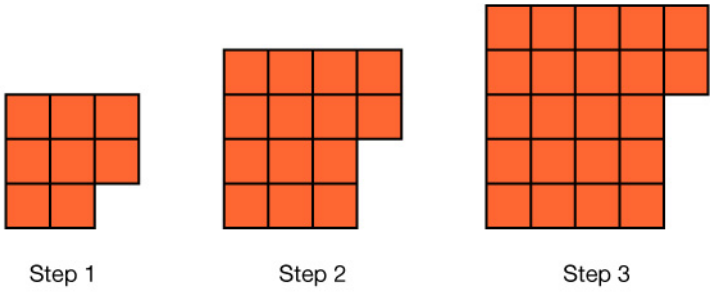
   a. Pattern A
   b. Pattern B

2. Each expression represents the total number of dots in a pattern where n represents the step number.

Select two expressions that represent a quadratic relationship between the step number and the total number of dots. (If you get stuck, consider sketching the first few steps of each pattern as described by the expression.)

a. n^2
b. $2n$
c. $n \cdot n$
d. $n + n$
e. $n + 2$
f. $n \div 2$

3. Use the pattern below to answer the questions.



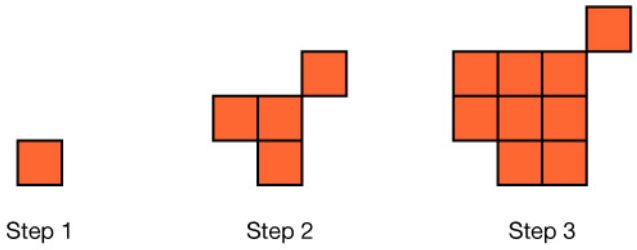
a. How many blocks are in Step 4?

b. How many blocks are in Step 5?

7.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Sketch or describe the figure in Step 4 and Step 15.



a. How many small squares are in Step 4?

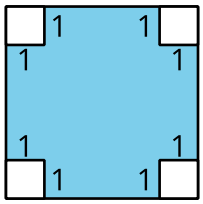
b. How many small squares are in Step 15?

c. Which equation represents the relationship between the step number, n, and the number of small squares, y, in each step?

a. $y = n^2 - n$
b. $y = n^2 - 1$
c. $y = n^2 + 1$
d. $y = n^2$

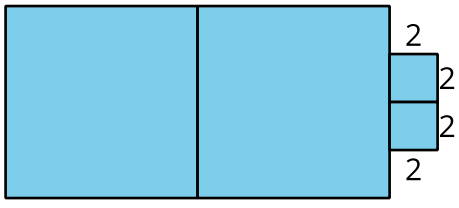
2. Each figure is composed of large squares and small squares. The side length of the large square is x. Choose the correct expression that represents the area, y, of the shaded part of each figure.

Figure A



a. $y = x^2 - 8$
b. $y = x^2 - 4$
c. $y = x^2 - 1$
d. $y = x^2$

Figure B



a. $y = 2x^2$
b. $y = x^2 + 8$
c. $y = 2x^2 + 8$
d. $y = x^2$

3. Use the sequence below to answer the following two questions.

7.4 COMPARING QUADRATIC AND EXPONENTIAL FUNCTIONS

7.4.0 Lesson Overview

In this lesson, you will investigate how quantities that grow quadratically compare to those that grow exponentially. You will use different mathematical models to show the relationship between the functions to determine which one grows faster.

When you finish this lesson, you will be able to:

- Explain using graphs, tables, or calculations how exponential functions eventually grow faster than quadratic functions.

Here are the **activities** that will help you reach those goals:

- 7.4.1: Comparing Expressions
- 7.4.2: Exploring Linear, Exponential, and Quadratic Growth
  - 7.4.2: Self Check
  - 7.4.2: Additional Resources
- 7.4.3: Comparing Exponential and Quadratic Functions
  - 7.4.3: Self Check
  - 7.4.3: Additional Resources
- 7.4.4: Comparing Exponential and Quadratic Expressions

After that, you'll **practice and review**.

- 7.4.5: Practice
- 7.4.6: Lesson Summary

7.4.1 Comparing Expressions

Warm Up

List these quantities in order, from least to greatest, without evaluating each expression. Be prepared to explain your reasoning.

2^{10}

10^2

2^9

9^2

7.4.2 Exploring Linear, Exponential, and Quadratic Growth

In this activity, you will compare two different patterns. In each pattern, the number of small squares is a function of the step number, n.

1. In Pattern A, the length and width of the rectangle grow by 1 small square from each step to the next.

Pattern A

7.4.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following to answer questions 1 – 7.

The table shows values of the expressions $10x^2$ and 2^x.

x	$10x^2$	2^x
1	10	2
2	40	4
3	90	8
4	160	16
8		
10		
12		

1. Complete the statement.

a. When the value of x increases by 1, the expression $10x^2$:

a. Changes logarithmically. It has decreased by one-half.
b. Grows by decreasing amounts but by increasing growth factors.
c. Changes exponentially. It doubles.
d. Grows by increasing amounts but by decreasing growth factors.

b. When the value of x increases by 1, the expression 2^x:

a. Changes logarithmically. It has decreased by one-half.
b. Grows by decreasing amounts but by increasing growth factors.
c. Changes exponentially. It doubles.
d. Grows by increasing amounts but by decreasing growth factors.

2. Will the value of $10x^2$ be greater than or less than 2^x when x is 8?

a. greater than
b. less than

3. Will the value of 2^x be greater than or less than $10x^2$ when x is 10?

a. greater than
b. less than

4. Will the value of $10x^2$ be greater than or less than 2^x when x is 12?

a. greater than
b. less than

5. Determine the following values of $10x^2$.

a. What is the value of $10x^2$ when x is 8?

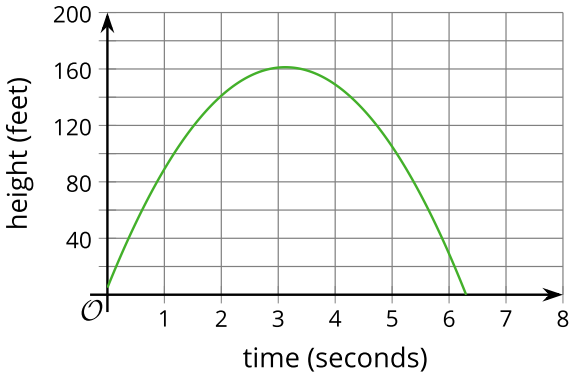
b. What is the value of $10x^2$ when x is 10?

7.5.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. A rocket is launched in the air and its height, in feet, is modeled by the function h.

Here is a graph representing h.



Select three true statements about the situation.

a. The rocket is launched from a height less than 20 feet above the ground.
b. The rocket is launched from about 20 feet above the ground.
c. The rocket reaches its maximum height after about 3 seconds.
d. The rocket reaches its maximum height after about 160 seconds.
e. The maximum height of the rocket is about 160 feet.

2. A baseball travels d meters t seconds after being dropped from the top of a building. The distance traveled by the baseball can be modeled by the equation $d = 5t2$.

Use this table to answer questions a – g.

t (seconds)	d (meters)
0	
0.5	
1	
1.5	
2	

a. How many meters (d) did the baseball travel in 0 seconds (t)?
b. How many meters (d) did the baseball travel in 0.5 seconds (t)?
c. How many meters (d) did the baseball travel in 1 second (t)?
d. How many meters (d) did the baseball travel in 1.5 seconds (t)?
e. How many meters (d) did the baseball travel in 2 seconds (t)?
f. Is this graph an accurate representation of the data in the table?

7.6 BUILDING QUADRATIC FUNCTIONS TO DESCRIBE SITUATIONS, PART 2

7.6.0 Lesson Overview

In this lesson, you will build on your knowledge of using simple quadratic functions to describe how an object falls over time, given the effect of constantly changing speed. You will construct quadratic functions to represent projectile motions, objects that are launched directly upward. Working through the activities, you will discover that gravity affects the speed of a projectile, and the linear model will need to be represented differently to adjust for the change.

When you finish this lesson, you will be able to:

- Create quadratic functions and graphs that represent a situation.
- Relate the vertex of a graph and the zeros of a function to a situation.
- Know that the domain of a function can depend on the situation it represents.

Here are the **activities** that will help you reach those goals:

- 7.6.1: Using Linear Functions to Describe Constant Speed
- 7.6.2: The Force of Gravity Change in Quadratic Functions
  - 7.6.2: Self Check
  - 7.6.2: Additional Resources
- 7.6.3: Using Quadratic Functions to Describe Height
  - 7.6.3: Self Check
  - 7.6.3: Additional Resources
- 7.6.4: Interpreting Graphs of Quadratic Functions

After that, you'll **practice and review.**

- 7.6.5: Practice
- 7.6.6: Lesson Summary

7.6.1 Using Linear Functions to Describe Constant Speed

Warm Up

A cannon is 10 feet off the ground. It launches a cannonball straight up with a velocity of 406 feet per second.

Imagine that there is no gravity and that the cannonball continues to travel upward with the same velocity.

1. Answer questions a–f using the table with the heights of the cannonball at different times.

Seconds	Distance above ground (feet)
0	10
1	a. _____
2	b. _____
3	c. _____
4	d. _____
5	e. _____
t	f. _____

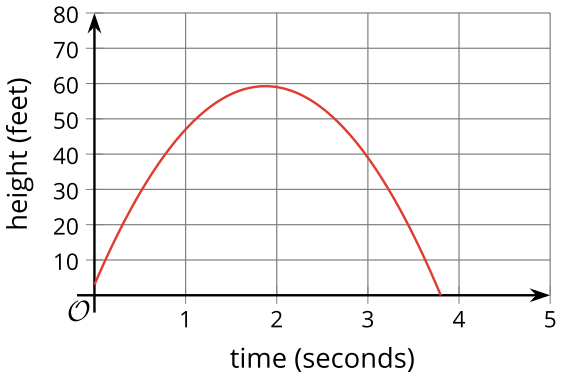
7.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The height of a diver above the water is given by $h(t) = -5t^2 + 10t + 3$, where t is time measured in seconds and $h(t)$ is measured in meters. **Select two** statements that are true about the situation.

a. The diver begins 5 meters above the water.
b. The diver begins 3 meters above the water.
c. The function has 1 zero that makes sense in this situation.
d. The function has 2 zeros that make sense in this situation.
e. The graph that represents h starts at the origin and curves upward.
f. The diver begins at the same height as the water level.

2. The height of a baseball, in feet, is modeled by the function h given by the equation $h(t) = 3 + 60t - 16t^2$. The graph of the function is shown.



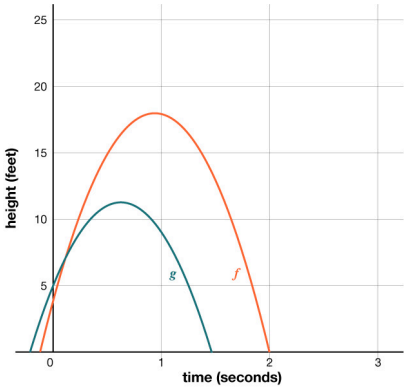
a. About when does the baseball reach its maximum height? Round to the nearest second.

b. About how high is the maximum height of the baseball? Round to the nearest ten.

c. When does the baseball hit the ground? Round to the nearest second.

3. Two rocks are launched straight up in the air. The height of Rock A is given by the function f, where $f(t) = 4 + 30t - 16t^2$. The height of Rock B is given by g, where $g(t) = 5 + 20t - 16t^2$. In both functions, t is time measured in seconds and height is measured in feet.

The following is a graph of the functions. Use the graph to answer the questions a – c.



a. Which graph is the height of Rock A, $f(t) = 4 + 30t - 16t^2$?

a. Green Graph
b. Red Graph

7.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following scenario to answer questions 1 – 5.

Based on past musical productions, a theater predicts selling $400 - 8p$ tickets when each ticket is sold at p dollars.

1. Answer questions a–g to complete the middle column in the table to find out how many tickets the theater expects to sell and what revenues it expects to receive at the given ticket prices.

Ticket price (dollars)	Number of tickets sold	Revenue (dollars)
5	360	1800
10	a. ____	
15	b. ____	
20	c. ____	
30	d. ____	
45	e. ____	
50	f. ____	
p	g. ____	

a. When the price is $\$10$ per ticket, what is the number of tickets sold?

b. When the price is $\$15$ per ticket, what is the number of tickets sold?

c. When the price is $\$20$ per ticket, what is the number of tickets sold?

d. When the price is $\$30$ per ticket, what is the number of tickets sold?

e. When the price is $\$45$ per ticket, what is the number of tickets sold?

f. When the price is $\$50$ per ticket, what is the number of tickets sold?

g. When the price is p per ticket, what is the number of tickets sold?

a. $p(8p - 400)$
b. $p(400 - 8p)$
c. $400 - 8p$
d. $392p$

2. Answer questions a-g to complete the right column in the table to find out what revenues the theater expects to receive at the given ticket prices.

7.8.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Choose the expression that best describes the diagram.

	$2x$	5
x	$2x^2$	$5x$
3	$6x$	15

a. $(2x + 5)(x + 3)$ is equivalent to $30x^2$.
b. $(2x + 5)(x + 3)$ is equivalent to $2x^2 + 11x + 15$.
c. $(2x + 5)(x + 3)$ is equivalent to $2x^2 + 14x + 23$.
d. $(2x + 5)(x + 3)$ is equivalent to $2x^2 + 5x + 8$.

In questions 2a–2d, match each expanded quadratic expression with an equivalent expression written as a product.

2a. $x^2 + 12x + 32$

a. $(x + 2)(x + 3)$
b. $(x + 8)(x + 4)$
c. $(x + 4)(x + 2)$
d. $(x + 2)(x + 6)$

2b. $x^2 + 5x + 6$

a. $(x + 2)(x + 3)$
b. $(x + 8)(x + 4)$
c. $(x + 4)(x + 2)$
d. $(x + 2)(x + 6)$

2c. $x^2 + 6x + 8$

a. $(x + 2)(x + 3)$
b. $(x + 8)(x + 4)$
c. $(x + 4)(x + 2)$
d. $(x + 2)(x + 6)$

2d. $x^2 + 8x + 12$

a. $(x + 2)(x + 3)$
b. $(x + 8)(x + 4)$
c. $(x + 4)(x + 2)$
d. $(x + 2)(x + 6)$

3. **Select three** expressions that are equivalent to $x^2 + 4x$.

a. $x(x + 4)$
b. $(x + 2)^2$
c. $(x + x)(x + 4)$
d. $(x + 2)^2 - 4$
e. $(x + 4)x$

Tyler drew a diagram to expand $(x + 5)(2x + 3)$.

7.9 STANDARD FORM AND FACTORED FORM

7.9.0 Lesson Overview

In this lesson, you will draw and interpret area diagrams and use the distributive property to write quadratic equations in factored and standard forms.

When you finish this lesson, you will be able to:

- Rewrite quadratic expressions given in factored form in standard form using either the distributive property or a diagram.
- Know the difference between "factored form" and "standard form."

Here are the **activities** that will help you reach those goals:

- 7.9.1: Solving Equations Using Opposite Operations
- 7.9.2: Finding Products of Difference
  - 7.9.2: Self Check
  - 7.9.2: Additional Resources
- 7.9.3: Standard and Factored Forms of Quadratic Expressions
  - 7.9.3: Self Check
  - 7.9.3: Additional Resources
- 7.9.4: Expressions in Standard Form

After that, you'll **practice and review**.

- 7.9.5: Practice
- 7.9.6: Lesson Summary

7.9.1 Solving Equations Using Opposite Operations

Warm Up

Solve each equation mentally. Then, enter your answer to check it.

1. $40 - 8 = 40 + n$

2. $25 + -100 = 25 - n$

3. $-3 - \frac{1}{2} = 3 + \frac{1}{2}n$

4. $72 - n = 72 + 6$

7.9.2 Finding Products of Differences

Activity

1. Show that $(x - 1)(x - 1)$ and $x^2 - 2x + 1$ are equivalent expressions by drawing a diagram or applying the **distributive property**. Show your reasoning.

2. For each expression, write an equivalent expression. Show your reasoning.

a. $(x + 1)(x - 1)$

b. $(x - 2)(x + 3)$

c. $(x - 2)^2$

Video: Finding Products of Differences

Watch the following video to learn more about finding products of differences.

Finding Products of Differences (https://openstax.org/r/products-differences)

7.9.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Choose the standard form of the quadratic expression.

a. $(x+4)(x-1)$

a. $-x^2-3x+4$
b. x^2+3x-4
c. $-x^2+3x-4$
d. x^2-3x+4

b. $(2x-1)(3x-1)$

a. $-6x^2+5x-1$
b. $-6x^2-5x+1$
c. $6x^2-5x+1$
d. $6x^2+5x-1$

2. Consider the expression $8-6x+x^2$.

a. Is the expression in standard form?

a. Yes
b. No

b. Is the expression equivalent to $(x-4)(x-2)$?

a. Yes
b. No

3. Find the standard form of the quadratic expression $(x-2)(x+8)$. Use the standard form to answer questions a, b, and c.

a. What is the value of a in this expression ax^2+bx+c?

b. What is the value of b in this expression ax^2+bx+c?

c. What is the value of c in this expression ax^2+bx+c?

7.10.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. A quadratic function f is defined by $f(x) = (x - 7)(x + 3)$.

a. Without graphing, identify the x-intercepts of the graph of f.

a. $(0, -7), (0, 3)$
b. $(0, 7), (0, -3)$
c. $(-7, 0), (3, 0)$
d. $(7, 0), (-3, 0)$

b. Use the standard form of $(x - 7)(x + 3)$ to identify the y-value of the y-intercept of the graph of f.

2. Answer these questions for the equation $y = x^2 - 5x + 4$.

a. What is the y-value of the y-intercept of this equation's graph?

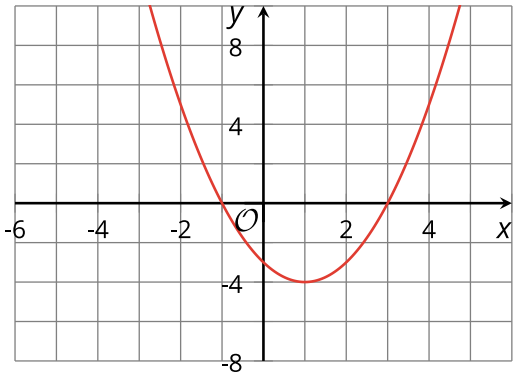
b. An equivalent way to write this equation is $y = (x - 4)(x - 1)$. What are the x-intercepts of this equation's graph?

a. $(0, -1), (0, -4)$
b. $(0, 1), (0, 4)$
c. $(-1, 0), (-4, 0)$
d. $(1, 0), (4, 0)$

3. Noah said that if we graph $y = (x - 1)(x + 6)$, the x-intercepts will be at $(1, 0)$ and $(-6, 0)$. Is Noah correct?

a. Yes
b. No

4. **Select two** x-intercepts of this graph.



a. 1
b. -1
c. 3
d. -3

5. Use the standard form of $y = (x - 2)(x - 4)$ to identify the y-value of the y-intercept of the graph.

7.11.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. **Select two** true statements about the graph that represents $y = 2x(x - 11)$.

   a. Its x-intercepts are at $(-2, 0)$ and $(11, 0)$.
   b. Its x-intercepts are at $(0, 0)$ and $(11, 0)$.
   c. Its x-intercepts are at $(2, 0)$ and $(-11, 0)$.
   d. It has only one x-intercept.
   e. The x-coordinate of its vertex is -4.5.
   f. The x-coordinate of its vertex is 11.
   g. The x-coordinate of its vertex is 4.5.
   h. The x-coordinate of its vertex is 5.5.

2. **Select two** equations whose graphs have a vertex with x-coordinate 2.

   a. $y = (x - 2)(x - 4)$
   b. $y = (x - 2)(x + 2)$
   c. $y = (x - 1)(x - 3)$
   d. $y = x(x + 4)$
   e. $y = x(x - 4)$

3. Three different equations are given. Refer to these equations to complete questions a–f.

Equation	x-intercept	x-coordinate of the vertex
$y = x(x - 2)$	a. _____	b. _____
$y = (x - 4)(x + 5)$	c. _____	d. _____
$y = -5x(3 - x)$	e. _____	f. _____

   a. What are the x-intercepts of the graph that represents $y = x(x - 2)$?

      a. $(0, 0)$ and $(-2, 0)$
      b. $(0, 0)$ and $(0, -2)$
      c. $(0, 0)$ and $(2, 0)$
      d. $(0, 0)$ and $(0, 2)$

   b. What is the x-coordinate of the vertex of the graph that represents $y = x(x - 2)$?

   c. What are the x-intercepts of the graph that represents $y = (x - 4)(x + 5)$?

      a. $(0, 5)$ and $(0, -4)$
      b. $(5, 0)$ and $(-4, 0)$
      c. $(0, -5)$ and $(0, 4)$
      d. $(-5, 0)$ and $(4, 0)$

   d. What is the x-coordinate of the vertex of the graph that represents $y = (x - 4)(x + 5)$?

   e. What are the x-intercepts of the graph that represents $y = -5x(3 - x)$?

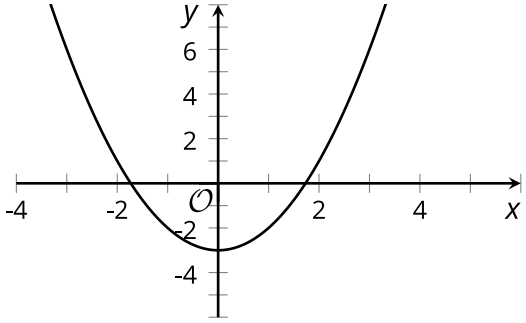
      a. $(0, 0)$ and $(0, 3)$
      b. $(0, 0)$ and $(3, 0)$
      c. $(0, 0)$ and $(-3, 0)$
      d. $(0, 0)$ and $(0, -3)$

   f. What is the x-coordinate of the vertex of the graph that represents $y = -5x(3 - x)$?

7.12.6 Practice

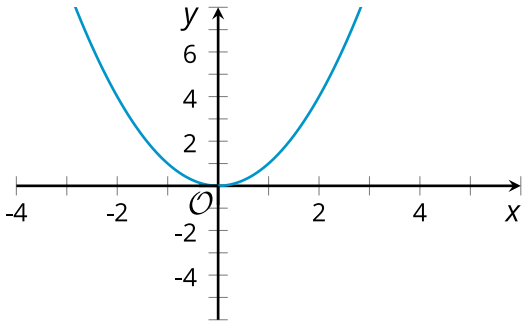
Complete the following questions to practice the skills you have learned in this lesson.

1. Choose the quadratic equation that is represented by the graph.



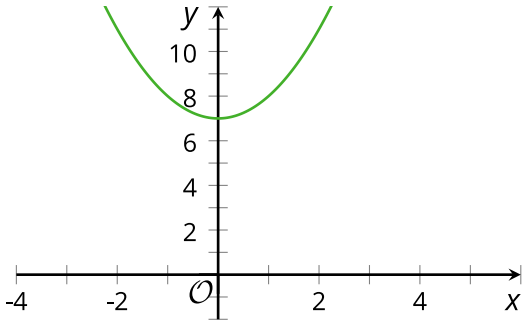
a. $f(x) = x^2 + 7$
b. $f(x) = x^2$
c. $f(x) = x^2 - 3$
d. $f(x) = x^2 + 5$

2. Choose the quadratic equation that is represented by the graph.



a. $f(x) = x^2 + 7$
b. $f(x) = x^2$
c. $f(x) = x^2 - 3$
d. $f(x) = x^2 + 5$

3. Choose the quadratic equation that is represented by the graph.



a. $f(x) = x^2 + 7$
b. $f(x) = x^2$
c. $f(x) = x^2 - 3$
d. $f(x) = x^2 + 5$

7.13 GRAPHING THE STANDARD FORM, PART 2

7.13.0 Lesson Overview

In this lesson, you will work on activities that focus on the coefficient of the linear term, the b in $ax^2 + bx + c$, and how changes to it affect the graph. You will discover that adding a linear term to the squared term translates the graph in both horizontal and vertical directions. This understanding will show that writing an expression such as $x^2 + b$ in factored form can help us reason how to sketch the graph.

When you finish this lesson, you will be able to:

- Explain how the b in $ax^2 + bx + c$ affects the graph of the equation.
- Match equations given in standard and factored form with their graph.

Here are the **activities** that will help you reach those goals:

- 7.13.1: Equivalent Expressions
- 7.13.2: The Linear Term in a Quadratic Expression
  - 7.13.2: Self Check
  - 7.13.2: Additional Resources
- 7.13.3: Writing Equations that Represent a Graph
  - 7.13.3: Self Check
  - 7.13.3: Additional Resources
- 7.14.4: Writing Quadratic Equations from Real Solutions
  - 7.14.4: Self Check
  - 7.14.4: Additional Resources
- 7.13.5: Graphing Quadratic Equations

After that, you'll **practice and review**.

- 7.13.6: Practice
- 7.13.7: Lesson Summary

7.13.1 Equivalent Expressions

Warm Up

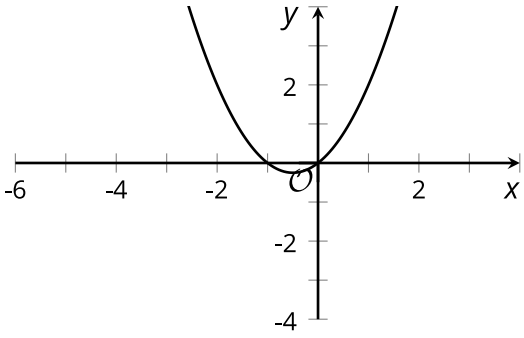
1. Complete each row with an equivalent expression in standard form or factored form.

Standard Form	Factored Form
x^2	a. ____
d. ____	$x(x + 9)$
$x^2 - 18x$	b. ____
e. ____	$x(6 - x)$
$-x^2 + 10x$	c. ____
f. ____	$-x(x + 2.75)$

7.13.6 Practice

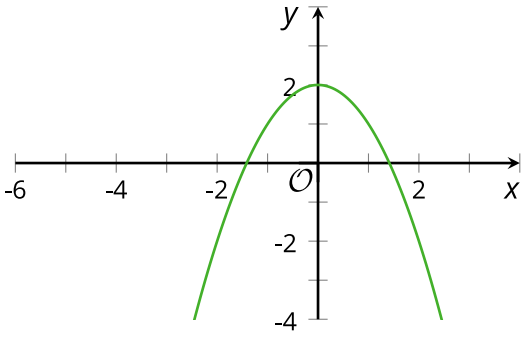
Complete the following questions to practice the skills you have learned in this lesson.

1. Choose the quadratic equation that represents each graph.



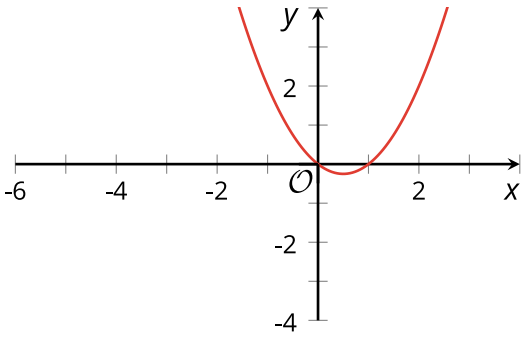
a.

a. $y = x^2 + 3x$
b. $y = x^2 - x$
c. $y = -x^2 + 2$
d. $y = x^2 + x$



b.

a. $y = x^2 + 3x$
b. $y = x^2 - x$
c. $y = -x^2 + 2$
d. $y = x^2 + x$



c.

a. $y = x^2 + 3x$
b. $y = x^2 - x$
c. $y = -x^2 + 2$
d. $y = x^2 + x$

7.14 GRAPHS THAT REPRESENT SITUATIONS

7.14.0 Lesson Overview

In this lesson, you will interpret equations and graphs of quadratic functions in terms of the situations they represent. Additionally, you will use your analyses to solve problems and to compare quadratic functions given in different representations.

When you finish this lesson, you will be able to:

- Explain how a quadratic equation and its graph relate to a situation.

Here are the **activities** that will help you reach those goals:

- 7.14.1: Evaluating Quadratic Functions in the Real World
- 7.14.2: Interpreting a Functional Relationship Between Two Quantities
  - 7.14.2: Self Check
  - 7.14.2: Additional Resources
- 7.14.3: Analyzing Functions Using Different Representations
  - 7.14.3: Self Check
  - 7.14.3: Additional Resources
- 7.14.4: Using Quadratic Functions to Solve Problems
- 7.14.5: Identifying Key Features of Graphed Functions

After that, you'll **practice and review**.

- 7.14.6: Practice
- 7.14.7: Lesson Summary

7.14.1 Evaluating Quadratic Functions in the Real World

Warm Up

The height, in inches, of a frog's jump is modeled by the equation $h(t) = 60t - 75t^2$, where the time t, after the frog jumps, is measured in seconds.



1. For $h(t) = 60t - 75t^2$:

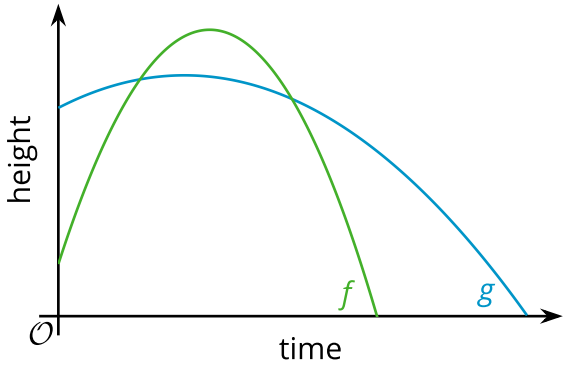
a. Find $h(0)$.

b. Find $h(0.8)$.

7.14.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here are graphs of functions f and g.
Each represents the height of an object being launched into the air as a function of time.



a. Which object was launched from a higher point?

   a. The object represented by function g
   b. The object represented by function f

b. Which object reached a higher point?

   a. The object represented by function g
   b. The object represented by function f

c. Which object was launched with the greater upward velocity?

   a. The object represented by function g
   b. The object represented by function f

d. Which object landed last?

   a. The object represented by function g
   b. The object represented by function f

2. The function h given by $h(t) = (1 - t)(8 + 16t)$ models the height of a ball in feet, t seconds after it was thrown.

a. Find one of the zeros of the functions using $(8 + 16t)$.

b. Find one of the zeros of the functions using $(1 - t)$.

c. What do the zeros tell us in this situation?

   a. The negative zero represents the time, in seconds, when the basketball is on the ground or hits the ground.
   b. The positive zero represents the time, in seconds, when the basketball reaches the maximum height.
   c. The negative zero represents the height, in feet, when the time is zero.
   d. The positive zero means the time, in seconds, when the basketball is on the ground or hits the ground.

d. Are both zeros meaningful?

   a. Yes
   b. No

e. From what height, in feet, is the ball thrown?

f. When, in seconds, does the ball reach its highest point?

g. How high, in feet, does the ball go?

3. The height in feet of a thrown football is modeled by the equation $f(t) = 6 + 30t - 16t^2$, where time t is measured in seconds.

7.15.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. **Select three** quadratic expressions written in vertex form.

   a. $y = (x - 2)^2 + 1$
   b. $y = x^2 - 4$
   c. $y = x(x + 1)$
   d. $y = (x + 3)(x + 3)$
   e. $y = (x - 4)^2 + 6$

Examine the following two equations and use them to answer questions 2 – 5. One defines function m and the other defines function p.

- $m(x) = x(x + 6)$
- $p(x) = (x + 3)^2 - 9$

2. How do you know that the expressions defining m and p are equivalent?

   a. Both expressions are equal to $x^2 - 3$.
   b. Both expressions are equal to $2x + 6$.
   c. Both expressions are equal to $x^2 - 6x$.
   d. Both expressions are equal to $x^2 + 6x$.

3. Recall $m(x) = x(x + 6)$ and $p(x) = (x + 3)^2 - 9$. What is the x-coordinate of the vertex of the graph of m?

4. Recall $m(x) = x(x + 6)$ and $p(x) = (x + 3)^2 - 9$. What is the y-coordinate of the vertex of the graph of m?

5. Recall $m(x) = x(x + 6)$ and $p(x) = (x + 3)^2 - 9$. What are the x-intercepts of the graph of p?

   a. $(0, 0)$ and $(0, 6)$
   b. $(0, 0)$ and $(0, -6)$
   c. $(0, 0)$ and $(6, 0)$
   d. $(0, 0)$ and $(-6, 0)$

For questions 6 – 11, enter the coordinates of the vertex of the graph that represents the equation.

6. Find the coordinates of the vertex for the equation $y = (x - 3)^2 + 5$.

   a. What is the x-coordinate of the vertex?

   b. What is the y-coordinate of the vertex?

7. Find the coordinates of the vertex for the equation $y = (x + 7)^2 + 3$.

   a. What is the x-coordinate of the vertex?

   b. What is the y-coordinate of the vertex?

8. Find the coordinates of the vertex for the equation $y = (x - 4)^2$.

   a. What is the x-coordinate of the vertex?

   b. What is the y-coordinate of the vertex?

9. Find the coordinates of the vertex for the equation $y = x^2 - 1$.

   a. What is the x-coordinate of the vertex?

   b. What is the y-coordinate of the vertex?

10. Find the coordinates of the vertex for the equation $y = 2(x + 1)^2 - 5$.

    a. What is the x-coordinate of the vertex?

    b. What is the y-coordinate of the vertex?

7.16.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. What is the x-coordinate of the vertex of the graph that represents $y = (x - 2)^2 - 8$?

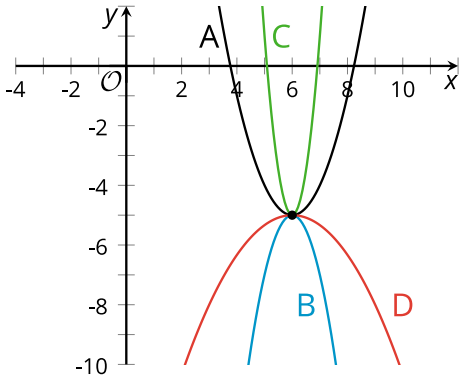
2. What is the y-coordinate of the vertex of the graph that represents $y = (x - 2)^2 - 8$?

3. Where is the y-intercept for the equation $y = (x - 2)^2 - 8$ located?

4. Which of the following is another point on the graph of the equation $y = (x - 2)^2 - 8$?

a. $(-4, -4)$
b. $(4, 4)$
c. $(-4, 4)$
d. $(4, -4)$

5. Which graph is a representation of $y = (x - 6)^2 - 5$?

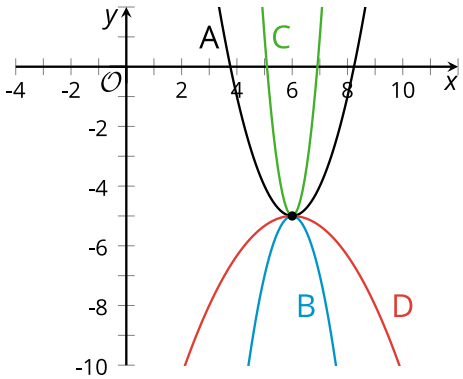


a. Graph A
b. Graph B
c. Graph C
d. Graph D

6. The function v is defined by $v(x) = 12(x + 5)^2 - 7$.
Without graphing, determine if the vertex of the graph representing v shows the minimum or maximum value of the function.

a. The vertex shows the minimum value.
b. The vertex shows the maximum value.

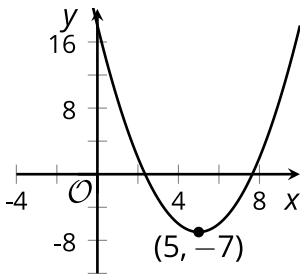
7. Choose the equation that matches the graph of parabola D .



7.17.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here is the graph of quadratic function f.

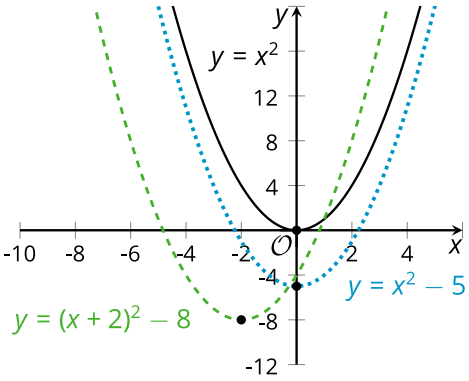


- Andre uses the expression $(x-5)^2+7$ to define f.
- Noah uses the expression $(x+5)^2-7$ to define f.

Do you agree with either of them?

a. Yes
b. No

2. Here are the graphs of $f(x)=x^2$, $f(x)=x^2-5$, and $f(x)=(x+2)^2-8$.



a. How do the three graphs compare?

a. All three graphs contain the point $(0,0)$.
b. They all have a negative a coefficient in the vertex form of the equation.
c. The shape is the same but they are in different locations.
d. They all have the same vertex.

b. Compare the graphs of $f(x)=x^2$ and $f(x)=x^2-5$. What role does the -5 play in the comparison?

a. Subtracting 5 from the squared term shifts the graph down by 5 units.
b. Subtracting 5 from the squared term shifts the graph up by 5 units.
c. Subtracting 5 from the squared term shifts the graph right by 5 units.
d. Subtracting 5 from the squared term shifts the graph left by 5 units.

c. Compare the graphs of $f(x)=x^2$ and $f(x)=(x+2)^2-8$. What roles do the $+2$ and -8 play in the comparison?

a. Adding 2 to x before squaring shifts the graph of $f(x)=x^2$ to the down by 2 units and subtracting 8 from the squared term shifts the graph right by 8 units.
b. Adding 2 to x before squaring shifts the graph of $f(x)=x^2$ to the right by 2 units and subtracting 8 from the squared term shifts the graph up by 8 units.
c. Adding 2 to x before squaring shifts the graph of $f(x)=x^2$ to the up by 2 units and subtracting 8 from the squared term shifts the graph left by 8 units.

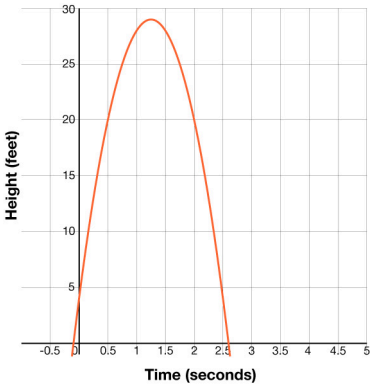
8.1.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

Use the following scenario to answer questions 1 – 3.

A water balloon is launched by a catapult into the air. The water balloon is launched from 4 feet above the ground, with an initial vertical velocity of 40 feet per second.

The function $h(t) = -16t^2 + 40t + 4$ models the height of the water balloon over the ground, in feet, t seconds after launch.



1. What is the height of the water balloon 1 second after launch?

a. 34 feet
b. 28 feet
c. 22 feet
d. 18 feet

2. Will the water balloon still be in the air 2 seconds after the launch?

a. No
b. Yes

3. Using the graph, estimate when the water balloon hit the ground.

a. 2.6 seconds
b. 2.1 seconds
c. 1.8 seconds
d. 1.2 seconds

Use the following scenario to answer questions 4 – 5.

Suppose you want to frame a different picture that measures 12 inches by 9 inches. The frame is created from 46 square inches of framing material. The frame must be uniform in width, and all of the framing material must be used.

4. Which of the following is true about the situation described?

a. The perimeter of the framed picture is greater than 42 inches.
b. The area of the framed picture is 148 square inches.
c. The perimeter of the framed picture is 42 inches.
d. The perimeter of the picture is 40 inches.

5. What is the total area of the framed picture?

a. 176 square inches
b. 154 square inches
c. 108 square inches
d. 46 square inches

8.2.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The expression $9t + 7.25$ represents the cost to purchase tickets for a concert, where t is the number of tickets. A group of friends paid $79.25 for tickets. How many tickets were bought?

a. 12 tickets
b. 10 tickets
c. 8 tickets
d. 6 tickets

Use the following scenario to answer questions 2 – 4.

A model rocket is launched from a launchpad. The function modeling the height of the rocket, in feet, t seconds after being launched is shown.

$f(t) = -15t^2 + 58t + 8$

2. Which equation would you solve to find the time at which the rocket hits the ground?

a. $15t^2 + 58t + 8 = 0$
b. $-15t^2 + 58t = 0$
c. $-15t^2 + 58t = 8$
d. $-15t^2 + 58t + 8 = 0$

3. Use a non-graphing method to find the solution to the previous equation to determine the time that has passed when the rocket hits the ground.

a. 5 seconds
b. 4 seconds
c. 3 seconds
d. 2 seconds

4. How high off the ground is the launchpad from which the rocket lifts off?

a. 15 feet
b. 8 feet
c. 2 feet
d. 0 feet

Use the following scenario to answer questions 5 – 8.

The expressions $t(90 - 3t)$ and $-3t^2 + 90t$ define the same function. The function models the number of lumens emitted by a spotlight in an art exhibit over the course of t seconds.

5. Which equation represents the time when the spotlight is not emitting any light?

a. $-3t^2 + 90t = 30$
b. $-3t^2 + 90t = 0$
c. $-3t^2 + 90t = 675$
d. $t(90 - 3t) = -3t^2 + 90t$

6. At what time(s) does the spotlight not emit any light?

a. 15 seconds
b. Only 0 seconds
c. 10 and 25 seconds
d. 0 and 30 seconds

7. The artist planned for the spotlight to emit 600 lumens at 10 seconds from the start. At what other time will the spotlight emit 600 lumens of light?

a. 30 seconds
b. 25 seconds

8.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 2, determine the number of solutions. Then solve the equation.

1. $3x^2 = 48$

a. 2 solutions; $x = 12$ and $x = -12$
b. 2 solutions; $x = 4$ and $x = -4$
c. 1 solution; $x = 4$
d. 0 solutions

2. $(x + 2)(x + 2) = 0$

a. 2 solutions; $x = 2$ and $x = -2$
b. 1 solution; $x = 2$
c. 1 solution; $x = -2$
d. 0 solutions

For questions 3 – 10, solve the quadratic equation.

3. $4m^2 = 36$

a. $m = 2$ and $m = -2$
b. $m = 9$ and $m = -9$
c. $m = 3$
d. $m = 3$ and $m = -3$

4. $d^2 - 25 = 0$

a. $d = 25$
b. $d = -5$
c. $d = 5$
d. $d = 5$ and $d = -5$

5. $\frac{2}{3}z^2 = 54$

a. $z = 7$ and $z = -7$
b. $z = 9$ and $z = -9$
c. $z = 7$ and $z = -9$
d. $z = 6$ and $z = -6$

6. $4p^2 + 10 = 266$

a. $p = 8$ and $p = -8$
b. $p = 16$ and $p = -16$
c. $p = 12$ and $p = -12$
d. $p = 18$ and $p = -18$

7. $(a - 7)^2 + 5 = 105$

a. $a = 3$ and $a = -3$
b. $a = -10$ and $a = 10$
c. $a = -17$ and $a = 3$
d. $a = 17$ and $a = -3$

8. $(b - 1)^2 - 9 = 55$

a. $b = -9$ and $b = 7$
b. $b = 9$ and $b = -7$
c. $b = 8$ and $b = -8$
d. $b = 9$ and $b = -9$

9. $\frac{1}{7}(x + 3)^2 - 5 = 2$

8.4.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, solve each equation. Identify every correct answer.

1. $x(3x + 12) = 0$

a. $x = 0$ and $x = -4$
b. $x = 4$ and $x = -4$
c. $x = 0$ and $x = -3$
d. $x = 0$ and $x = -12$

2. $(x - 7)(x + 9) = 0$

a. $x = -7$ and $x = 9$
b. $x = 7$ and $x = -9$
c. $x = 0$, $x = 7$, and $x = -9$
d. $x = 7$ and $x = -7$

3. $(x - 8)(4x + 15) = 0$

a. $x = 8$ and $x = \frac{15}{4}$
b. $x = 8$ and $x = \frac{4}{15}$
c. $x = 8$ and $x = -\frac{4}{15}$
d. $x = 8$ and $x = -\frac{15}{4}$

4. $x(x - 6)(5x - 20) = 0$

a. $x = 0$ and $x = 6$
b. $x = 4$ and $x = 6$
c. $x = 0$, $x = 4$, and $x = 6$
d. $x = 0$, $x = -4$, and $x = 6$

For questions 5 - 6, use the following.

The functions below model the height of a projectile in feet, t seconds after launch.

- $h(t) = -7t^2 + 34t + 48$
- $h(t) = (-7t - 8)(t - 6)$

5. Without graphing, determine at what time the height of the object is 0 feet.

a. 0 seconds
b. $-\frac{8}{7}$ seconds
c. 6 seconds
d. $\frac{8}{7}$ seconds

6. Determine the height of the object before it is launched.

a. 48 feet
b. 6 feet
c. $\frac{8}{7}$ feet
d. 0 feet

For questions 7 – 10, solve each equation. Identify every correct answer.

7. $x(6x - 18) = 0$

a. $x = -3$
b. $x = 0$ and $x = 3$
c. $x = 3$
d. $x = 0$ and $x = -3$

8.5.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 2, decide whether each statement is true or false.

1. $x(x-5)=0$ has only one solution.

a. false
b. true

2. A solution to $x^2-49=0$ is $x=-7$.

a. false
b. true

3. Use graphing technology to determine the number of solutions for the equation $(x-4)(x+5)=-20$. Then find any solutions.

a. Two solutions; $x=-1$ and $x=0$
b. Two solutions; $x=1$ and $x=0$
c. One solution; $x=1$
d. No solution

4. Use graphing technology to determine the number of solutions for the equation $(x+2)(x+5)=-2$. Then find any solutions.

a. Two solutions; $x=3$ and $x=4$
b. Two solutions; $x=-3$ and $x=-4$
c. Two solutions; $x=-3$ and $x=4$
d. No solution

For questions 5 – 8, choose the best method to solve the equation and then use the method to find the solution.

5. $x(x-6)=0$

a. $x=6$ and $x=-6$
b. $x=0$ and $x=6$
c. $x=6$
d. $x=0$ and $x=-6$

6. $\frac{1}{2}(x-4)^2=32$

a. $x=12$ and $x=-4$
b. $x=-8$ and $x=8$
c. $x=-12$ and $x=4$
d. $x=-4$ and $x=4$

7. $(x+7)(x-2)=-8$

a. $x=-6$ and $x=1$
b. $x=-8$ and $x=-1$
c. $x=-7$ and $x=2$
d. No solutions

8. $(x-4)(x+9)=-42.25$

a. $x=-5$ and $x=-1$
b. $x=-3$ and $x=-2$
c. $x=-2.5$
d. No solutions

9. Which statement is true of the equation $x^2+36=0$?

a. The solutions are $x=6$ and $x=-6$.

8.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Write $(x + 6)(x + 5)$ in standard form.

a. $x^2 - 11x + 30$
b. $x^2 + 30$
c. $x^2 + 11x + 30$
d. $x^2 + 10x - 30$

2. Write $(x - 3)(x - 7)$ in standard form.

a. $x^2 - 10x + 21$
b. $x^2 + 10x + 21$
c. $x^2 + 10x - 21$
d. $x^2 + 21$

3. An expression in standard form is shown. The diagram below shows it is equivalent to which expression in factored form?

	x	2
x	x^2	$2x$
9	$9x$	18

$x^2 + 11x + 18$

a. $x(x + 18)$
b. $(x + 2)(x - 9)$
c. $(x + 2)(x + 9)$
d. $(x - 2)(x - 9)$

4. An expression in standard form is shown. The diagram below shows it is equivalent to which expression in factored form?

	x	-5
x	x^2	$-5x$
-7	$-7x$	35

$x^2 - 12x + 35$

a. $(x + 5)(x + 7)$
b. $x(x - 35)$
c. $(x - 5)(x + 7)$
d. $(x - 5)(x - 7)$

5. Find the factored form of $x^2 + 13x + 36$.

a. $(x + 9)(x + 4)$
b. $(x + 18)(x + 2)$
c. $(x + 12)(x + 1)$
d. $(x - 12)(x - 3)$

8.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 7, find the factored form of the quadratic expression.

1. $x^2 + 4x - 12$

a. $(x - 2)(x + 6)$
b. $(x + 2)(x - 6)$
c. $(x + 3)(x - 4)$
d. $(x - 2)(x - 6)$

2. $x^2 - 3x - 18$

a. $(x - 7)(x + 3)$
b. $(x - 18)(x + 1)$
c. $(x - 6)(x + 3)$
d. $(x - 9)(x + 2)$

3. $x^2 - 12x - 45$

a. $(x - 15)(x + 3)$
b. $(x - 9)(x + 5)$
c. $(x - 8)(x - 4)$
d. $(x - 45)(x + 1)$

4. $x^2 - x - 20$

a. $(x + 5)(x - 4)$
b. $(x - 20)(x + 1)$
c. $(x - 5)(x + 4)$
d. $(x - 10)(x + 2)$

5. $x^2 + 2x - 35$

a. $(x - 5)(x - 7)$
b. $(x + 5)(x - 7)$
c. $(x + 5)(x + 7)$
d. $(x - 5)(x + 7)$

6. $x^2 - 23x - 50$

a. $(x - 10)(x - 13)$
b. $(x - 25)(x + 2)$
c. $(x - 50)(x + 1)$
d. $(x - 10)(x + 5)$

7. $x^2 + 7x - 60$

a. $(x - 4)(x + 15)$
b. $(x - 10)(x + 3)$
c. $(x - 20)(x + 3)$
d. $(x - 5)(x + 12)$

8. Consider the expression $x^2 + bx - 80$, where the leading coefficient equals 1. Identify all the pairs of factors of –80 that would result in positive values of b.

a. 80 and -1, 40 and -2, 20 and -4, 16 and -5, 10 and -8
b. -80 and -1, -40 and -2, -20 and -4, -16 and -5, -10 and -8
c. -80 and 1, -40 and 2, -20 and 4, -16 and 5, -10 and 8
d. 80 and 1, 40 and 2, 20 and 4, 16 and 5, 10 and 8

9. Consider the expression $x^2 + bx - 80$, where the leading coefficient equals 1. Identify all the pairs of factors of

8.8 REWRITING QUADRATIC EXPRESSIONS IN FACTORED FORM, PART 3

8.8.0 Lesson Overview

In this lesson, you will encounter quadratic expressions without a linear term and consider how to write them in factored form. We will begin by studying numerical examples and noticing that certain expressions are equivalent.

Remember for some digital tools, when you want to enter a number with an exponent, use the ^ symbol. It will look like "x^2" when you want to enter x^2. To make the ^ symbol, push shift and the numeral 6 at the same time.

When you finish this lesson, you will be able to:

- Explain why multiplying binomials that are a sum and a difference, $(x + m)(x - m)$, results in a quadratic expression with no linear term.
- Rewrite a quadratic expression of the form $x^2 - c$ into its factored form.

Here are the **activities** that will help you reach those goals:

- 8.8.1: Evaluating Expressions Using Mental Math
- 8.8.2: Recognizing the Expanded Product of the Difference of Two Squares
  - 8.8.2: Self Check
  - 8.8.2: Additional Resources
- 8.8.3: Factoring Quadratic Equations without a Linear Term
  - 8.8.3: Self Check
  - 8.8.3: Additional Resources
- 8.8.4: Determining if an Expression Can Be Rewritten in Factored Form

After that, you'll **practice and review**.

- 8.8.5: Practice
- 8.8.6: Lesson Summary

8.8.1 Evaluating Expressions Using Mental Math

Warm Up

Finding a product mentally can be done by changing the way you look at a multiplication problem.

For example, $9 \cdot 39$ can be thought of as $9 \cdot (40 - 1)$. For some people, this might be easier to calculate mentally since you can imagine the solution as $360 - 9$, or 351. This method leverages the distributive property of multiplication over addition.

For the following questions, find each product mentally. Then, explain how you changed the way you looked at each multiplication problem to use the distributive property.

1. $9 \cdot 11$

2. $19 \cdot 21$

3. $99 \cdot 101$

4. $109 \cdot 101$

8.8.2 Recognizing the Expanded Product of the Difference of Two Squares

Activity

1. Clare claims that $(10 + 3)(10 - 3)$ is equivalent to $10^2 - 3^2$ and $(20 + 1)(20 - 1)$ is equivalent to $20^2 - 1^2$. Do you agree? Show your reasoning.

2. You can use your observations from the first question to mentally solve an expression such as $105 \cdot 95$.

a. First, think about the problem differently. Evaluate $(100 + 5)(100 - 5)$ using the distributive property, or FOIL. Show

8.8.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Which of the following is equivalent to $10^2 - 12^2$?

a. $(10 + 12)(10 - 12)$
b. $(10 - 12)^2$
c. $(10 - 12)(10 - 12)$
d. $(10 + 12)(10 + 12)$

For questions 2 – 10, select the expression in factored form.

2. $c^2 - 25$

a. $(c + 5)(c + 5)$
b. $(c - 5)(c + 5)$
c. $(c - 5)(c - 5)$
d. $c(c - 25)$

3. $9w^2 - 121$

a. $(3w - 11)(3w - 11)$
b. $w(9w - 121)$
c. $(3w + 11)(3w + 11)$
d. $(3w - 11)(3w + 11)$

4. $h^2 - 121$

a. $(h + 11)(h + 11)$
b. $(h - 11)(h - 11)$
c. $(h - 11)(h + 11)$
d. $(h - 11)^2$

5. $64 - d^2$

a. $(8 + d)(8 + d)$
b. $(8 - d)(8 + d)$
c. $(8 - d)(8 - d)$
d. Not possible

6. $4b^2 - 25$

a. $(2b + 5)(2b + 5)$
b. $b(4b - 25)$
c. $(2b - 5)(2b - 5)$
d. $(2b - 5)(2b + 5)$

7. $49 - 81z^2$

a. $(7 - 9z)(7 + 9z)$
b. $z(49 - 81z)$
c. $(7 - 9z)(7 - 9z)$
d. $(7 + 9z)(7 + 9z)$

8. $100y^2 - 4$

a. $(10y - 2)(10y - 2)$
b. $(10y - 2)(10y + 2)$
c. $(10y + 2)(10y + 2)$
d. Not possible

9. $x^2 + 9$

a. $(x + 3)(x + 3)$
b. $(x - 3)(x + 3)$

8.9.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 5, solve each quadratic equation by rewriting it in factored form and using the zero product property.

1. $x^2 + 11x + 35 = 7$

a. $x = -5$ and $x = -7$
b. $x = 14$ and $x = -3$
c. $x = -6$ and $x = -7$
d. $x = -4$ and $x = -7$

2. $x^2 - 14x + 20 = -25$

a. $x = 9$ and $x = 5$
b. $x = -10$ and $x = -4$
c. $x = 10$ and $x = 2$
d. $x = 3$ and $x = 15$

3. $x^2 - 6x - 52 = -12$

a. $x = -8$ and $x = 5$
b. $x = 10$ and $x = -4$
c. $x = 8$ and $x = -8$
d. $x = -16$ and $x = 10$

4. $x^2 - x - 64 = -8$

a. $x = 4$ and $x = -14$
b. $x = 28$ and $x = -2$
c. $x = -7$ and $x = 8$
d. $x = 8$ and $x = -8$

5. $x^2 + 9x + 13 = -5$

a. $x = 3$ and $x = 6$
b. $x = 8$ and $x = 1$
c. $x = -9$ and $x = -2$
d. $x = -3$ and $x = -6$

6. Without graphing, identify the function whose graph has only one zero.

a. $f(x) = x^2 - 8x + 16$
b. $f(x) = x^2 + 6x + 5$
c. $f(x) = x^2 - 13x + 42$
d. $f(x) = x^2 - 9$

7. Without graphing, identify the function whose graph has only one zero.

a. $f(x) = x^2 - 10x + 9$
b. $f(x) = x^2 + 12x + 36$
c. $f(x) = x^2 + x - 30$
d. $f(x) = x^2 + 7x - 30$

8. Without graphing, identify the function whose graph has two zeros.

a. $f(x) = x^2 - 12x + 36$
b. $f(x) = x^2 + 8x + 16$
c. $f(x) = x^2 - 8x + 16$
d. $f(x) = x^2 - 81$

9. Without graphing, identify the function whose graph has two zeros.

8.10.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, find the factored form of the quadratic expression.

1. $3x^2 + 16x + 5$

a. $(3x + 1)(x + 5)$
b. $(3x - 1)(x - 5)$
c. $(3x + 1)(x - 5)$
d. $(3x - 5)(x - 1)$

2. $2x^2 + 11x - 6$

a. $(2x + 1)(x - 6)$
b. $(2x - 1)(x + 6)$
c. $(2x - 6)(x + 1)$
d. $(2x + 6)(x - 1)$

3. $4x^2 - 13x + 3$

a. $(4x + 1)(x + 3)$
b. $(4x - 3)(x - 1)$
c. $(4x + 3)(x + 1)$
d. $(4x - 1)(x - 3)$

4. $2x^2 - 3x - 2$

a. $(2x - 1)(x + 2)$
b. $(2x - 2)(x + 1)$
c. $(2x + 2)(x - 1)$
d. $(2x + 1)(x - 2)$

5. The height, in meters, of a baseball being hit is modeled by the function $h(x) = -2x^2 + 12x + 3$ where x is the number of seconds since being hit.

Approximately how many seconds pass before the baseball lands on the ground? Use graphing technology if needed.

a. 7.56 seconds
b. 7.26 seconds
c. 6.24 seconds
d. 4.38 seconds

6. The height, in meters, of a roller coaster is modeled by the function $h(x) = -2x^2 + 20x + 5$ where x is the number of seconds after starting.

Approximately how many seconds pass before the roller coaster reaches the ground? Use graphing technology if needed.

a. 10.244 seconds
b. 8.365 seconds
c. 7.963 seconds
d. 5.221 seconds

For questions 7 – 10, find the factored form of the quadratic expression.

7. $3x^2 + 2x - 5$

a. $(3x - 5)(x + 1)$
b. $(3x + 5)(x - 1)$
c. $(x + 5)(3x - 1)$

8.11.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 – 4, identify a quadratic function in standard form, $f(x)$, with the given zeros.

1. 1 and 9

a. $f(x) = x^2 - 10x + 9$
b. $f(x) = x^2 + 8x + 9$
c. $f(x) = x^2 - 8x + 9$
d. $f(x) = x^2 + 10x + 9$

2. $\sqrt{5}$ and $-\sqrt{5}$

a. $f(x) = x^2 + 5$
b. $f(x) = x^2 - 5$
c. $f(x) = x^2 + 5x - 5$
d. $f(x) = x^2 - 5x - 5$

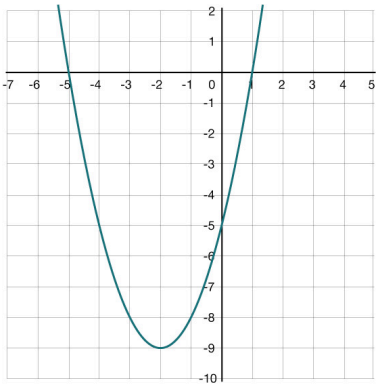
3. -7

a. $f(x) = x^2 - 14x + 49$
b. $f(x) = x^2 - 49$
c. $f(x) = x^2 - 14x - 49$
d. $f(x) = x^2 + 14x + 49$

4. -3

a. $f(x) = x^2 - 9$
b. $f(x) = x^2 + 6x + 9$
c. $f(x) = x^2 + 9$
d. $f(x) = x^2 - 6x + 9$

5. Examine the graph below to identify the zeros of the function. Then, write a quadratic function in standard form with the given zeros.



a. $f(x) = x^2 + 6x - 5$
b. $f(x) = x^2 - 4x - 5$
c. $f(x) = x^2 - 6x - 5$
d. $f(x) = x^2 + 4x - 5$

6. Examine the graph below to identify the zeros of the function. Then, write a quadratic function in standard form with the given zeros.

8.12 USING TECHNOLOGY TO FIND THE QUADRATIC REGRESSION

8.12.0 Lesson Overview

Scientists and mathematicians frequently collect data from their experiments. These data sets can be analyzed to better understand a situation. In this lesson, you will use graphing technology to analyze data sets so that you can identify a curve of best fit for any given data. You will then use the curve of best fit to make accurate predictions about real-world situations.

When you finish this lesson, you will be able to:

- Graph a quadratic data set using technology.
- Find a regression curve, or curve of best fit, using a quadratic data set.
- Use a regression curve to create a quadratic model for making predictions.
- Make predictions about a real-world situation using a quadratic data set.

Here are the **activities** that will help you reach those goals:

- 8.12.1: Graphing a Quadratic Data Set Using Technology
- 8.12.2: Finding the Curve of Best Fit
  - 8.12.2: Self Check
  - 8.12.2: Additional Resources
- 8.12.3: Making Predictions Using a Quadratic Model
  - 8.12.3: Self Check
  - 8.12.3: Additional Resources
- 8.12.4: Finding the Missing Data in the Set

After that, you'll **practice and review**.

- 8.12.5: Practice
- 8.12.6: Lesson Summary

8.12.1 Graphing a Quadratic Data Set Using Technology

Warm Up

1. Click on the "+" sign and choose to add a table in the Desmos graphing tool. This will generate a table for entering a data set.

Desmos: Quadratic Data Set Table

Access multimedia content (http://openstax.org/books/algebra-1/pages/8-12-1-graphing-a-quadratic-data-set-using-technology)

2. Enter the data set shown in the table below into the corresponding x_1 and y_1 columns.

(If preferred, the data can be copied and pasted into Desmos from a spreadsheet using keyboard shortcuts. Windows: Ctrl+C for copy and Ctrl+V for paste. Mac: Command+C for copy and Command+V for paste.)

x_1	y_1
1	2
2	5
3	10

8.12.5 Practice

Access the Desmos guide PDF (https://openstax.org/r/desmos-guide) for tips on solving problems with the Desmos graphing calculator.

Complete the following questions to practice the skills you have learned in this lesson.

Access multimedia content (http://openstax.org/books/algebra-1/pages/8-12-5-practice)

1. Enter the given data into Desmos, then calculate the regression approximation to determine the curve of best fit.

x_1	y_1
2	56
5	116
8	155
11	205
17	230

What is the equation for the curve of best fit that models the data?

a. $y = -0.93x^2 + 22.63x + 10.48$
b. $y = -0.69x^2 + 24.83x + 8.07$
c. $y = -3.11x^2 + 16.32x + 8.94$
d. $y = -0.7x^2 + 35.93x + 8.19$

2. Enter the given data into Desmos, then calculate the regression approximation to determine the curve of best fit.

x_1	y_1
3	105
9	250
14	349
23	400
37	258

What is the equation for the curve of best fit that models the data?

a. $y = -0.74x^2 + 34.32x + 7.44$
b. $y = -0.62x^2 + 17.45x + 9.64$
c. $y = -2.31x^2 + 24.711x + 3.25$
d. $y = -0.56x^2 + 38.74x + 5.31$

Access multimedia content (http://openstax.org/books/algebra-1/pages/8-12-5-practice)

9.1 WHAT ARE PERFECT SQUARES?

9.1.0 Lesson Overview

In this lesson, you will gain understanding of the structure of perfect-square expressions. This lesson will also help you see that perfect squares can be handy for solving equations because we can find their square roots.

When you finish this lesson, you will be able to:

- Recognize perfect-square expressions written in different forms.
- Recognize quadratic equations that have a perfect-square expression and solve the equations.

Here are the **activities** that will help you reach those goals:

- 9.1.1: Equations with Quadratic Expressions on Both Sides of the Equal Sign
- 9.1.2: Recognizing Structure in Perfect-Square Expressions
  - 9.1.2: Self Check
  - 9.1.2: Additional Resources
- 9.1.3: Solving Multi Step Quadratic Equations Using Perfect Squares
  - 9.1.3: Self Check
  - 9.1.3: Additional Resources
- 9.1.4: Explain Why an Expression Is a Perfect Square

After that, you'll **practice and review**.

- 9.1.5: Practice
- 9.1.6: Lesson Summary

9.1.1 Equations with Quadratic Expressions on Both Sides of the Equal Sign

Warm Up

In each equation, what expression could be substituted for a so the equation is true for all values of x?

1. $x^2 = a^2$

2. $(3x)^2 = a^2$

3. $a^2 = 7x \cdot 7x$

4. $25x^2 = a^2$

5. $a^2 = \frac{1}{4}x^2$

6. $a^2 = (x+1)^2$

7. $(2x-9)(2x-9) = a^2$

9.1.2 Recognizing Structure in Perfect-Square Expressions

Activity

1. Each expression is written as the product of factors. Write an equivalent expression in standard form.

a. $(3x)^2$

b. $7x \cdot 7x$

c. $(x+4)(x+4)$

d. $(x+1)^2$

e. $(x-7)^2$

f. $(x+n)^2$

2. Why do you think the following expressions can be described as **perfect square**s?

9.1.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. **Select three** expressions that are perfect squares.

a. $(x+5)(x+5)$
b. $(-9+c)(c-9)$
c. $(y-10)(10-y)$
d. $(2x-1)(2x+1)$
e. $(4-3x)(3-4x)$
f. $(a+b)(b+a)$

2. Each diagram represents the square of an expression or a perfect square. Complete the cells in the last table by answering questions a–d.

Diagram 1

$(n+7)^2$

	n	7
n	n^2	$7n$
7	$7n$	7^2

Diagram 2

$(5-m)^2$

	5	$-m$
5	5^2	$-5m$
$-m$	$-5m$	$(-m)^2$

Diagram 3

$(h+\frac{1}{3})^2$

	h	$\frac{1}{3}$
h	a	b
$\frac{1}{3}$	c	d

a. What is the answer for Cell a in Diagram 3?

a. $(\frac{1}{3})^2$
b. $(\frac{1}{3}h)^2$
c. h^2
d. $\frac{1}{3}h$

b. What is the answer for Cell b in Diagram 3?

a. $(\frac{1}{3})^2$
b. $(\frac{1}{3}h)^2$
c. h^2
d. $\frac{1}{3}h$

c. What is the answer for Cell c in Diagram 3?

a. $(\frac{1}{3})^2$
b. $(\frac{1}{3}h)^2$
c. h^2
d. $\frac{1}{3}h$

d. What is the answer for Cell d in Diagram 3?

a. $(\frac{1}{3})^2$
b. $(\frac{1}{3}h)^2$
c. h^2
d. $\frac{1}{3}h$

9.2.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For questions 1 - 5, the standard form of an equation is $ax^2 + bx + c$. For each expression, find the missing constant c that would make the expression a perfect square.

1. $x^2 - 6x + __c__$
2. $x^2 + 2x + __c__$
3. $x^2 + 14x + __c__$
4. $x^2 - 4x + __c__$
5. $x^2 + 24x + __c__$

6. If $x^2 - 6x + c$ is factored to be $(x_____)^2$, what is the missing value of the factored form of this expression? Remember to include the sign.

7. If $x^2 + 2x + c$ is factored to be $(x_____)^2$, what is the missing value of the factored form of this expression? Remember to include the sign.

8. If $x^2 + 14x + c$ is factored to be $(x_____)^2$, what is the missing value of the factored form of this expression? Remember to enter the sign.

9. If $x^2 - 4x + c$ is factored to be $(x_____)^2$, what is the missing value of the factored form of this expression? Remember to enter the sign.

10. If $x^2 + 24x + c$ is factored to be $(x_____)^2$, what is the missing value of the factored form of this expression? Remember to enter the sign.

For each equation in questions 11 - 14, use the following scenario.

Jada looks at Mai's work and is confused. She doesn't see how Mai got her answer.

Complete Mai's missing steps by answering the questions to help Jada see how Mai solved the equation.

$x^2 + 12x = 13$

Step 1 - $x^2 + 12x + _____ = 13 + _____$

Step 2 - $(x + 6)^2 = 49$

Step 3 - $\sqrt{(x + 6)^2} = \sqrt{49}$

Step 4 - $x + 6 = \pm _____$

Step 5 - $x + 6 = _____$

Step 6 - $x + 6 = - _____$

$x = 1$ or $x = -13$

11. What is the missing term in Step 1?
12. What is the missing term in Steps 4, 5, and 6?
13. What is the value of x in Step 5?
14. What is the value of x in Step 6?

For questions 15 - 20, solve choose the equivalent equation with a perfect square on one side.

9.3.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For the expressions in questions 1 – 4, choose which number would make the expression a perfect square. Then, choose an equivalent expression in factored form.

1. $x^2 + 3x$

a. What number can be added to make $x^2 + 3x$ a perfect square?

a. $-\frac{2}{3}$
b. $\frac{9}{4}$
c. $\frac{4}{9}$
d. $\frac{2}{3}$

b. Which expression is equivalent to $x^2 + 3x$?

a. $3(x + 1)$
b. $(x + 3)(x + 3)$
c. $x(3x)$
d. $x(x + 3)$

2. $x^2 + 0.6x$

a. What number can be added to make $x^2 + 0.6x$ a perfect square?

a. 0.03
b. 0.36
c. 0.81
d. 0.09

b. Which expression is equivalent to $x^2 + 0.6x$?

a. $0.6(x + 1)$
b. $x(0.6x)$
c. $x(x + 0.6)$
d. $x(x + 0.6x)$

3. $x^2 - 11x$

a. What number can be added to make $x^2 - 11x$ a perfect square?

a. $\frac{121}{4}$
b. $\frac{5}{2}$
c. $\frac{25}{2}$
d. $\frac{11}{2}$

b. Which expression is equivalent to $x^2 - 11x$?

a. $11x(x - 1)$
b. $11(x - 1)$
c. $x(x - 11)$
d. $x(x - 11x)$

4. $x^2 - \frac{5}{2}x$

a. What number can be added to make $x^2 - \frac{5}{2}x$ a perfect square?

a. $\frac{6}{8}$
b. $\frac{5}{8}$
c. $\frac{25}{16}$

9.4 COMPLETING THE SQUARE, PART 3

9.4.0 Lesson Overview

In this lesson, you will learn to complete the square to solve quadratic equations, in which the squared term has a coefficient other than 1.

When you finish this lesson, you will be able to:

- Complete the square for quadratic expressions of the form $ax^2 + bx + c$ when a is not 1 and explain the process.
- Solve quadratic equations in which the squared term coefficient is not 1 by completing the square.

Here are the **activities** that will help you reach those goals:

- 9.4.1: Perfect Squares with Coefficients Other than 1
- 9.4.2: Rewrite Squared Expressions
  - 9.4.2: Self Check
  - 9.4.2: Additional Resources
- 9.4.3: Standard Form and Squared Factors
  - 9.4.3: Self Check
  - 9.4.3: Additional Resources
- 9.4.4: Three Different Methods for Solving an Equation
- 9.4.5: Perfect Square and Squared Factors

After that, you'll **practice and review**.

- 9.4.6: Practice
- 9.4.7: Lesson Summary

9.4.1 Perfect Squares with Coefficients Other than 1

Warm Up

Elena says, "$(x + 3)^2$ can be expanded into $x^2 + 6x + 9$. Likewise, $(2x + 3)^2$ can be expanded into $4x^2 + 6x + 9$."

Find an error in Elena's statement and correct the error. Show your reasoning.

9.4.2 Rewrite Squared Expressions

Activity

1. Write each expression in standard form:

a. $(4x + 1)^2$

b. $(5x - 2)^2$

c. $(\frac{1}{2}x + 7)^2$

d. $(3x + n)^2$

e. $(kx + m)^2$

2. Decide if each expression is a perfect square. If so, write an equivalent expression of the form $(kx + m)^2$. If not, suggest one change to turn it into a perfect square.

9.4.6 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. **Select three** expressions that are perfect squares.

   a. $9x^2 + 24x + 16$
   b. $2x^2 + 20x + 100$
   c. $(7 - 3x)^2$
   d. $(5x + 4)(5x - 4)$
   e. $(1 - 2x)(-2x + 1)$
   f. $4x^2 + 6x + 94$

2. Find the missing number that makes the expression a perfect square.

$49x^2 - _____x + 16$
$36x^2 + _____x + 4$
$4x^2 - _____x + 25$
$9x^2 + _____x + 9$
$121x^2 + _____x + 9$

3. Choose the completely factored form of the expression.

   a. $49x^2 - 56x + 16$

      a. $(28x - 11)^2$
      b. $(7x - 4)^2$
      c. $(49x - 16)^2$
      d. $(7x + 4)^2$

   b. $36x^2 + 24x + 4$

      a. $4(3x + 1)^2$
      b. $(6x + 2)^2$
      c. $(36x + 4)^2$
      d. $(6x - 2)^2$

   c. $4x^2 - 20x + 25$

      a. $(2x + 5)^2$
      b. $(4x - 25)^2$
      c. $(2x - 5)^2$
      d. $(10x + 7)^2$

   d. $121x^2 + 66x + 9$

      a. $(11x - 3)^2$
      b. $(3x - 3)^2$
      c. $(121x - 9)^2$
      d. $(11x + 3)^2$

4. Find the missing number that makes the expression a perfect square.

$9x^2 + 42x + _____$
$49x^2 - 28x + _____$
$25x^2 + 110x + _____$
$64x^2 - 144x + _____$
$4x^2 + 24x + _____$

5. Based on your responses in question 4 above, choose the factored form of the expression.

   a. $9x^2 + 42x + _____$

9.5.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Solve each equation and write the solutions using the ± notation, then choose the correct solution from the list.

a. $x^2 = 144$

   a. $x = \pm 24$
   b. $x = \pm 12$
   c. $x = \pm 72$
   d. $x = \pm 144$

b. $x^2 = 5$

   a. $x = \pm \sqrt{10}$
   b. $x = \pm \frac{5}{2}$
   c. $x = \pm 5$
   d. $x = \pm \sqrt{5}$

c. $4x^2 = 28$

   a. $x = \pm \sqrt{7}$
   b. $x = \pm \frac{7}{2}$
   c. $x = \pm 7$
   d. $x = \pm \sqrt{\frac{7}{2}}$

d. $x^2 = \frac{25}{4}$

   a. $x = \pm \frac{5}{4}$
   b. $x = \pm \frac{10}{2}$
   c. $x = \pm \frac{5}{2}$
   d. $x = \pm \frac{25}{4}$

e. $2x^2 = 22$

   a. $x = \pm \sqrt{11}$
   b. $x = \pm \frac{11}{2}$
   c. $x = \pm 11$
   d. $x = \pm \sqrt{\frac{11}{2}}$

f. $7x^2 = 16$

   a. $x = \pm \frac{\sqrt{16}}{7}$
   b. $x = \pm 8$
   c. $x = \pm 16$
   d. $x = \pm \sqrt{\frac{16}{7}}$

2. For each expression, choose an equivalent expression.

a. $10 \pm \sqrt{4}$

   a. $4 + \sqrt{10}$ and $4 - \sqrt{10}$
   b. 3 and 5
   c. 8 and 12
   d. $4 + \sqrt{2}$ and $4 - \sqrt{2}$
   e. -17 and 5

9.6.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

For each equation in questions 1 – 5, identify the values of a, b, and c that you would substitute into the quadratic formula to solve the equation.

1. $3x^2 + 8x + 4 = 0$

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

2. $2x^2 - 5x + 2 = 0$

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

3. $-9x^2 + 13x - 1 = 0$

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

4. $x^2 + x - 11 = 0$

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

5. $-x^2 + 16x + 64 = 0$

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

6. For each equation, use the quadratic formula to solve the equation, then choose the correct **two** solutions for x.

a. $x^2 + 9x + 20 = 0$

a. 4
b. -4
c. -5
d. 5
e. 1
f. 9
g. 20

b. $x^2 - 10x + 21 = 0$

a. 3
b. -7
c. 21
d. -10
e. 1
f. -4
g. 7

9.7.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Mai and Jada are solving the equation $2x^2 - 7x = 15$ using the quadratic formula.

Answer the following questions about the values of a, b, and c if this equation is written in standard form, $ax^2 + bx + c = 0$.

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

d. Mai and Jada are solving the equation $2x^2 - 7x = 15$ using the quadratic formula but found different solutions.

Mai's Method	Jada's Method
$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(-15)}}{2(2)}$ $x = \frac{-7 \pm \sqrt{49 - (-120)}}{4}$ $x = \frac{-7 \pm \sqrt{169}}{4}$ $x = \frac{-7 \pm 13}{4}$ $x = \frac{-7+13}{4} = \frac{6}{4} = \frac{3}{2}$ $x = \frac{-7-13}{4} = \frac{-20}{4} = -5$	$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(2)(-15)}}{2(2)}$ $x = \frac{7 \pm \sqrt{49 - (-120)}}{4}$ $x = \frac{7 \pm \sqrt{71}}{4}$

Who has the correct solution?

a. Mai
b. Jada
c. Both of them
d. Neither one

2. The equation $h(t) = -16t^2 + 80t + 64$ represents the height, in feet, of a potato t seconds after it was launched from a mechanical device.

a. Choose the equation that would allow us to find the time the potato hits the ground.

a. $0 = t^2 - 5t - 4$
b. $0 = -16t^2 + 80t + 64$
c. $0 = -t^2 - 5t + 4$
d. $0 = 16t^2 + 80t + 64$

b. Solve the equation without graphing and write it down to find the approximate time the potato hits the ground rounded to one decimal place.

3. Priya found $x = 3$ and $x = -1$ as solutions to $3x^2 - 6x - 9 = 0$.

a. If you check Priya's answer by substitution, is the answer correct?

a. No
b. Yes

b. If you check Priya's answer by graphing, is the answer correct?

a. No
b. Yes

9.8.4 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The quadratic equation $x^2 + 7x + 10 = 0$ is in the form of $ax^2 + bx + c = 0$.

a. What is the value of a?

b. What is the value of b?

c. What is the value of c?

Examine the solution method that has been started below and use it to answer the remaining three parts of question 1.

Original equation
$x^2 + 7x + 10 = 0$

Step 1 - Subtract 10 from each side.
$x^2 + 7x = -10$

Step 2 - Multiply each side by 4.
$4x^2 + 4(7x) = -4(10)$

Step 3 - Rewrite $4x^2$ as $(2x)^2$ and $4(7x)$ as $2(7)2x$.
$(2x)^2 + 2(7)(2x) + ____^2 = ____^2 - 4(10)$

Step 4 - _____
$(2x + ____)^2 = ____^2 - 4(10)$

Step 5 - _____
$2x + ____ = \pm\sqrt{____^2 - 4(10)}$

Step 6 - _____
$2x = ____ \pm \sqrt{____^2 - 4(10)}$

Step 7 - _____
$x = ____$

d. In Step 2, what might be a good reason for multiplying each side of the equation by 4?

a. Multiplying by 4 makes the coefficient of the squared term a perfect square, which makes it easier to complete the square.
b. Multiplying by 4 makes the coefficient of the b term a perfect square, which makes it easier to complete the square.
c. If the value of c is positive and you multiply by 4, it will be easier to divide the b term by 3 before you square it.
d. You must always multiply both sides by 4 when completing the square.

e. Complete the unfinished steps, then **select two** solutions.

a. -10
b. -5
c. -2
d. -4

f. Substitute the values of a, b, and c into the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, but do not evaluate any of the expressions. Explain how this expression is related to solving $x^2 + 7x + 10 = 0$ by completing the square.

a. For this equation, the quadratic formula and completing the square are not the same and therefore not related.
b. They are related because neither one can have a negative c value.
c. Rather than evaluating at each step, the calculation is done all at once, at the end.
d. In both forms of solving the equation, you need to list the values of a, b, and c to determine the solutions.

2. Consider the standard form of the equation $x^2 - 39 = 0$.

a. What is the value of a?

9.9.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Which of the following points is the vertex of the quadratic function $f(x) = x^2 + 8x + 22$?

a. $(4, 6)$
b. $(-4, 6)$
c. $(0, 22)$
d. $(8, 22)$

For questions 2 – 4, use the function $f(x) = (x - 4)(x + 2)$.

2. Write the function in standard form.

3. Write the function from question 2 in vertex form.

4. Identify the vertex of the quadratic function $f(x) = (x - 1)^2 - 9$.

a. $(-9, -1)$
b. $(-1, 9)$
c. $(1, 9)$
d. $(1, -9)$

For questions 5 - 6, write an equation in vertex form of the quadratic function that has a vertex at $(-4, 1)$ and also goes through the point $(6, -1)$.

5. Find the value of a for vertex form in simplest form.

6. Write the equation in vertex form. Hint: Be sure to use $y =$ to denote your equation.

For questions 7 and 8, write an equation in vertex form of the quadratic function that has a vertex at $(-2, -3)$ and also goes through the point $(4, 2)$.

7. Find the value of a for vertex form in simplest form.

8. Write the equation in vertex form. Hint: Be sure to use $y =$ to denote your equation.

For questions 9 - 10, use the function $f(x) = x^2$ that is shifted to create the function $g(x)$ that has a vertex at $(-4, 6)$

9. Write $g(x)$ in vertex form.

10. Write $g(x)$ in standard form.

Additional Resources

Standard to Vertex Form

Different forms are useful for finding different characteristics of quadratic functions.

Practice writing $x^2 + 8x - 7$ into **vertex form (of a quadratic expression)**.

Step 1 - Work to complete the square:

$b = 8$, so $(\frac{b}{2})^2 = (\frac{8}{2})^2 = 16$.

Step 2 - Add the value to create the perfect square, and subtract the value from the original constant (this is like adding 0).

$x^2 + 8x + 16 - 7 - 16$

Step 3 - Factor the trinomial and simplify.

$(x + 4)^2 - 23$

TRY IT **Standard to Vertex Form**

Convert the quadratic expression $x^2 + 16x - 10$ to vertex form.

9.10.4 Rewriting Expressions in Vertex Form

Activity

1. Here is one way to rewrite $3x^2 + 12x + 9$ in **vertex form (of a quadratic expression)**.

a. Study the steps and write a brief explanation of what is happening at each step.

Original expression $3x^2 + 12x + 9$

Step 1 - $3(x^2 + 4x + 3)$

Step 2 - $3(x^2 + 4x + 4 + 3 - 4)$

Step 3 - $3(x^2 + 4x + 4 - 1)$

Step 4 - $3((x + 2)^2 - 1)$

Step 5 - $3(x + 2)^2 - 3$

b. What is the x-coordinate of the **vertex (of a graph)** that represents this expression?

c. What is the y-coordinate of the vertex of the graph that represents this expression?

d. Does the graph open upward or downward? Explain how you know.

2. Rewrite $-2x^2 - 4x + 6$ in vertex form.

3. Rewrite $4x^2 + 24x + 20$ in vertex form.

4. Rewrite $-x^2 + 20x$ in vertex form.

Are you ready for more?

Extending Your Thinking

1. Fill in the missing parts of the function $f(x) = 2(x - 3)(x - 9)$ in vertex form $f(x) =$ **a.** $(x$ **b.** **c.** $)^2$ **d.** **e.** without completing the square. (Hint: Think about finding the zeros of the function.)

a. What is the value of a?

b. What is the sign of b?

c. What is the value of c?

d. What is the sign of d?

9.10.7 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. The following quadratic expressions all define the same function.

- $(x+5)(x+3)$
- $x^2+8x+15$
- $(x+4)^2-1$

Select three statements that are true about the graph of this function.

a. The y-intercept is $(0,-15)$.
b. The vertex is $(-4,-1)$.
c. The x-intercepts are $(-5,0)$ and $(-3,0)$.
d. The x-intercepts are $(0,5)$ and $(0,3)$.
e. The x-intercept is $(0,15)$.
f. The y-intercept is $(0,15)$.
g. The vertex is $(4,-1)$.

2. The following expressions all define the same quadratic function.

- $(x-4)(x+6)$
- $x^2+2x-24$
- $(x+1)^2-25$

a. What is the value of the y-coordinate of the y-intercept $(0,___)$ of the graph of the function?

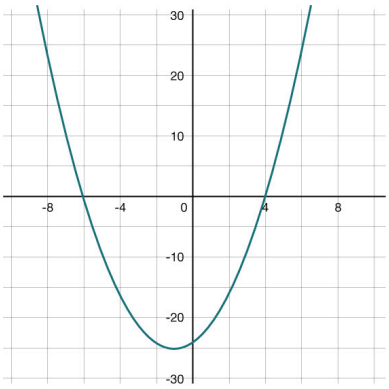
b. **Select two** x-intercepts of the graph.

a. $(6,0)$
b. $(-4,0)$
c. $(-6,0)$
d. $(5,0)$
e. $(4,0)$

c. What is the x-coordinate of the vertex of the graph?

d. What is the y-coordinate of the vertex of the graph?

e. Is this an accurate graph of the function?



a. No
b. Yes

3. Here is one way an expression in standard form is rewritten into vertex form.

Original expression
x^2-7x+6

9.11 USING QUADRATIC EXPRESSIONS IN VERTEX FORM TO SOLVE PROBLEMS

9.11.0 Lesson Overview

In this lesson, you will learn how to use the vertex form of a quadratic to determine the maximum or minimum value of a function and to solve problems.

When you finish this lesson, you will be able to:

- Find the maximum or minimum of a function by writing the quadratic expression that defines it in vertex form.
- Explain why the vertex is a maximum or minimum when given a quadratic function in vertex form.

Here are the **activities** that will help you reach those goals:

- 9.11.1: Maximum and Minimum Value of a Function
- 9.11.2: Does the Vertex Represent the Minimum or Maximum Value?
  - 9.11.2: Self Check
  - 9.11.2: Additional Resources
- 9.11.3: Comparing Maximums between Quadratics
  - 9.11.3: Self Check
  - 9.11.3: Additional Resources
- 9.11.4: Maximum, Minimum, and Analyzing the Vertex

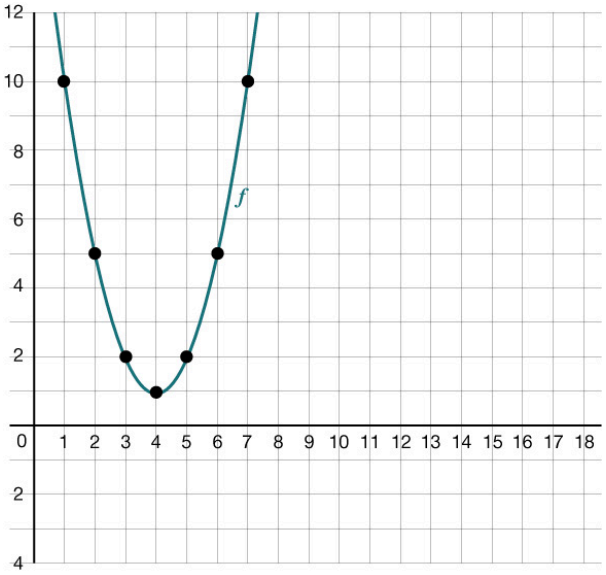
After that, you'll **practice and review**.

- 9.11.5: Practice
- 9.11.6: Lesson Summary

9.11.1 Maximum and Minimum Value of a Function

Warm Up

Examine the graph that represents the function, f, defined by $f(x) = (x-4)^2 + 1$.



Use this graph to answer questions 1 – 2.

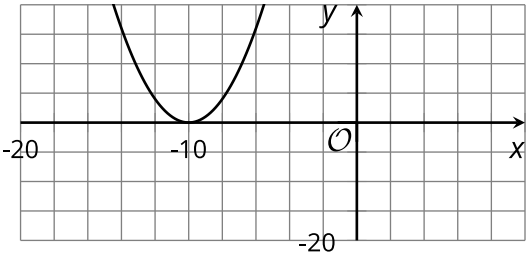
1. $f(1)$ can be expressed in words as "the value of f when x is 1." Find or compute:

a. The value of f when x is 1.

9.11.5 Practice

Complete the following questions to practice the skills you have learned in this lesson.

1. Here is a graph of a quadratic function $f(x)$. What is the minimum value of $f(x)$?



2. The graph that represents $f(x) = (x + 1)^2 - 4$ has its vertex at $(-1, -4)$.

Explain how we can tell from the expression $(x + 1)^2 - 4$ that -4 is the minimum value of f rather than the maximum value.

a. The coefficient of the quadratic term is positive; therefore, the vertex will be a minimum value.
b. We cannot tell from the expression; we have to graph it.
c. The expression $(x + 1)^2$ is a squared expression, so its value will always be positive or zero.
d. The constant is always the minimum value.

3. $g(x) = (x - 5)^2 + 6$ defines a quadratic function. Find the vertex of the graph of the function. Then, state whether the vertex corresponds to the maximum or the minimum value of the function.

a. What is the x-coordinate of the vertex?

b. What is the y-coordinate of the vertex?

c. Does the vertex correspond to the maximum or the minimum value of the function?

a. Minimum
b. Maximum

4. $h(x) = (x + 5)^2 - 1$ defines a quadratic function. Find the vertex of the graph of the function. Then, state whether the vertex corresponds to the maximum or the minimum value of the function.

a. What is the x-coordinate of the vertex?

b. What is the y-coordinate of the vertex?

c. Does the vertex correspond to the maximum or the minimum value of the function?

a. Minimum
b. Maximum

5. $f(x) = -2(x + 3)^2 - 10$ defines a quadratic function. Find the vertex of the graph of the function. Then, state whether the vertex corresponds to the maximum or the minimum value of the function.

a. What is the x-coordinate of the vertex?

b. What is the y-coordinate of the vertex?

c. Does the vertex correspond to the maximum or the minimum value of the function?

a. Minimum
b. Maximum

6. $p(x) = 3(x - 7)^2 + 11$ defines a quadratic function. Find the vertex of the graph of the function. Then, state whether the vertex corresponds to the maximum or the minimum value of the function.

PROJECT 9: USING QUADRATIC EQUATIONS TO MODEL SITUATIONS AND SOLVE PROBLEMS

Project 9 Overview

In this culminating project, you will synthesize methods of solving quadratic equations and graphing quadratic functions to answer questions about within a context. You will use tools learned throughout the unit to grapple with solving problems about a quadratic function that represents a context. Since this work requires using what you know to tackle an unfamiliar problem, you will need to make sense of problems and demonstrate perseverance.

Technology isn't required for this lesson, but there are opportunities for you to choose to use appropriate technology to solve problems.

Project Activities

1. Equations of Two Lines and a Curve
2. Analyzing a Dive
3. A Linear Function and a Quadratic Function
4. Profit from a River Cruise

Learning Targets

When you finish this project, you will have demonstrated how to:

- Interpret information about a quadratic function given its equation or a graph.
- Rewrite quadratic functions in different but equivalent forms of my choosing and use that form to solve problems.
- Decide which quadratic form to use depending on the questions being asked.

Equations of Two Lines and a Curve

1. Write an equation representing the line that passes through each pair of points.
   a. $(3, 3)$ and $(5, 5)$
   b. $(0, 4)$ and $(-4, 0)$
2. Solve this equation: $x + 1 = (x - 2)^2 - 3$. Show your reasoning.

Analyze a Dive

The function h, defined by $h(t) = -5t^2 + 10t + 7.5$, models the height of a diver above the water (in meters) t seconds after the diver leaves the board. For each question, explain your reasoning.

1. How high above the water is the diving board?
2. When does the diver hit the water?
3. At what point during her descent toward the water is the diver at the same height as the diving board?
4. When does the diver reach the maximum height of the dive?
5. What is the maximum height the diver reaches during the dive?

A Linear Function and a Quadratic Function

Here are graphs of a linear function and a quadratic function. The quadratic function is defined by the expression

$(x - 4)^2 - 5$

Find the coordinates of P, Q, and R without using graphing technology. Show your reasoning.